

MENDEL UNIVERSITY IN BRNO
Faculty of Forestry and Wood Technology

**Assessment of Socio-Economic and Environmental Impacts of
Timber Harvesting on Inhabitants of Kambui Hills Forest Reserve
in Sierra Leone.**

Dissertation.

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CONTENTS

1	INTRODUCTION.....	12
1.1	<i>Identification of the problem.....</i>	<i>15</i>
1.2	<i>Statement of the problem.....</i>	<i>17</i>
1.3	<i>Significance of the study</i>	<i>17</i>
1.4	<i>Research objectives.....</i>	<i>18</i>
1.5	<i>Research questions.....</i>	<i>18</i>
1.6	<i>Hypothesis.....</i>	<i>18</i>
2	LITERATURE REVIEW.....	19
2.1	<i>Global timber products trade.....</i>	<i>19</i>
2.2	<i>Literature on timber business from African countries</i>	<i>26</i>
2.3	<i>An overview of timber harvesting business in West Africa.....</i>	<i>31</i>
2.4	<i>Theories and models of sustainable impact assessment</i>	<i>41</i>
2.5	<i>Pre-colonial and colonial forest management development in Sierra Leone</i>	<i>48</i>
2.6	<i>Post-colonial Forest Management Strategies in Sierra Leone</i>	<i>55</i>
2.6.1	<i>The Earth Summit.....</i>	<i>58</i>
2.6.2	<i>The Kyoto Protocols, and the Paris Agreement</i>	<i>62</i>
2.7	<i>An overview of current forestry sector development and local actors.....</i>	<i>63</i>
2.7.1	<i>The Ministry of Environment, Sierra Leone</i>	<i>67</i>
2.7.2	<i>The Environmental Protection Agency in Sierra Leone (EPA-SL).....</i>	<i>69</i>
2.7.3	<i>The National Protected Area Authority, Serra Leone (NPAA-SL)</i>	<i>69</i>
2.7.4	<i>The National Forestry Division, Sierra Leone.....</i>	<i>70</i>
2.7.5	<i>The Gava/Forestry Industries Limited.</i>	<i>74</i>
2.7.6	<i>Exim International</i>	<i>76</i>
2.7.7	<i>Agricultural Community for Timber Development Association (ACOTIDA)....</i>	<i>77</i>
2.7.8	<i>Promoting Agriculture, Governance and Environment (PAGE)</i>	<i>77</i>
3	METHODOLOGY	79
3.1	<i>Study area.....</i>	<i>79</i>
3.2	<i>Methodological framework for descriptive statistical analysis</i>	<i>82</i>
3.3	<i>Design and Pre-testing of questionnaire.</i>	<i>83</i>
3.4	<i>Sampling procedure</i>	<i>83</i>
3.5	<i>Identification and definition of variables used in the descriptive statistical analysis .</i>	<i>84</i>

3.5.1	High labour costs	84
3.5.2	Loss of income and revenue	85
3.5.3	Increase cost of wood and timber products	85
3.5.4	Silting of rivers and disruption of Hydrological cycle.....	86
3.5.5	Damaging of trees and non-wood products	86
3.5.6	Loss of Biodiversity.....	87
3.5.7	Disappearance of vegetation.....	87
3.5.8	Climate change and global warming.....	88
3.6	<i>Data collection and model specification for descriptive analysis.....</i>	<i>89</i>
3.7	<i>Demographic characteristics of respondents.....</i>	<i>89</i>
3.8	<i>Methodological framework of regression analysis model.....</i>	<i>89</i>
3.9	<i>Identification and definition of variables in regression analysis.....</i>	<i>90</i>
3.9.1	Natural Resource Rent.	90
3.9.2	Exports.....	91
3.9.3	Gross Domestic Product.	91
3.9.4	Unemployment.....	92
3.9.5	Inflation.....	92
3.9.6	Carbon emissions.....	93
3.9.7	Primary enrolment and literacy Level.....	93
3.10	<i>Data collection and model specification in Regression analysis.....</i>	<i>94</i>
4	RESULTS AND DISCUSSIONS.....	96
4.1	<i>Descriptive statistical analysis.....</i>	<i>96</i>
4.1.1	The Environmental impact.....	100
4.1.2	The economic impact.....	101
4.1.3	The social impact.	101
4.1.4	Firms' compliance to SFM Principles.....	102
4.1.5	Discussions from the descriptive statistical analyses.....	108
4.2	<i>Regression model results.</i>	<i>112</i>
4.2.1	The environmental impacts.....	113
4.2.2	The economic impacts.....	116
4.2.3	The Social impacts.....	119
4.2.4	Discussions from regression model.	122
5	SUMMARY AND CONCLUSIONS.....	126
6	Policy Recommendations	128
7	REFERENCES.....	134

LIST OF TABLES

table 2-1 global producers of industrial round wood, 2022.	19
table 2-2 global importers of timber products, 2022	21
table 2-3 west african countries by forest cover, 2020.	32
table 2-4 wood production in selected west african countries, 2020.	33
table 2-5 selected west african countries, population, land area, forest area, gdp per capita and economic freedom score, 2022	34
table 2-6 selected west african countries, import, export, and trade balance for 2002, 2012 and 2022	36
table 2-7 protected areas	56
table 2-8 comparative climate indicators of sierra leone, guinea, and liberia in 2020.	62
table 2-9 sierra leone's production and trade of primary wood products in 2002, 2012 and 2022	63
table 4-1 demographic characteristics of the respondents in frequencies and %	96
table 4-2 household characteristics of the respondents	98
table 4-3 confirmatory factors analysis of socio-economic and environmental impact.	99
table 4-4 regression analysis results using software evIEWS 10.	113
table 4-5 coefficients of determination for the models.	113
table 4-6 variables and coefficients for model 1.	114
table 4-7 environmental impact statistical tests of significance and coefficients of determination	115
table 4-8 variables and coefficients for model 2	117
table 4-9 economic impacts statistical tests of significance and coefficients of determination	117
table 4-10 variables and coefficients for model 3	119
table 4-11 social impact statistical tests of significance and coefficients of determination	120
table 4-12 hypothesis results	122

LIST OF FIGURES

Figure 1-1: Silt water infested with timber waste.	15
Figure 1-2: Soil erosion	16
Figure 1-3: Disruption of Hydrological Circle	16
Figure 1-4: tree stump from timber harvesting	16
Figure 2-1: Map of West Africa	32
Figure 2-2: Organogram of Decentralization in the forestry sector	65
Figure 2-3: Organogram of The Environmental Ministry's decentralization.	69
Figure 2-4: Machinery abandoned by Gava company.	75
Figure 3-1: Map of Gola forest showing Kambui Hills Forest Reserve.	82
Figure 4-1: Firms CSR Policy formulation processes.	102
Figure 4-2: Laid-down policies of firms CSR.	103
Figure 4-3: Firms CSR Objectives	104
Figure 4-4: Duration of Firms Operations.	104
Figure 4-5: Life span of Firms CSR.	105
Figure 4-6: Measures to ensure CSR Sustainability.	105
Figure 4-7: Social challenges	106
Figure 4-8: Economic challenges	107
Figure 4-9: Environmental Challenges	107

LIST OF ABBREVIATIONS

ACGSF	Agricultural Credit Guarantee Scheme Fund
ACP	African, Caribbean and Pacific Countries
ADF	Augmented Dickey-Fuller Test
AFDB	African Development Bank
AMS	Aggregate Measurement of Support (in the Agreement on Agriculture)
AoA	Agreement on Agriculture
ASEAN	Association of Southeast Asian Nations
AU	African Union
AVE	Ad Valorem Equivalent
CAADP	Comprehensive African Agriculture Development Program
CBN	Central Bank of Nigeria
DDA	Doha Development Agenda
DR	Doha Round
EC	European Communities
ECOWAS	Economic Community of West African States
EU	European Union
EUROSTAT	Statistical Office of the European Communities
FAO	Food and Agriculture Organization of the United Nations
FAOSTAT	The Food and Agriculture Organization Corporate Statistical Database
FDI	Foreign Direct Investment
FMARD	Federal Ministry of Agriculture and Rural Development
GATT	General Agreement on Tariffs and Trade
GDP	Gross Domestic Product
HS	Harmonized Commodity Description and Coding System
IMF	International Monetary Fund
IRF	Impulse Response Function
ITC	International Trade Centre
LDCs	Least Developed Countries
MFA	Arrangement Regarding International Trade in Textiles
NAMA	Non-Agricultural Market Access
n.a	not available
NBS	National Bureau of Statistics
NEPC	Nigerian Export Promotion Council
n.e.s	not elsewhere specified
NTL	National Tariff Line
OECD	Organisation for Economic Co-operation and Development
OLS	Ordinary Least Square Regression
PP	Phillips-Perron Test
PSE	Producer Subsidy Equivalent (Agriculture)
REER	Real Effective Exchange Rate
RTAs	Regional Trade Agreements
SAP	Structural Adjustment Programme
SDT	Special and Differential Treatment (for developing countries)

SITC	Standard International Trade Classification
SPS	Sanitary and Phytosanitary Measures
SSA	Sub-Saharan Africa
SSG	Special Safeguard (in Agreement on Agriculture)
TDMs	Trade-distorting Measures
ST	Special Treatment (in Annex 5, Agreement on Agriculture)
TFA	Trade Facilitation Agreement
TI	Transparency International
TRQ	Tariff Rate Quota
UK	United Kingdom
UN	United Nations
UNCTAD	United Nations Conference on Trade and Development
UNDESA	United Nations Department for Economic and Social Affairs
UNEP	United Nations Environment Programme
UNIDO	United Nations Industrial Development Organization
URAA	Uruguay Round Agreement on Agriculture
USA	United States of America
VAR	Vector Autoregression
VDA	Variance Decomposition Analysis
World Bank	International Bank for Reconstruction and Development
WTO	World Trade Organization

1 INTRODUCTION

Forests generally form a major resource hub for the African continent. This could be partly traced to the socio-economic activities that dominate the continent, notably agriculture. Wood products are prominent among the resources harvested, although non-wood products form a substantial quantum of its resources. The primary wood products harvested include round wood for timber, fuelwood and poles used for construction and other purposes (Cerutti et al., 2015). Fuelwood accounts for 80 % of household energy requirements in rural areas (Shackleton and Kurunerichitepo, 2011), and thus accounts for close to 10 % of net energy consumption on the continent (Munro and Horst, 2012). In 2017, round wood production alone accounted for 746,340 metric tons in export value from the continent (FAO, 2017). In South Africa alone, it is estimated that each year, households use 11 million tons of fuelwood (Obasi et al., 2015). On the entire continent, it is difficult to quantify the benefits of the forest sector, partly due to poor data hub. The contribution of forests to the wellbeing of countries cannot only be measured in monetary terms for a number of indirect effects such as environmental benefits, social benefits, and free access rights to farm and graze, which are equally important (May et al., 2017). Thus, the assessment of such indirect benefits is crucial. In terms of the environmental benefits, services provided by forests are seldom fully valued or adequately reflected in forest planning and management decisions (FAO, 2017). Environmental services as provided by forests are currently receiving increasing attention on climate change mitigation debates. Such services include biodiversity conservation, watershed protection and carbon dioxide sequestration (Cerutti et al., 2015). Forest provide habitat for diverse number of insects, birds and animals. The forests also play a significant role in climate change mitigation due to their ability to sequester CO_2 . Moreover, the social benefits of forests across the continent are enormous. It provides beautiful sites for tourism, recreation, spiritual healing, leisure and religious practices (Obasi et al., 2015). The beauty of the forest, its diverse species and waterfalls, provide invaluable social benefit to many people. The sector is also crucial for educational purpose, as it attracts local and international students to do their forestry research. It also provides free grazing for livestock as abundance of open grassland is deemed valuable to the villagers for rearing their cattle (Ajake and Enang, 2012).

However, forest management on the continent is besieged with challenges, and have resulted in adverse consequences in most African countries. Failure to overcome these challenges have often led to loss of biodiversity wherein harvesting of forest wood products have resulted to adverse effects on the population and variety of plant and animal species. The removal of forest cover during logging has in some instances resulted in scarcity or out-right extinction of many important plant and animal species. Some wild animals have also been observed to migrate from areas where tree cover had been removed to undisturbed vegetations. Largely, some plant and animal genetic resources that could be used in producing new pharmaceuticals or traditional medicine are lost because of the destruction of forest cover.

Furthermore, soil erosion has also led to loss of forest cover, and ultimately, loss of protection the plant cover gives the soil (Butler, 2012). Timber harvesting also interrupts the normal nutrient cycle of the forest, promotes nitrification, and increases nutrient leaching, leaving the topsoil impoverished and susceptible to erosion. Eroded soil is often deposited in rivers and fishponds, causing siltation and contamination of such water bodies. This hurts the aquatic ecosystem and disrupts biodiversity by killing fish and other organisms (Cronin, 2009). Other notable effects across the continent include environmental pollution, soil compaction, disruption of the hydrological cycle (May et al., 2017).

In Sierra Leone, like in many African countries, forests have been a major source of livelihood for most people. The forestry sector is one of the main pivots on which the nation's welfare was built and continues to rest upon its resources. It is crucial to note that forests are not only important for material goods but also as a valuable ecological and cultural resource. The sector can be generally classified into formal and informal enterprises. The formal sector enterprise comprises of organized wood-based industries such as sawmills, plywood mills, particleboard mills and furniture factories while the informal enterprises include small forest-based enterprises operating without any formal corporate entity. They are mostly engaged in the production of firewood, charcoal, chewing stick and sculptured wood items. Notably, there have been changes in wood-based industries in Sierra Leone between the eve of the rebel war in 1990 and towards the end of the war in 2002. Specifically, there is a dramatic shift in the study area under review, partly because Kenema district, home to the KHFR, was one of the regions that suffered enormously from the war (Humper et al., 2004). The reduction in wood mills between 1990 and 2000 was largely attributed to the

displacement of residents and indigenous business enterprises. This number picked up again immediately after the war.

Forestry resources, especially wood products, have contributed hugely to the socio-economic development of Sierra Leone, particularly in the Kenema district. In 2018, Sierra Leone alone produced and exported 275,000 metric tons of industrial round wood, a total value of 105,699 US\$ (FAOSTAT, 2018). Apparently, it ranks among one of the highest revenue-generating sectors in the district. It also serves as a resource base for both formal and informal forest industries to harvest wood products that boost socio-economic activities. Recently, the demand for raw wood products from these forest sector players far outstripped the production or regeneration capacity of the KHFR in Kenema, eastern Sierra Leone.

Thus, adverse environmental effects will inevitably result from harvesting wood products for industrial purposes. It is therefore, compelling and essential to evaluate the impact of the activities of these formal and informal forest industry players on the environment. The protection of the forests has enormous benefits today and in the future. Forest maintenance can result in the amelioration of weather patterns, provision of clean air, protection of biological diversity, protection of watersheds, soil and food crops and provision of recreational facilities that will be crucial to the survival of future generations. Inversely, the indiscriminate logging in the KHFR and uncontrolled felling of trees for fuelwood and other socio-economic purposes should be a cause of concern. It has exposed the community to natural hazards like mudslides and seems to be a catalyst that will destroy resources for future generations. The adverse effects include loss of biodiversity, migration of wildlife, ecological imbalance, soil erosion, flooding, desert encroachment and disruption in the hydrological cycle of water catchment area. Harvesting timber products has both positive and negative effects on the people and environment within the Kambui hills metropolis. It contributes to the district's socio-economic development and Sierra Leone at large. For instance, it generates income for the community through taxes, provides employment for the indigenes, eases the energy burden for daily activities like cooking and heating, and contributes a decent percentage to the country's overall Gross Domestic Product (GDP). The adverse environmental effects are seemingly the consequences of the socio-economic activities through the operations of formal timber-based industries such as sawmills, plywood and veneer mills, particleboard mills, pulp & paper mills, and other human activities.

From this backdrop, this thesis assesses the gains and losses of harvesting timber products for socio-economic purposes. Focus will be placed on measures to minimize, if not eradicate, the adverse environmental effects of harvesting timber from the study area. Other works before this research have mainly been limited to identifying, and in many cases in isolation, either the gains of wood products harvest or have tried to identify some of the environmental effects of such activities. However, this work goes beyond that by focusing on the socio-economic characteristics of formal forest industry players within the study area and examining the challenges the formal wood-based industries face in harvesting timber products from the study area. The role of formal industry players is quite relevant in evaluating the effects of timber harvesting in the country generally and within the Kenema district. This assessment is based on qualitative and quantitative research, wherein a structured questionnaire is developed and executed within the research area. Data from this process is processed to give descriptive statistical socio-economic and environmental impact results. A quantitative empirical method is employed through a regression model analysis to process a time series of data derived from United Nations (UN) databases such as the Food and Agricultural Organization and other affiliate organizations.

1.1 Identification of the problem

The forestry sector is a major sector in Sierra Leone's economic development. It is pivotal to the nation's welfare. This is why the KHFR has always been a source of livelihood for the residents and the entire nation. Timber harvesting has led to a host of problems within the KHFR. See figures 1 to 4 below.



Figure 1-1: Silt water infested with timber waste.
Source: Field survey, 2022



Figure 1-2: Soil erosion
Source: field survey, 2022



Figure 1-3: Disruption of Hydrological Circle
Source: Primary survey, 2022



Figure 1-4: tree stump from timber harvesting
SOURCE: FIELD SURVEY, 2022

Recent visit to the forest reveals real environmental damage that has impacted the socio-economic activities of the locals. From the above photos (figures 1-1 to 1-4), it is revealing that silting of rivers with timber wastes, soil erosion, the impacts of socio-economic activities, the rate of deforestation, and hydrological disruption are among the visible impacts of timber harvesting in the area.

1.2 Statement of the problem

In Sierra Leone, like in many African countries, forests have been a major source of livelihood for most people. Forest reserves are not only important for material goods but also as a valuable ecological and cultural resource. These essential relevance of forest reserves holds same for the Kambui Hills Forest Reserve in the Kenema district, Eastern Sierra Leone. However, the utilization of forest reserves for socio-economic development of society is not devoid of environmental challenges. Over the most recent two decades, successive government policies on timber logging have been enacted without tangible results. In fact, the KHFR has gone from bad to worse in the court of public opinion. Although higher timber harvesting should correspond to increased tax gains by government, the reverse seems to be the case in Sierra Leone. The relevance of this research is based on the premise that the Kambui Hills Forest Reserve is confronted with sustainability challenges. Many socio-economic activities undertaken in the forest can only have unrepairable damages on the environment and society. Highlights of these unfortunate activities include excessive lumbering activities, rampant illegal logging, reckless firewood burning, lack of sustainable projects like tree planting, and ineffective government policies. The impacts of these activities are adverse on the environment and the country as it leads to climate change, loss of biodiversity, migration of wildlife, soil erosion, flooding, desert encroachment and disruption in hydrological cycle of water catchment area and a huge revenue loss to the state.

1.3 Significance of the study

Sierra Leone is globally recognized as a biodiversity hotspot and a part of the Upper Guinea Rainforest. Rich indigenous flora and fauna include important endemic species and other internationally recognized rare species. Unfortunately, poor economic growth and environmental degradation, mainly due to deforestation remain the major challenges for the country. One of the effects of environmental degradation has been frequent mudslides that have resulted in the loss of

lives and property. Therefore, the country must ensure a balance between SFM, socio-economic aspirations and environmental protection. This research will also review Sierra Leone's compliance status with the Kyoto Protocol and the Paris Agreement, to which the country is a signatory. It will provide a synopsis of the United Nations Conference on the Environment and Development (The Earth Summit) held in Rio de Janeiro, Brazil, in 1992.

1.4 Research objectives.

The main objective is to evaluate harvesting timber's socio-economic and environmental impacts on the residents within the Kambui Hills Forest Reserve, Kenema, Eastern Sierra Leone. Other specific objectives are to:

- (i) Provide an empirical assessment of socio-economic and environmental factors that impact residents within the study area.
- (ii) Provide an assessment of the external influence shaping timber business concerning Corporate Social responsibility (CSR).
- (iii) To make recommendations to stakeholders about the impacts of timber harvesting without full compliance with sustainable policies and CSR.

1.5 Research questions.

This research will seek to answer the following research questions.

1. What are the resultant social impacts of timber harvesting from KHFR?
2. What are the resultant economic impacts of timber harvesting from KHFR?
3. What are the resultant environmental impacts of timber harvesting from KHFR?

1.6 Hypothesis

The research will test the following hypothesis:

H₀₁: Environmental impacts of timber harvesting are caused by certain identified factors

H₀₂: Economic impacts of timber harvesting are caused by certain identified factors.

H₀₃: Social impacts of timber harvesting are caused by certain identified factors.

2 LITERATURE REVIEW

Forests cover 31 % of the global land area (FAO Global Forest Resources Assessment, 2020). Approximately half the forest area is relatively intact, and more than one-third is primary forest, that is, naturally regenerated forests of native species, where there are no visible indications of human activities at all (FAOSTAT, 2018). This can translate the total forest area to 4.06 billion hectares, or approximately 5000 m² (or 50 x 100 m) per person, but forests are not equally distributed around the globe. More than half of the world's forests are found in only five countries: the Russian Federation, Brazil, Canada, the United States of America, and China. Two-thirds (66 percent) of forests are found in ten countries (FAOSTAT, 2020). Since 1990, it is estimated that 420 million hectares of forest have been lost through conversion to other land uses, although the rate of deforestation has decreased over the past three decades. Between 2015 and 2020, the rate of deforestation was estimated at 10 million hectares per year, down from 16 million hectares per year in the 1990s (ITTO, 2020).

2.1 Global timber products trade

Africa occupies one-quarter of the world's remaining tropical forests, producing 4 % of the world's total tropical industrial round wood (ITTO, 2022). Even though it is less than that of Asia-Pacific (24 %) and Latin America (12 %), it is quite significant. More than 80 % of Asia-Pacific's share comes from China, Indonesia and India. The bulk of Latin America's share from Brazil (Ibid). The largest producers in Africa are Nigeria, Gabon, and Cameroon (Ceesay, 2018). It is interesting to note that irrespective of being home to more than half of the Congo Basin forests and the second-largest block of tropical rainforests after the Amazon, the DRC's official production figures for industrial round wood are only about one-tenth of those of countries like Gabon and Cameroon. Table 2-1 below shows the highest producers of industrial round wood globally in 2022.

TABLE 2-1 GLOBAL PRODUCERS OF INDUSTRIAL ROUND WOOD, 2022.

Wood products	Producers	Quantity (1 000 m ³)
Industrial round wood	Global production	2011135.99
	Asia-Pacific	486050.29

Latin America	246675.99
Africa	79758.99
EU+	390389.38
Indonesia	80800.00
Brazil	142989.00
India	48154.00
China	180237.00
Nigeria	9810.00
South Africa	16604.49
Cameroon	5460.00

Source: ITTO, 'Annual Review Statistics Database', 2022.

It is apparent that formal timber production and trade figures from Africa should be treated with caution, as their accuracy can be disputed. For instance, ITTO statistics for the DRC were the same for 2009 and 2010. These figures also differ substantially from FAO statistics for the same year (2009), which placed industrial round wood exports at 51,440 m³ and sawn wood at 20,906 m³, much lower than the ITTO figures (Kamara, 2020). There could be several explanations for this, including differences in definitions or measurements or challenges with data collection. Disclaimers aside, concerns about the absence of exact figures or disparities among those from different sources need not prevent one from identifying broad trends in the international timber trade and their probable impact on the continent. Perhaps, this explains why official timber exports from the DRC are relatively small compared with countries like South Africa, Nigeria, and Cameroon. However, there is also no evidence that the Great Lakes is home to large-scale illegal timber trading syndicates reported in countries like Russia, China, and other parts of East Asia (ibid).

Furthermore, factors that have prevented investors, be it reputable ones or adventure capitalists, from accessing the African continent's vast timber resources include insecurity and a lack of infrastructure. These factors have combined with other governance challenges to contribute to Africa's reputation as a notoriously risky and costly place to do business (Ferreira et al. 2019). Improvements on any of these challenges could unlock the continent's timber potential, a reality acknowledged with some trepidation by those who advocate for the conservation of the continent's forests. From the great lakes, exports from the eastern DRC for instance, would most likely flow

along the main trading corridors toward seaports such as Mombasa and Dar es Salaam. Already those areas of the DRC that are more accessible, such as parts of the western DRC close to the port of Matadi, saw higher rates of deforestation linked to international timber exports (Ceesay, 2018). It nevertheless remains interesting to consider the markets supplied by formal Congolese timber exports alone. According to FAO figures for 2019, the DRC exports its round wood to countries like France (17,000 m³), Portugal (12,000 m³) and China (6,000 m³); and its sawn wood to destinations including Morocco (3,987 m³), Portugal (3,987 m³) and Belgium (2,997 m³) (De Mendonca and Zhou, 2019). Within Africa, this can be contrasted with a country like Mozambique, a former Portuguese colony that today exports more than 80 % of its round wood to China. In 2019 alone, Mozambique's export to China amounted to 9,010 m³ out of a total of 10,793 m³ for timber exports combined. Mozambique's timber trade to China was criticized in a 2016 report provocatively titled, *Forest Governance in Zambézia, Mozambique: Chinese Takeaway* (Cheng et al. 2019).

As shown by the Mozambican example, Africa's past relations with the East have been given new impetus by the rise of 'emerging powers' like India and China. A lot has been made of these powers' growing appetite for raw materials, especially timber products. The growth in timber exports to emerging powers is driven in part by factors like a booming construction industry and consumer markets in these countries. Moreover, domestic consumption in these Eastern markets will most likely continue to grow, especially when flexible conditions about governance in domestic African politics are guaranteed by China in its engagement with the continent (Mansaray et al. 2015). These factors have contributed to the wave of alarmist reports from western media in the late 2000s. However, closer scrutiny of global tropical timber chains reveals that there is more to this story. See table 2-2 below, it details the 2022 import figures for unprocessed and semi-processed industrial timber products as they reflect the global tropical timber trends. Although China is not the only timber-importing country, it provides an interesting and relevant case study.

TABLE 2-2 GLOBAL IMPORTERS OF TIMBER PRODUCTS, 2022

Timber products	Importers	Quantity (1000 m ³)
Industrial round wood	Global imports	123504.09
	Asia-Pacific	57981.78
	EU	56217.92
	Latin America and the Caribbean	316.56

Sawn wood	China	43690.00
	india	4923.00
	Global Imports	138729.90
	Asia–Pacific	43514.04
	EU	35731.07
	Africa	3248.24
	Latin America and the Caribbean	4447.63
Plywood	China	26660.00
	Taiwan	1319.07
	Global Imports	29106.89
	Asia–Pacific	9271.57
	EU	6179.33
	Latin America and the Caribbean	1456.41
	Japan	2265.46
	Malaysia	1167.60

Source: ITTO Annual Review Statistics Database, 2022.

Chinese influence is clearly shown from the table above as the country that dominates the import of unprocessed tropical timber. This further demonstrates her place in the timber value chain. For example, China's import of tropical plywood as a value-added product is dwarfed by that of a country like Japan. This is because, rather than being a large importer of plywood, China is a producer. In 2018, its production of plywood accounted for 5,955 000 m³ out of a world total of 18,293,180 m³, larger share than that of other large, tropical, forested countries like Indonesia (3,200,000 m³), Malaysia (3,544 000 m³) and India (2,130 000 m³). Thus, in the minds of many experts, China is aptly referred to as 'the wood workshop of the world' (Lau, 2019). The sheer size of these retailers allows their relatively small price mark-ups to translate into huge profits at the very end of the value chain, while squeezing the timber suppliers in the developing world. China plays a central intermediary role in these globalized chains. Currently, official figures show that the DRC exports relatively little timber; and that most of the timber officially exported from the DRC still goes to traditional markets in Europe. This could change in future. Mozambique on the east coast of Africa, with easier access to its forests and close ties with China, serves as a case in point. In the context of the trade in tropical timber from the DRC, the first step towards better governance is to establish exactly how much is being traded internationally and to where. Beyond that, international initiatives aim to regulate the existing trade as well as to put in place institutions for regulating the global trade in timber, including a growth in supply from the DRC when it takes off. The discussion does not go into any detail about domestic initiatives in the DRC aimed at

regulating this trade. Instead, it focuses on international initiatives that either directly involve the DRC (such as FLEGT) or that have an implication on the trade from the DRC (like the US Lacey Act Amendment). Important lessons can be learnt from these international initiatives for the regional trade in timber.

Empirical evidence points that the African continent can meet the wood product needs of its growing population, projected to rise to 2.5 billion metric tons by 2050 and to 4 billion metric tons by the turn of the century (Oke, 2018). This is a discussion that spurred the African Natural Resources and Investment Centre (ANRC), a non-lending knowledge building entity of the ADB, to organize a webinar on timber's role in industrialization and regional integration. Pertinent to the discussion are ways to capitalize on forestry resources to support industrialization and boost the intra-African trade of wood products that will create jobs, wealth, and value addition across Africa. Prior to this, it is suggested that reports should be prepared to collect and analyse quantitative and qualitative data to fill existing knowledge gaps on the performance of the timber industry in the different sub-regions of Africa, including West Africa. The report will guide policymakers formulating industrial policies and strategies for the timber sector in West Africa.

Diversity apart, Chinese actors and their business models highlights some common aspects of their engagement in Africa's timber sector. The distinction between the characteristics of the Chinese investors versus other western actors is a necessary one to highlight. While the former seems to ignore political or human rights standing of their counterparts, the west seems to consider it as a necessary trait of engagement (Mansaray et al. 2015). The sustainability attributes of Chinese investment and trade are based on limited evidence. China in Africa debate suggests a uniqueness of Chinese engagement in comparison to Western engagement, thus prompting the all-important discussions on what differentiates Chinese engagement from the rest.

Across many prominent economies in the world, Chinese market demand consistently display one important trait: a clear preference for unprocessed timber products, as indicated by previous studies (Jegatheswaran et al. 2018). Due to the developed timber-processing industry, sometimes counter-intuitively, sawn wood can fetch a lower price than logs in China, leaving little financial incentives for companies in producer countries to engage in further processing (Grassi et al. 2012). Such a strong demand for unprocessed timber, however, is constrained by the forest legislation of several African countries. In Zambia, the government has imposed a quota and incremental tax

increase on unprocessed timber products since 1996 (Feng et al. 2020). The same law also maintains a ban on the export of raw timber in the form of round logs. In 1999, Cameroon enforced a partial export ban of logs of selected species while Mozambique implemented a similar ban in 2007. Gabon and Sierra Leone were the newest to join this list with export bans of all logs in 2010. In addition to addressing illegal logging and trade issues, the governments adopted these policies to encourage the development of the domestic timber processing industry, with the hope of capturing more value from natural resource extraction and trade. These policies have severely affected the profitability of timber export to China. Indeed, the profit of the five main Chinese companies in Gabon decreased by 60 % on average with one of them reporting 90 % reduction in its profit since the log export ban of 2010 (Fonseca, 2015).

The Chinese owned timber companies have responded to these regulations by concentrating on sawn-wood exports. However, many interpret sawn wood to include the most rudimentary processed timber products, and very few Chinese companies move beyond primary processing. Apparently, when a Chinese company produces plywood products, they are destined for India, North Africa, and Europe, and not China (Mansaray et al. 2015). The Chinese companies in several economies across Africa cite three reasons for which further processing is not profitable: lack of skilled workers, high transportation cost due to unreliable and inadequate infrastructure, and limited opportunities to recycle waste, as opposed to the developed Chinese processing industry where most waste can be sold for further manufacturing. When one Chinese company in Mozambique experimented with further processing, it proved too costly. Another Chinese company in Mozambique recently made a large investment to upgrade its facility for further processing; however, the products are not destined for the Chinese market, but to supply the burgeoning Mozambican domestic market for construction materials (Feng et al. 2020). Similarly, a Chinese company in Zambia also manufactures furniture, doors and school desks for the local market using timber that is left over from exports.

Characteristics of Chinese actors operating within several African countries exhibit a few noteworthy and unique traits: competitive and dynamic supply chains; upstream operations; active participation of local loggers, and occasional illegal practices. For competition and dynamic value chains, Chinese trading companies operate in a competitive dynamic market with fierce competition among themselves. These competitive sentiments are especially common among

informants from trading companies in some African countries, notably Mozambique and Cameroon (Kamara, 2020). This appears to refute the common perception held by other timber companies that the Chinese timber buyers reorganized into a group of cooperating oligarchs fixing prices at a lower level (Jegatheswaran et al. 2018).

In Cameroon, Chinese traders complained that there were too many Chinese competitors. One remarked that „now too many Chinese know about this. In the past, I could earn a profit of CFA5000 (ca. \$10/m³), but now, some Chinese take it even at CFA2000 (approx. \$4) “(Zhang et al. 2016). When local operators were asked to estimate the number of Chinese buyers in the informal illegal trade, Cameroonian informants repeatedly expressed that there were too many to estimate. In Gabon and Mozambique, the competition among Chinese companies is also evident in the fact that small loggers frequently engaged inside-selling to companies that placed higher bids (Kamara, 2020).

In Zambia, competition was also rife for the most preferred species, Zambian rosewood (*Guibourtia coleosperma*), at both production sites and central timber markets. Competition in production areas was observed most in Kaoma area where the number of Chinese concessions increased from one to four during the period. At the central timber markets in Lusaka, the number of timber trading companies increased from fewer than 10 in 2010 to 90 in 2018 (Feng et al. 2020). Of the total, 77 % were Chinese-owned companies registered in Zambia, 7.7 % were Zambian owned and the rest belong to other nationals (GRZ 2010–2020).

The increased number of Chinese timber trading companies also increased the demand for the rosewood species, resulting in price increase especially in the timber markets in Lusaka. Most Chinese companies in Gabon and Mozambique initially started as trading companies downstream but recently have chosen to move upstream closer to the extraction stage by acquiring concessions. For example, a Chinese manager in Gabon comments, “We were purely exporters. We became operators and then we acquired factories” (Uzu et al. 2022). The primary reason for moving upstream is to secure stable supplies of timber because local supplies are unreliable. In Mozambique for example, a Chinese furniture manufacturer had initially placed orders with small-scale loggers but the supply was unreliable because the small operators would frequently sell the timber to other traders who offered better prices, causing shortages in their manufacturing facility in China (Zhang et al. 2016).

Similarly, in Zambia, a Chinese trading company became a concessionaire by taking over a Zambian-owned concession to secure timber supplies. A Chinese manager in Gabon also confirms the insecurity they face in dealing with suppliers and explains the reason behind moving upstream that “the main objective was, above all, to obtain forests to secure consistent wood supplies” (Kamara, 2020).

Both in Cameroon and Mozambique, a lack of trust between Chinese buyers and local suppliers increasingly led Chinese buyers to abandon forward financing to small operators (Obasi et al. 2015). This system holds true for Chinese businesses in the Mano River Union countries of Sierra Leone, Guinea, and Liberia as well. Some have cited cost reduction by skipping intermediaries as a motivation behind the move. In addition to Chinese logging companies moving upstream to acquire concessions, Chinese companies also operate informally at the extraction stage in some of these African countries. One Chinese company established a semi-permanent facility in the forests, employing local villagers and few Chinese workers as loggers, sawyers, and carriers to source wood from nearby forests illegally (Macdonagh et al. 2019). While it is difficult to observe, it is also important to note that companies may sometimes use concession licenses to launder illegal timber logged elsewhere, as observed among companies operating in Peru (Macdonagh et al. 2019).

2.2 Literature on timber business from African countries

There is an enormous amount of literature on the subject matter. Some authors have attempted to discuss the methods of assessing the impacts of timber products on the environment while others have basically sought to identify the socio-economic impact of timber products. A publication by Cerruti et al. (2015) on the socioeconomic and environmental impacts of wood energy value chains in Sub-Saharan Africa: a systematic map protocol, was also reviewed for this research. The work is founded on the premise that systematic assessments of the socio-economic and environmental consequences of wood energy has underestimated the significance of its impacts and thus influenced policy debates and engagements negatively. The objective is to collate evidence from several studies of environmental and socio-economic impacts of wood energy value chains.

Cerruti et al (2015) categorized the supply and demand of wood energy, which comprised of wood energy production, harvesting, processing, and consumption as the ‘exposure’ factors. The exposure factor is a subjective value that the person assessing risk must define. Their work further

identified those cases where the same wood energy activities occur with available or accessible alternative energy sources and regulatory frameworks that govern the sector as the ‘comparator’ factors. The justification for their research is that SSA lack access to modern energy sources such as kerosene, liquefied petroleum gas (LPG) and electricity, most of the people rely on traditional biomass like firewood and charcoal. This notion is justified on the grounds that SSA represents the world’s highest regional per capita wood energy consumption, with an average consumption of 0.69 m³/year in 2018, compared with a global average of 0.27 m³/year in the same year (Abayomi, 2021). The data used to determine the result of their work was extracted from selected studies, tables were designed to compile quantitative and qualitative data from each of the relevant study areas selected. The aim of such data extraction was to assess the existence of socio-economic, environmental, and other changes on the target population due to the wood fuel production, processing, trading, consumption, and policy implementation (ibid).

The results of the research itself shows that an estimated 93 % of households in SSA are solely reliant on wood energy for their daily cooking needs (Uzu et al. 2022). It further highlighted some of the negative impacts of harvesting wood from the forest on both socio-economic and environmental fronts. This was published as a CEE Systematic Map with an associated searchable database which is summarized in a CIFOR policy brief. This is a form of database with supporting narrative of all the relevant articles that have been reviewed, thus presenting descriptive statistics in tables, graphs and charts on quantity, study location, and target population of reviewed articles. The motive here was to disseminate results through the advisory group and relevant working groups to inform decision makers in government, civil society, research, and development organisations (Cerruti et al. 2015).

This was a brilliant work with a lot of effort by the researchers to uncover and identify lapses in works before theirs. It is a work that is full of factual insights which laid bare the socio-economic characteristics of livelihood in SSA with regards to catching up with global technological standards. However, the work is a product of ambiguous insinuations as it attempts to establish its research justification. This is apparent in their classification of ‘exposure’ and ‘comparator’ in attempt to collate evidence from several studies of environmental and socio-economic impacts of wood energy value chains. Moreover, the idea of sharing the result of the findings itself in CIFOR policy brief and through conferences, graphs, and tables instead of a detail analysis of its findings

in the article itself left readers wanting of a conclusion. Thus, these shortfalls will be taken into consideration to avoid similar lapses. The research finding to this work will be concise and precisely highlighted.

Another paper reviewed is one done by Obasi et al. in 2015 titled ‘Environmental and socio-economic effects of timber harvesting in Ebonyi State, Nigeria’ It’s main aim was specifically to evaluate the socio-economic and environmental effects of timber harvesting in Ebonyi State, Nigeria. Through a structured questionnaire, Obasi et al. (2015) used a set of unmeasured variables and process them into categories using the Principal Component Factor Analysis (PCFA) and maximum variables (varimax). A set of variables were used to determine the result of their work. These variables include Silting of rivers and lake; High cost of farm labour; Occurrence of disputes and crises over land and compensation; Damaging of immature trees and non-wood forest product; Loss of income and revenue; Reduction in soil fertility and crop output; Loss of biodiversity; Loss of forestland; Disappearance of forest cover; Rural-urban migration; High cost of living; Increased cost of wood and timber products; Climate change/global warming. While the effort of Obasi and fellow authors should be commended for their choice of methodological technique to analyse their variables, one is left dumbfounded with the way the context of their research changed comparatively to the main objective of the research. To evaluate effects of certain actions on a community will only yield a result with both the advantages and disadvantages of that action on the said community, anything less than that will apparently render the result bias.

This is exactly what befell Obasi et al. in 2015 as their results only emphasized the negatives of timber harvesting. Furthermore, the authors’ error of mistaking timber harvesting for illegal logging proved self-conflicting as they fell victim of misrepresenting timber harvesting with illegal logging. Also, it does seem that the recommendations fell short of the quality of research they undertook. Amongst the few made, nothing is mentioned about sustainable efforts through reforestation, community sensitization or even corporate social responsibilities of stake holders. Although similar methodological approach in the form of Confirmatory Factor Analysis (CFA) through the Principal Component Factor Analysis is used for this work, it will be guided against the shortfalls highlighted above.

Another paper reviewed is the Medical Consumer Price Index (MCPI) on the Environmental Indicators for the Evaluation of Wood Products in Consideration of Site-Dependent Aspects. In

this work, May et al. (2017) identified many indicators and methods essential for the assessment of wood products' impact on the environment. They used a Qualitative Content Analysis (QCA) to code a total of 78 indicators and 20 site-dependent aspects from 118 reviews to identify existing environmental indicators and methods for an effective evaluation of the sustainability of wood products. Some of the notable indicators and methods were the Forest Wood Chain (FWC) which is often used as a comprehensive concept that focused on the production and eventual consumption of different primary and secondary wood products; Life Cycle Assessment (LCA) which evaluates the quantification of material and energy flows along a product's life cycle; Material Flow Analysis (MFA), Carbon Footprint (CF), or Life Cycle Sustainability Assessment (LCSA). They further used the Pressure-State-Response (PSR) framework, originally developed by the Organization for Economic Co-operation and Development (OECD). Its usefulness was to examine human activities in relation to impacts on the environment. The PSR has since been further developed by the European Environment Agency (EEA) and it is now known as the Driving Force-Pressure-State-Impact-Response (DPSIR) framework (EEA, 2018).

The motive was to better address the individual contributions of different economies to current and future environmental problems. This is a well thought research apparently, but the shortcomings seem quite glaring. Due to the large volume of methods and indicators reviewed, the authors fell victim of self-contradictions. Their conclusion cannot specifically point to what the best combination of indicators and methods are. Thus, the work lacked clarity in conclusion. It is important to always recognize the fact that there is no standard practice in the selection, prioritization, and grouping of indicators (European Parliament and The Council. 2018) and this proves to be an apparent pitfall of the researchers. Thus, it left the entire work open to an uncertainty as to which indicators to hold valid in a cause-and-effect relationship or whether the indicators are set in relation to existing causal chain concepts. As much as a lot could be learnt from the work itself, this research will guide itself against the underlying shortcomings as identified.

Another relevant literature reviewed is research conducted by Energy For Opportunity (EFO) and compiled by Paul Munro and Horst, (2012), titled 'the governance and trade of wood-based products in and around the Kambui Hills North Forest Reserve (KHNFR). The research sought to analyze the dynamics surrounding wood-based commercial trade relating to the Kambui Hills

North Forest Reserve and the factors that influence this trade. Its findings recommended different approaches for donors and policy makers to improve both the reserve's management and the livelihoods of those reliant on its resources. The methodology involved two phases of interviews (Munro and Horst, 2012).

The first focused on village level group interviews in 43 different communities located near the KHNFR and these communities later became part of a co-management programme for the forest reserve. Through a village level snowball sampling process and review of local area maps, the target villages and communities were identified. The second phase of the interview focused on the elite group and state actors like the Paramount Chief (PC), urban vendors, the Sierra Leone Water Company (SALWACO), district council, Kenema city council, Kenema police, the forestry division office and forest guards, Gava Forest Industries Corporation (GFIC), Exim, Capitol Trading Company, Agricultural Community for Timber Development Association, (ACOTIDA). The second phase interviews were basically a follow up on issues raised during the village focus group phase. Although this research was done extensively to cover a substantial portion of the Kambui Hills Reserve, its conclusion could not be taken as a representation of concrete facts. Firstly, several of the 43 sample communities, particularly in the Lower Bambara and Dodo Chiefdoms, seem to simply have no relationship with the KHFR. For them, the Kambui Hills are just an abstract area that they have little practical knowledge of. It is apparent that a key measure of a community's relationship and knowledge of the KHFR is distance.

It does seem that those within three to five kilometers of the reserve tends to either use the reserve or have some knowledge of its boundaries and the laws that govern it, while those beyond this distance had little or no interaction with the reserve. Several sizeable communities near the reserve seemed to be missing from the target community list, thus making the selection process a misrepresentation of what would have been the real target group. Secondly, some of the communities selected were ghost towns at the time of the research, with empty buildings that are only occupied during the farming season, and therefore essentially acted as temporary satellite settlements for larger nearby towns. This shows real lack of substance and reliability in the results. Furthermore, another risk associated with the conclusion rests with the strategy of co-management wherein the selected communities form part of the management structure of the reserve. This might only result to adding an extra layer of governance to an already complex and bloated governance

structure. It would have been quite relevant for the authors to recognize that heterogeneity of the communities. The smallest village on the selected sample communities had a population of 21 inhabitants while the largest selected village community had an estimated population of about 7,600 people (Munro and Horst, 2012). Thus, it is evident that the researchers' approach of establishing effective Village Forest Management Associations (VFMAs) differed greatly between two unequal selected sample village communities. Ignoring this huge difference is a deficiency for research of such enormous importance. Given the above shortcomings, this current research will be mindful of committing similar mistakes and rather embrace full incorporation of them as a guideline. Besides, the primary researcher for this work hails from the Kenema district, precisely from Nongowa chiefdom where the KHFR is located, thus making it easier to understand some of the problems identified above and is better placed to understand the communities that are essentially important to be considered as sample groups.

2.3 An overview of timber harvesting business in West Africa

There are 16 countries within the West Africa sub region under the Economic Community of West African states (ECOWAS): Benin, Burkina Faso, Cote D'Ivoire, Ghana, Gambia, Nigeria, Sierra Leone, Liberia, Mali, Senegal, Togo, Guinea, Guinea Bissau, Mauritania, Niger, and Cape Verde (ECOWAS Treaty, 1975). Over the years, the population of these countries have increased to 391.4 million people (World Bank, 2018). Uzu et al. (2022) did a review of timber trade in seven west African countries and the result shows similar patterns in terms of raw materials export while importing manufactured timber products for all selected countries. The seven countries selected were based on the difference in percentage of forest cover area with the plausible assumption that countries with high forest cover may have a competitive advantage over those with minimal to low forest cover (Uzu et al. 2022).

Based on this criteria, the following seven countries were selected for assessment: Nigeria, Ghana, Liberia, Sierra Leone, Gambia, Benin, and Guinea Bissau. The selected countries have a population of 260.2 million people which is 67 % of the total population of ECOWAS countries (Waqas and Magda, 2022). Data was derived from forest statistics, trade information for the last two decades and publications from FAO, African Development Bank (ADB), ECOWAS bulletins and the World Bank. See figure 2-1 below showing the map of West Africa with all member countries.



Figure 2-1: Map of West Africa

Source: World Bank, 2021 a.

The objective was to assess the state of forest wood business in the selected countries and to identify factors limiting the development of the sector. This objective was based on the premise that low wood processing capacity within the selected countries is a recurrent theme, and the selected countries are net importers of wood pulp and paper; that all selected countries have an over reliance on forest wood products and a desire for an influx of private investment to stimulate wood processing capacities. For the statistical analysis, a fixed effect model suited the work and was used to examine the critical drivers of the forestry sector in the sub region (Hoepner and Schopohl, 2020). A statistical method is used to estimate unprocessed wood with wood volume (m^3) and the estimate for processed wood using the processed wood volume (m^3) as the dependent variable of the seven countries for the entire period. The independent variables were the socio-political and institutional changes, demographic, economic, environmental, technological changes, Economic freedom scores, population, and GDP per capita for each year in the 20year period. See table 2-3 below that illustrates forest cover among the West African countries with categories 1 to 6 ranging from forest crisis to very high forest cover.

TABLE 2-3 WEST AFRICAN COUNTRIES BY FOREST COVER, 2020.

CATEGORY	1	2	3	4	5	6
Status	Forest	Low	Modest	Adequate	High	Very high
	Crisis	Forest	Forest	Forest	Forest	Forest

		Cover	Cover	Cover	Cover	Cover
Forest cover (%)	0-5	6-10	11-20	21-30	31-50	51+
Countries	Mauritania	Togo	Mali	Burkina Faso	Ghana	Guinea-Bissau
	Niger	Benin	Cape-Verde	Nigeria	Liberia	
				Ivory Coast	Senegal	
				Gambia	Sierra Leone	
				Guinea		

Source: World Bank 2021a; World Bank 2021b.

See table 2-4 below where results from Uzu et al. (2022) shows that the fixed effects for some countries are more meaningful relative to others. For instance, it suggests that if Nigeria and Gambia had the same GDP per capita, Nigeria's unprocessed and processed wood production would be higher than Gambia's. Furthermore, Ghana and Liberia had the second and third highest fixed effects for processed and unprocessed wood which also suggests that Ghana's wood production is relatively higher than Gambia but lesser than both Nigeria and Gambia. The other countries like Benin, Sierra Leone and Liberia shows similar differences in fixed effect results.

TABLE 2-4 WOOD PRODUCTION IN SELECTED WEST AFRICAN COUNTRIES, 2020.

Country	Economic Freedom	GDP per capita	Fixed effect
Unprocessed			
Nigeria	0.00323	0.00009	17.540
Sierra Leone			15.297
Benin			15.309
Ghana			16.959
Gambia			13.262
Liberia			15.674
Guinea-Bissau			14.532
Processed			
Nigeria	-0.02350	0.00018	15.040
Sierra Leone			9.914
Benin			12.120
Ghana			14.477
Gambia			8.394
Liberia			12.532

Source: World Bank 2021 a.

The results for the fixed effects further suggest that Nigeria and Ghana are better positioned to have higher production volumes of unprocessed and processed wood, based on certain variables specific to them. Some of these unobservable factors could include land tenure system, long term government policies, and culture (Uzu et al. 2022). Specifically, Nigeria was once a major global supplier of roundwood and even though its forestry sector has gone through several phases, it remains one of the highest suppliers of industrial roundwood globally (Diez-Busto 2021). In addition, local furniture industries are a major market for unprocessed wood in the country and in certain states of Nigeria, a wood-based industry can be found in nearly every local government area (Dogan and Inglesi-Lotz 2017, Uzu et al. 2022). Estimates are that between 2002 and 2022, wood charcoal production increased by 48.9% resulting in a 2019 output volume of 4,595,112 tons (FAO, 2021). In general, wood production outputs come from both Nigeria's naturally regenerating forests and its plantation forests, yet plantation forests make up only 1% of Nigeria's total forest area (World Bank, 2021). Uzu et al. (2022) also noted that although there is adequate legal framework for the forestry sector, prevalent corruption, and disagreements over land ownership in rural areas make the sector ineffective and redundant (Brown and Pearce, 2020). Table 2-5 below highlights the population, forest area, GDP, and economic freedom score for the selected West African countries.

TABLE 2-5 SELECTED WEST AFRICAN COUNTRIES, POPULATION, LAND AREA, FOREST AREA, GDP PER CAPITA AND ECONOMIC FREEDOM SCORE, 2022

Forest cover category	Country	Population (millions)	Land area (000s km ³)	Forest area (% of land area)	GDP per capita (\$)	Economic freedom score
3	Nigeria	235.9	923.8	23.7	5.363	58.7
3	Sierra Leone	8.2	72.3	35.1	1.794	51.7
4	Benin	11.1	114.8	27.8	3.432	59.6
4	Ghana	30.8	238.5	35.1	5.652	59.2
5	Gambia	2.3	11.3	24	2.321	58.8
5	Liberia	4.9	111.4	79.1	1.491	49.2

6	Guinea-Bissau	1.9	36.1	70.4	2.077	54.9
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Source: World Bank 2022.

Results for Ghana on the other hand, suggests that between 2002 and 2022, the production of primary wood products and wood charcoal increased by 81.5 % and 98.1% respectively (Akomaning et al. 2022). Remarkably, Ghana signed a Memorandum of Understanding (MOU) with Greenland Resources AB of Sweden for the establishment of 200,000-hectare wood-pulp forest plantation over the next seven years that will establish a proposed \$2.8 billion pulp mill (Assa, 2021). This venture may have come as a direct response to a major wood products trade deficit in 2019 from importation of pulp and paper products from specifically India (10.1%), China (9.8%), and Indonesia at 9.5% (Uzu et al. 2022). Ironically, although the production of processed wood increased by 20.3% between 2002 and 2022, the export of processed wood declined by 55.7% (Assa, 2021). As the first African country to sign a Voluntary Partnership Agreement with the European Union, Ghana can freely export her wood products to the EU market (EC/VPA Report 2009). Proportionately, revenue grew between 2002 and 2022 as a direct result of an increase in the quantity of wood charcoal exported. Accordingly, agriculture constitutes about 19.7% of the country's Gross Domestic Product in 2018 (Ghana Statistical Service, 2019). The forestry sub-sector constitutes 7.7% of the agriculture sector with an overall contribution of 1.5% to the country's GDP (Ghana Statistical Service, 2019, Ghana Statistical Service, 2020). This vital contribution of the forestry subsector to the nation's economy is not new as from 1990 to 2008, timber production consistently ranked as the fourth largest foreign exchange earner for Ghana (Grooters. and Bortelot, 2017).

Notwithstanding these advantages, acquiring clear title to land is usually challenging with corruption very much rampant in government establishments (Assa, 2021). For instance, concession holders pay very low social responsibility agreement fee relative to the international market value of the harvested timber (Dogan and Inglesi-Lotz, 2017). The actual amount generated by the social responsibility agreement fee, when fully complied with, has been estimated to now be 1 % of the market value of the harvested timber (Akomaning et al. 2022). See table 2-6 below that illustrates timber export, import and trade balance for the selected countries.

TABLE 2-6 SELECTED WEST AFRICAN COUNTRIES, IMPORT, EXPORT, AND TRADE BALANCE FOR 2002, 2012 AND 2022

2002				2012			2022		
Country	Export	Import	Trade Balance	Export	Import	Trade Balance	Export	Import	Trade Balance
Nigeria	33.3	(278.3)	(245.0)	98.7	(994.1)	(895.5)	134.9	(1.102.1)	(967.3)
Sierra Leone	2.5	(6.8)	(4.3)	12.7	(7.8)	4.9	132.2	(15.5)	(116.7)
Benin	1.4	(15.9)	(14.6)	16.1	(17.0)	(0.9)	33.9	(18.1)	15.7
Ghana	137.6	(37.8)	99.9	178.8	(103.9)	74.9	290.0	(406.7)	116.8
Gambia	0.3	(2.6)	(2.3)	1.0	(8.2)	(7.2)	77.2	(11.4)	65.8
Liberia	213.9	(0.5)	213.3	5.5	(4.6)	0.8	23.6	(11.9)	11.7
Guinea Bissau	1.0	(0.2)	0.8	1.9	(1.2)	0.7	9.2	(1.4)	7.8

Source: World Bank 2022.

Sierra Leone and Benin presents their own unique circumstances as revealed by Uzu et al (2022) in their work. Although, Sierra Leone is a smaller country given its population and land space, the country has positive timber trade balances in 2012 and 2022, by far the highest trade balance in 2022 of the seven countries examined, except for Nigeria and Ghana. Between 2002 and 2022, the volume and revenue of unprocessed wood export surged from 591 m³ to 1 4,911 m³ representing US\$4.3 million to US\$116.7 million, respectively (FAO, 2021). The quantity of wood charcoal produced, and wood processed in Sierra Leone increased by 55% and 2% respectively, between 2002 and 2022 (Uzu, 2022). The sad commentary is that Sierra Leone's forestry sector is guided by inadequate legal framework and absence of a land titling system (Jordan et al. 2018). In the last twenty years, timber trade in the country has varied remarkably from period to period because of policy changes and the civil war. The government of Sierra Leone has instituted bans on export of logs on at least four separate occasions: 2001 to 2002, 2007 to 2008, 2011 and 2018 (Kamara, 2020). While each of these bans had specific objectives, they were instituted to achieve at least one of the following: sustainability of forest resources, create employment opportunities, foster local processing, and minimize the potential loss of forest revenue from increased illegal logging

activities. However, it has been reported that exports of wood continued during some of these periods, nearly exclusively to China (World Bank Report, 2021). Benin also presents a robust timber business engagement alongside the other countries reviewed by Uzu et al. (2022). The result suggests that in 2012 alone, the estimated volume of primary wood products produced in Benin was 7 million m³, an 11.9% increase from 2002 while the volume of wood charcoal produced declined in the same period by 73.2%, from 183,197 tons in 2002 to 49,050 tons in 2012 (Kamara, 2020). The forest sector contributes 4–5 % of the gross domestic product in the country and less than 2 % of wood charcoal has been exported in the last twenty years (Villela et al. 2019). Most of wood charcoal produced are as well consumed in the country. Consequently, the World Bank funded the Gazette Forests Management Project to develop 15,000 hectares of acacia (*Acacia* spp. Mill.) plantations to reduce wood charcoal reliance (World Bank Report, 2018). In both 2002 and 2012, Benin had negative trade balances due to the significant importation of processed wood products and about 80% of the country's court cases are related to disagreements over land ownership (Uzu et al. 2022). Just like Sierra Leone, the absence of existing Voluntary Partnership Agreement with the European Union, may have also limited Benin's potential export destination.

The results for Gambia, Liberia, and Guinea Bissau were much alike but also with unique characteristics according to Uzu et al. (2022). The volume of wood products produced in Gambia increased by 22.8 % between 2002 and 2020 representing a volume of 879,911 m³. Gambia's forests do not produce much timber but with respect to wood charcoal, output increased in the twenty-year period by 48.1 % to about 69,000 tons in 2019 (FAO, 2021). In addition to threats from anthropogenic factors like population growth and agricultural practices, Gambian forests are currently at high risk from the impacts of climate change and are increasingly degraded. The country's export of unprocessed wood products was 171,250 m³ representing about US\$59.6 million in 2019. On the other hand, the quantity of processed wood imported increased by 80.7 % over the last twenty years (Uzu et al. 2022). Progress has been made after failing to produce wood pulp and paper over the last two decades. In 2019, Gambia exported unprocessed wood products to China (25.7 %), Morocco (23.7%), and Turkey (11.9%) as top three countries that supplied Gambia with paper and paperboard products in 2019. Overall, Gambia had negative wood products' trade balances in 2009, yet in 2019 they recorded a trade surplus (Kamara, 2020). Government focused priority for the forestry sector were outlined in its 2019 to 2028 National

Forest plan. Private sector establishment regulations were relaxed and other barriers limiting private forest plantation investments were eased off (Villela et al. 2019). Most of the land is held as customary land, State land and freehold land. Like in Ghana, Sierra Leone and Liberia, corruption has rendered enforcement ineffective and generally weak (Obasi et al, 2015). In addition, the intricate nature of the overlapping multiple land tenure systems results in poor land title enforcement and sometimes to conflicts between contesting clans (FAO, 2021).

In Liberia, results from Uzu et al. (2022) reveals that the wood processing industry in Liberia has historically been export-oriented, focusing on a few highly valued tree species. In 2019, the volume of wood products produced in Liberia was about 9.8 million m³, and 98.6% of these products were unprocessed logs. This distribution of wood products was consistent throughout the last two decades. The volume of wood products increased by 31.6 % between 2000 and 2019, while the volume of unprocessed logs and wood charcoal products within the same period increased by 64.9% and 96.7% respectively (FAO, 2021). This explains the positive trade balance in all focus years, relying on revenues from sale of logs. The result further reveals that while most of the wood exported in the period was unprocessed, there was a steady increase in the proportion of processed wood export from 2002 to 2022. For instance, only 0.7 % of the wood exported in 2000 was processed but this increased to 1.4 % and 7.7 % in 2009 and 2019 respectively (Uzu et al. 2022). Like other countries in this review, the top three destinations were China, France, and India. The country operates a dualistic land tenure system, one based on statutory law and the other on customary law (Kamara, 2020). Property rights are not very protected, and enforcement is riddled with corrupt long processes. Also, with over 80 % of the land in the country being unregistered, conflicting land ownership records are commonplace (Fonseca et al. 2021).

Guinea Bissau has a positive wood trade output between 2002 and 2022, according to results from Uzu et al. (2022). Production of unprocessed wood and wood charcoal products grew by 27.8 % and 49.5 % respectively. The low amount of processed wood might be indicative of the low timber processing capacity of the country. There is a provision in the country's Law that bans all exports of unworked wood in a bid to improve local value-adding capacity (Freeman and Dmytriiev, 2017). This increase in the export quantity resulted in an 81 % and 48 % growth in the revenue generated from unprocessed and processed wood respectively (Uzu et al. 2022).

This result may also suggest that most of the wood charcoal produced is consumed locally for fuel. The country did not import wood charcoal in any of the years, which may be indicative of the fact that the local production sufficiently met the country's demand. Regarding wood pulp and paper products importation, the import quantity was greater than the demand, suggesting a re-exportation of some of its wood pulp and paper products imported. The result also confirms that the cost of importing of wood pulp and paper products had the highest value of all wood products imported in 2012 and 2022. In relation to its overall contribution to the country's GDP, although the majority of the country's total export consists of unprocessed cashew nuts (69.2% in 2011 and 82.2% in 2015), the export of wood constituted 1.1% of the government's total export revenue in 2011 and grew to 6.2% in 2015 (World Trade Organization, 2017). Similar to results from other countries discussed, the protection of property rights and enforcement of those rights in the country is poor due to slow legal system, no independence of the judiciary, and endemic corruption in government places (Obasi et al, 2015).

Although the research by Uzu et al. (2022) was quite extensive and vigorous, there were limitations to the work that could otherwise be resolved to make the results further valid and reliable. One such limitation is the author's expectation or neglect that while the insignificant coefficients of the economic freedom scores in the fixed effect models may seem to indicate that prioritizing socio-political and institutional improvements with the goal of increasing the production volume of unprocessed and processed wood may not make any significant impact, it can be opined that any possible policy implication should be further tempered by the general uncertainty of data supplied to FAO from individual government agencies because weak national monitoring systems does result to exaggeration, error, and uncertainty in FAO forest-based activity information (Wei et al, 2023).

It is important to consider that sustainable development goals and other issues involving non-timber forest products might influence the forest management environment in these countries. This consideration is not reflected in the analyses. Moreover, the concept that capital investment in manufacturing industries might increase employment opportunities and improve the overall standard of living and domestic business opportunities in West African countries might be misleading. The influx of private investment is still required to stimulate interest in wood processing capacities of all the countries in this review but contemporary findings in this area

suggests that the low wood processing capacity of the focus West African countries is a recurrent theme, as evidenced by the disproportionate export of unprocessed logs compared to processed wood. Thus, improving wood processing capacity will not only increase potential revenue generated, but also employment opportunities (Villela et al, 2019).

Furthermore, Uzu et al. (2022) did not fully consider the influence of business atmosphere in the studied countries. A series of complicated public-private partnerships for management and utilization of forest resources might be beneficial for sustaining social systems through improving access to land and resources and through promoting innovation (Poponi et al. 2019). There is a willingness of private citizens to develop forest plantations, but issues related to land tenure, seedling availability and quality, extension assistance, and government policies seem to discourage such activities (Perunova and Zimmermannova, 2022).

Also, the lack of very clear financial and economic models of forest investments tailored to each country and thus an assessment of the whole seven countries based on the same parameters is worth reconsideration. The output of the forest wood products (timber, wood pulp, charcoal or mixed) and tree species, investment period/ cycle, tree maturation/harvest cycle and contemplated markets, all varied in the various countries studied. Moreover, the scenarios regarding how specific changes in macroeconomic factors affect the return on the investment are quite different. Ultimately, this type of work may be misleading to address what is observed in forest development projects in specific countries within West Africa (Purvis et al. 2019). Thus, it is important to guide current research works against these challenges and limitations. It is necessary to obtain standardized values measuring environmental and technological changes from 2000 to 2019. This would have substituted for the independent variables dropped from the analysis and may result to a two-model approach in the methodology. As major timber importing countries in the developed world commit themselves to the reduction of illegal timber into their markets through such mechanisms as the U.S. Lacey Act (16 U.S.C. 3371-3378) and the European Timber Regulation, West African timber producing countries had to invest in forest certification programs in order to access those markets (Obasi et al. 2015).

2.4 Theories and models of sustainable impact assessment

Definition of sustainable development has evolved over decades. The World Commission on Environment and Development (WCED), also known as the Brundtland Commission report in 1987 defined Sustainable Development (SD) as “development which meets the needs of the present without compromising the ability of future generations to meet their own needs”. The Rio Earth Summit defined it as “the right to development fulfilled so as to equitably meet developmental and environmental needs of present and future generations.” The concept has become prominent on the world political agenda and thus, the need for impacts assessment paradigms. As prime example, the European Union requires a sustainability Impact Assessment (SIA) of all larger policy proposals, including estimates of policy-induced economic, environmental, and societal impacts inside and outside the European Union (EC, 2001). The broad elements of SD proved it hard to define, it incorporates a normative equity dimension that is quite subjective and cannot be analysed scientifically (Sonny and Riko, 2016). A robust approach from the scientific community help to develop three dimensions of SD, i.e. environmental quality, economic performance and equity concerns, intertwined and subject to tradeoffs. There is a wide range of alternative quantitative approaches for assessing the causal chains between a proposed policy change and its potential economic, environmental, and social impacts.

Globally, we use 1.75 times the natural resources of this planet unsustainably (Salmivaara et al. 2018), 2.8 times in the European Union (Villeneuve et al. 2017). This is more natural resource usage than the planet can regenerate, which will lead to resource depletion, if not halted. The argument of lack of knowledge for such actions hold no more. Sustainable use of natural resources and climate change have brought relevant concern that we are devouring the resources of this planet beyond its regenerative capacity and hinder the planet’s ability to support human life. Sustainability assessment and sustainability science are seen as a way to address these challenges (Jing et al., 2023).

It is apparent that climate change, among other socio-economic crises, results from our past and present unsustainable patterns. The 2018 Intergovernmental Panel on Climate Change report (IPCC report, 2018) and the ensuing press coverage proclaimed what scientists have been saying for a long time: the need for radical changes to stay within 1.5 °C of global warming. It is thus

believed that natural resources such as forests can play a significant role in climate change mitigation (Grassi et al. 2012; Nabuurs et al. 2017).

Ultimately, we want to avoid the problem that emanates from using our natural resources because our survival depends on it. The primary aim of impact assessment is to prevent a degraded environment that is unhealthy and incapable of producing the resources necessary to sustain the present levels of human and non-human life (Gwandu et al. 2018). Following the Earth Summit in Rio de Janeiro, Brazil, in 1992, the broader societal orientation on sustainable development was also introduced with the advent of principles for sustainable forest management (SFM), encompassing the three pillars of sustainability: economic, social, and environmental (AGBO et al. 2015). Decision-making for sustainable development calls for scientific support by (i) anticipating the possible consequences of management options and (ii) identifying improved management solutions.

At the EU level, there has been a consistent movement toward the utilization of Sustainability Impact Assessments (SIA). First, the EU Strategy for Sustainable Development committed to perform impact assessments of all proposed major policies. The objective is to study how factors such as policy, management, or technological development affect the sustainability of a sector or value chain and helps to inform decision makers about consequences of decision alternatives. The SIA can be used to complement existing environmental assessment approaches such as Life Cycle Assessment (LCA), Material Flow Analysis and ecological, carbon, and water footprints. Further to these developments, the SENSOR EU project developed a Sustainability Impact Assessment Tool (SIAT) that causally links policy changes to land-use changes and the subsequent impacts on sustainability. SIAT implicitly includes impacts of forest resource management and uses a high level of aggregation in determining the impacts on production chains downstream from a forest.

While all these measuring tools exist, none of them bridge all the different scales. Thus, the EFORWOOD project (2005–2010) was to develop a tool that implements sustainability impact assessment by combining quantified indicators of sustainability with value chain (Aulisi et al. 2008; Arunachalam and Pandey, 2019). With this tool, EFORWOOD aimed at assessing the sustainability impacts of policies, technological development scenarios, or other such changes for the forest-based sector, or for value chains therein. The existing methodologies were not fully able to provide for a value chain-based assessment of the three pillars of sustainability, while covering

all different phases of the value chain from forest management to the products' end of life. In the same vein of achieving a more holistic assessment by connecting multiple sectors of activity and spanning different dimensions of sustainability lead to the development of the new concept: the Tool for Sustainability Impact Assessment (ToSIA).

Life Cycle Assessment (LCA) and Life Cycle Sustainability Assessment (LCSA) are reliable analytical alternatives in Impact Assessment literature. LCA evaluates the quantification of material and energy flows along a product's life cycle. Researchers and the public across the tropical domains are drawn to the use of LCA and LCSA.

The use of LCA is very widespread across topical domains, and it is the method that is mostly used to assess forest products usage and its accompanying environmental impacts (Appiah et al. 2018). LCA can be used, for example, to compare product or service alternatives to identify the product/service causing the least environmental effect. It provides a process to compare two alternative forest biomass supply chains. LCA can also be used to identify environmental sustainability threats and hotspots within a production chain. LCA methods have experienced strong development since the early 1990s (Chukwuone and Okorji, 2008). These development phases led to the operationalization for business and policy makers.

The United Nations Environment Program (UNEP) and Society of Environmental Toxicology and Chemistry (SETAC) agrees that Life Cycle Initiative is to pool together expansions of LCA to cover the three dimensions of sustainability under Life Cycle Sustainability Assessment (LCSA) (Ciroth et al. 2011). The traditional LCA is to assess the environmental impacts, but the LCSA consists of life cycle costing (LCC) for assessing the economic implications of a life cycle and social life cycle assessment (S-LCA) for the assessment of social consequences. While the need for LCSA studies is continuously relevant, its continued development processes is rewarding (Kovanda et al. 2012). The development of LCIA and LCSA broadens the capacity of LCA for SIA and is catching up to the issues that ToSIA was designed to solve. The idea of designing ToSIA was to create a new stand-alone method, to improve on previous efforts, ToSIA is frequently used to carry out LCSA.

Material Flow Analysis (MFA) as an analytical alternative aims to quantify the material flows and stocks of a selected substance in the anthroposphere, for example, for elements such as carbon (Dogan and Aslan, 2017). It is an established method for assessing resource-use efficiency. MFA

uses processes as places where the flows merge and divide, and also as places where stocks can accumulate. MFA has been adopted to analyse flows at a national/regional scale and at a company level (Kovanda et al. 2012). Applied to the life cycle of a product, it can be used to support the Life Cycle Inventory (LCI) as a variable part of LCA. However, in combination with LCA, as for example, in the study on increasing wood cascade by Bais-Moleman et al. (2018), more indicators can be covered, but still typically delimited to environmental aspects.

Forest Wood Chain (FWC) which is often used as a comprehensive concept where the production and eventual consumption of different primary and secondary wood products is linked to the original forest resource.

Pressure-State-Response (PSR) framework (originally developed by the Organization for Economic Co-operation and Development (OECD), its usefulness was to examine human activities in relation to impacts on the environment. The PSR has since been expanded by the European Environment Agency (EEA) to the **Driving Force-Pressure-State-Impact-Response (DPSIR)** framework. The motive was to better address the individual contributions of different economies to current and future environmental problems (Bhatt et al, 2021).

No specific modelling Framework could fit all requirements for comprehensive Sustainability Impact Assessment. This review is thus limited to specific paradigms and models that are common across the various dimensions of SIA. Accomplishing one objective frequently means backpedalling on another. Since economics is the study of trade-offs, this means that there is plenty for economists to contribute to make the concept of Sustainable Development operational. One important contribution over the last years has been the assessment of external costs as a prerequisite towards getting it right. Given full information on external costs, two aspects of Sustainable Development, namely economic performance (gross efficiency) and the environmental quality, can be merged to a comprehensive net efficiency dimension.

One of the models frequently used by business to determine impacts of sustainable development assessment is the **B Corp Certification**. This is built on the premise that a business is about meeting high standards of verified performance, accountability, and transparency on factors from employee benefits and charitable giving to supply chain practices and input materials (Diez-Busto et al. 2021). Certain criteria must be met to achieve certification including a demonstration of high social and environmental performance by either meeting the Impact Assessment score of 80

minimum or passing a risk review test. Another criterion is to make a legal commitment of changing corporate governance structure of the interested business entity. This is to ensure accountability to shareholders and achieve benefit corporation status. The interested business party must also meet transparency standards and allow information about their performance measured against B Corp's standards which will be publicly accessible from B Corp's website. This model was created in 2006 as 'B Lab' commonly known as the B Corporation Certification and is intended to be an instrument for evaluating the social, economic, and environmental performance of companies. The certification process depends on the size and complexity of the company and based on five indicators: governance, workers welfare, the community, environmental considerations, and customers satisfaction which are measured by a set of relevant variables (Lau, 2019). Focus is placed on this model due to the high growth and popularity amongst companies, especially in Europe. Furthermore, its relevance is associated with the common features it had with UN Impact Assessment practices.

In theory, several concepts support the adoption and sustainability of B Corp certification. For instance, the stakeholder theory (Freeman, 1984; Freeman and Dmytriiev, 2017) highlights the relevance of a firm's critical relationships with its stakeholders around integration of business and societal considerations fostering stakeholder value and leading to improved performance. Furthermore, the Resource-Based View (RBV) theory propounds that organizations should identify and use valuable, rare, difficult to copy, and non-substitutable resources to gain competitive advantages and abnormal profits (Berck et al, 2003). The adoption of B Corp certification can generate these resources and support integration with stakeholders in response to their demands. In addition to the two theories above, the institutional theory which relies on the adoption of models from successful organizations, do explain the destination of B Corp certification as a search for organizational legitimacy (Bailis et al, 2001)

There are some scepticisms about the financial impact resulting from implementing the B Corp certification, but the positive effects on the growth of companies' turnover in the short term, resulting from increased transparency and the positive socio-environmental impact observed, outweigh the adverse effects due to the strict audit procedure (Bamwesigye et al, 2021). However, critical analysis of the potential of this certification compared to the basic principles of the circular economy allows an opportunity to portray multiple dynamic viewpoints concerning stakeholders

(Poponi et al, 2019). It can lead to activation of the circular economy pathways because of different stakeholders in an international context, and, with the environment, for instance, being one of the evaluation indicators during the B audit process, companies' environmental impact processes have a direct and indirect association with their long-term development and financial performance (de Mendonca and Zhou, 2019).

Thus, B Corp certification can contribute fundamentally to economic growth of such companies. A study conducted on certified B Corporation companies in the first 10 years of certification identified that institutional and economic resources support the diffusion of this certification model. As a result of this study, companies will be more likely to become certified if the number of organizations covered by this model is greater, contributing to the social and environmental well-being of the planet (Rankin and Matthews, 2020). Other studies have investigated the most common reasons companies and organisations seek B Corp certification. According to the analysis, the main reasons for pursuing these certifications are that companies identify their mission with this model to enhance their values and identity in the markets (Moroz et al. 2018). This result also suggests that the image this certification conveys to the consumer is one of trust and effectiveness in the social, economic, and environmental fields. In a separate study, B Corp certification in sustainability is positively associated with company performance. It suggests further that organizations and companies with a strong brand in markets often believe they have already achieved high ethical standards (Villela et al. 2019). Mitigating and reducing the ecological footprint, certainly one of the aspects valued by B Corp, cannot be considered by companies as a sacrifice but rather as ensuring the quality of life and a more promising future. A growing trend of ecological consumerism has opened a vast market of opportunities for entrepreneurs to conduct more sustainable businesses (Feng et al., 2020).

Moreover, approaching sustainability without mentioning the importance of social responsibility is an incomplete assessment of the sustainable development of companies. Generally, governmental funds and financial investment choices are allocated to companies that incorporate corporate social responsibility policies in their mission (Hoepner and Schopohl, 2020). An organization with high status in terms of social responsibility and high ethical principles and practices, as is apparently the case with B Corp companies, is a reliable business partner and a reputable member of the business community (Iiyama et al. 2014). Other studies also highlighted

those stakeholders in business markets increasingly value incorporating social responsibility measures (Fonseca, 2015).

The computable general equilibrium (CGE) models is another analytical framework that is suitable for assessing the impacts of policy interference on environmental quality, economic performance and social factors of sustainable development assessment. The methodological elements of this model include the decomposition procedures of general equilibrium effects that deliver a better understanding of key determinants for policy effects; the embedding of large-scale general equilibrium models in an optimal policy framework that considerably widens the scope of policy analysis; and systematic sensitivity analysis to test the robustness of model results concerning uncertainties in the parameters (Christensen et al., 2019). The focus on CGE model is selectivity. Obviously, there is a wide range of alternative quantitative approaches for assessing the causal chains between a proposed policy change and its potential economic, environmental, and social impacts (Cronin and Pandya, 2009). The framework does not fit all requirements for comprehensive Sustainability Impact Assessment and thus requires a package of models (or methods) depending on the policy measure or issue to be assessed and the data availability. The model is an established analytical framework for evaluating the economic implications of policy intervention (Paelman et al. 2021). The main virtue is the micro-consistent representation of direct effects, indirect feedback, and spill overs induced by exogenous policy changes (Thissen et al. 2019).

Although the general contribution to actual policy analysis remains limited, the general equilibrium analysis provides important qualitative insights into the adjustment reactions by economic agents to exogenous policy constraints. This is because theoretical models are designed to keep analytical tractability. As soon as the complexities of a more detailed production structure are considered, analytical solutions are no longer available, and numerical solution methods are required. Thus, (CGE) models have become vital for applied analysis of measures in various policy domains, including fiscal policy, trade policy, and environmental policy (Persyn et al., 2020). These models incorporate many details and produce concrete numbers on policy-induced economic and ecological effects.

Furthermore, CGE models provide an open framework for the incorporation of new research strings, the new growth and trade theory that have important relationships to other disciplines that

use an integrated assessment approach (Caporale et al., 2021) This flexibility makes CGE models a valuable tool for Sustainability Impact Assessment. Ultimately, Sustainability Impact Assessment requires a global and intertemporal perspective. There are also various examples of models with overlapping generations (OLG), but an OLG framework with multiple regions and sectors still poses considerable challenges and requires trade-offs with the level of details that can be captured in the model.

Principal component analysis (PCA) is used for the analysis contained in this work. It is one of the oldest and most used methods for reducing multidimensional data. The basic idea of PCA is to find a set of linear transformations of the original variables such that the new set of variables could describe most of the variance in relatively fewer variables. The new set of variables is presented and derived, in a decreasing order of contribution. Additionally, the first new variable that is known as the first principal component contains the largest proportion of the variance of the original variable set, and the second principal component contains less proportion.

2.5 Pre-colonial and colonial forest management development in Sierra Leone

Sierra Leone has always been the subject of extreme estimates in recent discussions about forestry and wood products trade. This fact apparently holds true in pre-colonial and colonial periods. This has been attributed not just to farming and land use practices, but also to the timber trade, thus highlighting the influence of local actors and the supposed role of external influences. It is a case where particular ambiguity about the delimitation of idealized vegetation zones already had major interpretative ramifications. In the early 1900s, as much as 5,000,000 ha may still have featured to be undisturbed forest cover but a measure of the pervasive impact of human activities have reduced that amount of primary moist forest to a mere 290,000 ha (Mansaray et al. 2015).

Deforestation has been a recent development, and this is inferred by the World Conservation Monitoring Centre which suggests that 50 % of the country has climatic conditions suitable for tropical rainforest, but less than 5 per cent is still covered with mature, dryland closed forest. Deforestation is mainly a result of the rapidly increasing human population requiring more agricultural land and fuelwood (Myers, 1996). But as Obot, E. (2002) affirm, such figures suggests that post war devastation in Sierra Leone have little in common with turn of the century estimates

which suggest that the country had already lost most of almost all its forest cover (Zhang and Weng, 2016.). Earlier colonial observers erroneously assessed deforestation as having already happened. Thus, Sierra Leone's report to the first British Empire Forestry Conference stated that the 'primeval forest' is now "almost disappearing (Ceesay, 2020).

It was alleged that the natives have cleared up to about 90 per cent of the virgin forests by their system of shifting cultivation (Kazi, 2011). In fact, it was claimed that there is evidence that at no remote period in history that the whole country was covered with dense high forest. At the time of the conference, the land area covered by high forest is estimated to 3 % and this was likely to be a generous estimate. Many contemporary assessments are more informed of conditions around 1900 and to the early-mid nineteenth century, as did early colonial analysts. Debates that are based on the environmental impact of timber exports have been accredited such significance by recent studies and that Sierra Leone has come to be treated as a nineteenth century forerunner to deforestation elsewhere in West Africa (Vanessa et al, 2007). Thus, the nineteenth century question about timber trade was that, as rudimentary as it was then, could timber trade really have had such an environmental impact, and whether Sierra Leone had such an extensive forest cover in the first place.

Sierra Leone became an exporter of timber from the 1810s. The importance of this timber export led to analysis of several authors who linked it to veterinary epidemiology (Todaro and Smith, 2009). They argue that the epidemic horse disease of 1856-1870 in Sierra Leone was caused by the appearance of tsetse fly and the disease called trypanosomiasis (Tijani, 2007). This occurred, they argued, because of deforestation. Some change in the prevailing ecological conditions must have occurred which favoured the expansion and increase of the tsetse fly population and brought it into contact with the horses on the peninsula. Their argument is that before the colonial period, much of Sierra Leone was covered with tropical high forest in which the tsetse flies mainly associated with. Deforestation in the colony's immediate hinterland was principally due to felling of trees for timber during three phases: 1816 -1830s, when only first grade timber was removed; 1830-1850 when small merchants cleared what timber remained, and from the 1840s, when cleared lands were either cultivated or left to regenerate. This shift from the 1830s was necessary as the northern rivers had become less profitable than the more southern Ribi river, the Bumpe chiefdom on Yawri Bay, and Bagru in Sherbro land (Munro, 2012).

As Dorward and Payne themselves documented, the scarcity of preferred timber species (*Oldfieldia Africana*, and *Milicia excelsa*) was taken to mean deforestation but this was just how selective timber species extraction was. Thus, if the northern rivers produced less timber, this may not be because timber had already been felled on such a huge scale, but because there was not much of the preferred timber species. At that time, timber was conveniently measured in 'loads', the customary quantity which could be carried in a cart or wagon. Generally equatorial forest in West Africa could supply between 15 and 50 cubic metres/ha (Grassi et al. 2012). Nabuurs et al. (2017) estimated a production of 40 cubic metres/ha of timber suitable for ship building. While yields in some forests are considerably higher, for instance, 90 cubic metres/ha in some Ghanaian forests, modern extraction (mining), or already depleted forests can produce much smaller figures of between 3 and 8 cubic metres/ha (Vertessy et al, 2000).

Using timber export figures during this period, the area from which timber was taken from 1827-1843 at the height of the felling can be estimated at 9.258 ha (i.e., an area less than 10 km²), averaging 544 ha/year (Nabuurs et al. 2017). In total, between 1816 and 1870, it can be calculated that timber was taken from about 15,000 ha, (i.e., an area of 12 km x 12km), or less than 400 ha/annum (1820-1860). And up to the time when the timber trade moved south, timber had been extracted from perhaps some 5,000 ha, an area of 7 km². That timber was removed does not suggest that this area was 'deforested', first because mature forest regenerates rapidly after felling, and second because the felling was so selective. It is thus reasonable to argue that the rate and extent of nineteenth century timber felling was almost insignificant relative to the area of Sierra Leone. If there was a shortage of timber during the period of nineteenth century export, it seems more likely to have been because there was not much in the first place, than because the felling was so rapacious. And when it is stated that "timber has to be dragged several miles to a riverbank" this seems less likely to be because the intervening land had been deforested, as Nabuurs et al. (2017) wanted us to believe, but because either the intervening land did not hold forest of the desired quality, or any forest at all.

Nabuurs et al. (2017) went on to suggest that by the late 1860s the timber trade from the southern rivers (Ribi and Bagru) was also declining, partly due to the exhaustion of profitable reserves and partly due to competition from the growing palm kernel trade. They minimize how by then timber prices had collapsed, along with the admiralty demand after ship building switched from wood to

iron, making the role of supply problems in the decline highly questionable. Thus, the inference of dwindled forest due to timber felling and later agricultural clearance and replacement with palms must be wholly incorrect. The volume of timber extracted, and the area exploited, can indeed be reasonably accounted for by the extensive network of riverine forests found in this part of Sierra Leone. As summarized, it is almost an invariable rule that every river and stream possess a belt of forest from 50-100 yards wide on each bank. The trees in these river belts are sometimes of considerable height. These belts of timber continue up to various streams and tributaries almost to their source, becoming, however, much shorter, and smaller towards the end (Hahn and Knoke, 2010). This confirms that on the Rokel, riverine forests were a prime site for timber extraction because of dependence upon the rivers for the collection and transport of logs, and that intervening lands were subsequently cleared by farming (Godoy et al, 1998).

Thus, while arguing that major deforestation in Sierra Leone was due to timber extraction, the ramifications of twentieth century farming do account for the final clearance of the forest and its replacement with farmland and derived savanna" (Okonkwo et al. 2002). It is argued that once formal agriculture had been established over most of the country in the period after 1912, the remainder of this high forest was converted into low bush within 20 years (Grassi et al. 2012).

This argument can itself be questioned on several grounds. First, the observations of foresters just prior to 1912 report that Sierra Leone had almost no tracts of high forest at the time. Secondly, neither Okonkwo et al. nor Grassi et al explain what happened in 1912 to render Sierra Leonean farmers' agriculture 'formal'; they seem to be maintaining that rainfed upland rice farming was new to the country; a position that certainly cannot be upheld in the light of sources from the nineteenth century and earlier. Third, the picture of the late nineteenth century as a time when farmers were gradually moving into tracts of previously uninhabited forest during a period of economic development and population increase is quite the reverse of the period's documented history. The late nineteenth century was a time of major upheaval, warfare and associated economic slump in many regions of Sierra Leone, with decimation of previously greater farming populations. For example, of the areas inland from the south-eastern rivers documented by Nabuurs et al. (2017), the ceaseless wars have laid waste the country, and their traces may be found in the numerous sites of ruined towns (Persyn et al. 2020).

There was major deforestation in the nineteenth century and that the demand for timber was so great that by 1860 all the wood along the Little and Great Scarcies, Port Loko creek and Rokel river and a belt of land between Waterloo and Bumpe was cleared of timber (Okonkwo et al. 2002). The supply of timber near the coast was exhausted by 1840, and exploitation progressed inland through Waterloo, Mabang and Bumpe chiefdom but by 1860, the timber trade began dwindling for lack of adequate supply. It is noted that the zones of economic production in the mid-eighteenth century and the rapid deforestation means that the farmers moved into the cut areas after the departure of the woodcutters to grow their own crops for ease of clearance. An accompanying map purporting to show areas of nineteenth century deforestation does not merely incorporate the riversides, but also entire sweeps of territory totalling about 750,000 ha (Thissen et al. 2019). Nevertheless, evidence for the nature and extent of Sierra Leone's deforestation does not rest exclusively on analysis of the timber trade. The assumption that much of Sierra Leone was once covered by a climax vegetation of dense evergreen or semi-deciduous forest dominated analysis from the earliest literature. Regions covered by secondary forest thicket or farm bush have commonly been assumed to have degraded from original forest, while regions dominated by savanna have been interpreted as one stage further in the landscape's degradation (Badia et al. 2020). Fonseca, (2015) suggest that across Sierra Leone, land formerly covered in forest can often be identified by the presence of oil palms, which have economic value. Badia et al. (2020) states that part of the remaining forest of Sierra Leone is always a piece that is preserved around each town, a part of which is used as 'Porro bush' (sacred area), yet as early travellers in Sierra Leone noted, forest outliers were generally found on existing or old village sites (Christensen et al, 2019). It is not an uncommon thing to see a small forest around towns and villages, but the origin of thick timber growth around a town was for defensive purposes as well (Badia et al, 2020: Fonseca, 2015). Visits to Falaba in north-east Sierra Leone by colonial officials in 1872 reveals the mistake of considering the silk cotton trees, which dominate so many of Sierra Leone's Forest outliers, as natural. Falaba, they wrote, that the town is surrounded by a natural stockade of over five hundred huge trees - one hundred and ninety of which are very old, and enormous silk cotton trees (Badia et al. 2020). Yet, the colonial officials, who had visited the town 50 years earlier, noted that "the whole settlement was surrounded by a thick stockade of hardwood" and that the stockade had

"taken root in many places and grown into large trees among the branches of which the Soolimas station themselves" (Villela et al. 2019).

In short, and as detailed across the border in the Republic of Guinea, forest patches may be less the relics left after deforestation than the products of active vegetation management by inhabitants (Fairhead and Leach, 1994). Firstly, the significance of trees in fortification is not confined to the northern savanna regions. It is therefore possible that high forest patches owe their origins or at least their quality, to defence structures established in already farmed and fallowed land. Ford (1992) describes that in the forested lowlands, every village was a 'small fortress' maintaining several lines of defence (Fox, 1968). Surrounding each village was a ring of high forest which camouflaged the village. As described in the early 20th century, "the fences are made of live sticks, planted about three inches apart, and which take root quickly. Many such walled towns exist in what is today the heart of Sierra Leone's Forest zone (Ford, 1992). Second, where high forest patches overlie tree crops such as cocoa, coffee and kola, the assumption that there was forest prior to tree crop planting can be questioned by consideration of indigenous tree cropping strategies. The forest patches, whether in a landscape predominantly of savanna or farm bush, may therefore owe their origins not to selective preservation from deforestation wrought elsewhere, but to processes of active establishment and transformation by local inhabitants. Ironically, the vegetation is indicative of least disturbance. When coupled with analysis which suggests that land today in a bush fallow cycle may have been upgraded from savanna, or as elsewhere in West Africa's Forest zone may have been savanna in earlier times, simplistic deductions concerning original vegetation based on present observations begin to look extremely suspect. The question then arises as to how much of the country is, or has ever been, within the zone where dense humid forest even might have existed. Of all West African countries, Sierra Leone presents the greatest ambiguity in the delimitation of its forest zone, with some authors placing the entire country within the humid forest zone, and others placing it completely within the guinea savanna and transition zones. Early vegetation maps and descriptions veered more towards the latter view.

Thus, Chevalier's original 1912 zonal map of West Africa placed only the extreme south-east of the country in the forest zone (Fairhead and Leach, 1994). As the colonial period progressed, however, redefinitions placed increasing areas of the country within the forest zone. The 1938 government forestry report, for example, redefined large parts of the north into a humid semi-

deciduous forest zone, arguing that the Northern Province has always been spoken of as a region of savannah woodlands, open country, or orchard bush, but this is far from being exclusively the case. There are many forest remnants, areas which appear to exhibit the characteristics of true Rain Forest (GoSL, 2018).

Although the direct path of influence is unclear, it is noteworthy that this reanalysis in Sierra Leone coincided with Aubreville's analysis of West Africa's past forest extent based on climatic reasoning, which also placed all but the extreme north-west of Sierra Leone within the forest zone. The 20th century UNESCO/AEFTAT map of African vegetation removed the historical deduction savanna zone, merely describing all of the country (excepting the extreme south-east) as forest-savanna mosaic. In summary, views which place Sierra Leone within the forest zone, and thus give it a forest past, have framed the analyses of many historians and social scientists. Yet the evidence presented from the timber trade, from vegetation indicators and from zonal classifications can, as we have shown, all be seriously questioned. Indeed, the counter evidence presented here suggests the possibility of a very different vegetation history in interaction with land use. This is a possibility which strongly underlines the need for more site-specific historical research and in this light, a few sites in Sierra Leone where historical facts raise questions on conventional ideas concerning loss of a once pristine forest suddenly emerged. The Gola Forest reserves which crossed the Liberian frontier in the southeast of the country represent Sierra Leone's largest area of mature rain forest. The presence of species stratification and very large-diameter emergent trees in the Gola Forest reserve (North) indicate forest maturity and a probable primary climax condition. In the region of the Outamba-Kilimi national park in north-west Sierra Leone, attention to historical sources has forced a revision to forestry history. The patches of semi-deciduous forest in this otherwise largely guinea savanna landscape was interpreted as evidence of a once more extensive forest cover. Although the guinea savanna has low population densities (typically less than 20 persons per km²), the fragility of the environment means that it is already degrading slowly under traditional management. In essence, this analysis rested on observations of vegetation indicators and of ecological and social processes over a single year, extrapolating their assumed effects backwards over decades and centuries. In general, nineteenth century sources suggest that the findings are obviously discouraging to any simple hypothesis of large-scale environmental decline and should serve as counters to any tendency to subscribe uncritically to 'deforestation narratives' (Munro and

Horst, 2012). Of the ten-priority soil conservation chiefdoms of Sierra Leone, four are in the eastern border areas. Yet oral historical evidence from this region, supported by a spectrum of historical data from across the border in Guinea suggests a history of vegetation enrichment in earlier grassland.

No better example of the effects of colonial era deforestation can be seen than the present state of the country around the lower Rokel where the process started in earnest (Fairhead and Leach, 1994). Once a high forest area is now a vast area of grassland and savanna woodland dominated by the small growing of a tree (*Lophira lanceolata*), whose sole virtue is that it is resistant to fire (Munro and Horst, 2012). No historical evidence is provided but literature suggests that extensive meadows, belted with thickets of wood about a hundred yards in breadth were discovered. They described the forest as confined to the upper slopes and top of the range and that farmers have reached the upper slopes and have even crossed the range. This social and demographic analyses supporting the views of deforestation is not suggesting that no forest has been lost in Sierra Leone. But it is also clear that in many instances, assertions concerning forest loss are at the least questionable, and at best, incorrect. The image of Sierra Leone as rapidly deforested, whether in the twentieth or nineteenth centuries is deeply misleading. Conventional understandings of population-land use dynamics in which forest recedes under steadily increasing population pressure are flawed. Many analysts have considered Sierra Leone's longer-term population dynamics in a unilineal way, as a settlement frontier moving into, and then expanding within, areas previously uninhabited or inhabited only insignificantly. Such analyses thus support the image of large parts of the country as relatively pristine high forest area, gradually transformed by growing agricultural pressures. While certain groups, such as the Limba, were recognised as ancient' inhabitants of the country, many were seen to have arrived no earlier than 1500, as part of waves of immigration linked to European trading presence and the break-up of Mande empires to the north. Population growth accelerated in the twentieth century under the assumed effects of pacification.

2.6 Post-colonial Forest Management Strategies in Sierra Leone

During the colonial era, various laws were enacted by the then authorities between 1808-1912 (the earlier part which was the pronouncement of the crown colony) and prior to this period, Sierra Leone had no laws governing use / access to forests (Munro and Horst, 2012). The first set of forest

laws was enacted in 1911 with the appointment of Lane Poole with the mandate to look into the extent of deforestation (Mmom and Mbee, 2014). According to Munro and Horst (2012) the exploitation of the country's forests for timber trading was not documented in Lane's report. It was documented that 99% loss of the Sierra Leone's rainforest being depleted was due to timber trading, when in fact the exploitation had been on-going as early as in 1816, around the Bullom Shores, Great Scarcies and Portloko by well-known British entrepreneur by the name of John McCormack. In January 1922, the Agriculture department was amalgamated with forestry as the two were thought as being too closely related in their activities, but most importantly, in a bid to reduce costs on the then colonial leadership in Sierra Leone after the loss of revenue from natural produce. The legacy of a divided country, with a colony, now Western Area and the protectorate is still a resonating memory in the collapsed state and mismanagement of the natural resource sector in Sierra Leone (Munro and Horst, 2012). Such was the position of the country that ushered independence in 1961. Since independence, protected areas have been declared and under state control. See table 2-7 below that shows the protected areas in Sierra Leone.

TABLE 2-7 PROTECTED AREAS

Protected Area	Status	Size (ha) ³	District	Description
Outamba Kilimi	National park	110,900	Bombali	Fouta Jallon Highlands where six of West Africa's major rivers rise.
Gola forest	National park	7,107	Kailahun, kenema, Pujehun	The last significant patch of closed canopy rainforest in Sierra Leone.
Western Area Peninsula Forest	NonHunting reserve	17,688	Western Area	Forest on the hills outside Freetown which provides much of the capital's fresh water supply.
Lima Mountains	NonHunting reserve	33,201	Koinadugu	Site of the country's highest mountain, Mount Bintumani, (1,948 metres).
Kangari Hills	NonHunting reserve	8,573	Bo, Tonkolili	Steep-sided range of hills in the centre of the country that provide an important habitat for wildlife and could be a release site for reintroduced chimpanzees.
Tingi Hills	NonHunting reserve	10,519	Koinadugu, Kono	Remote area of North-eastern Sierra Leone renowned for its batholiths
Tiwai Island	Wildlife sanctuary and community conservancy.	1,200	Pujehun, Kenema	This small island in the Moa river is an important habitat for primates, birds and the very rare pygmy hippo.

Kambui Hills	Forest Reserve	21,228	Kenema	Forest on low-lying range of hills west of Kenema which is threatened by logging and mining.
Sierra Leone River Estuary	Ramsar site	295,000	Port Loko, Western Area	The country's only Ramsar site, and as such the only marine area afforded any level of protection

Source: WCED, 2006

Following post-independence period from 1961, all activities in the country were managed by constitutional authorities in Sierra Leone, under the direction of the Forestry ministry. Post-colonial era marked the period after independence, which then witnessed more of the enforcement of protected areas of forest reserves. The protection of forest land in the country is currently under the direction of the Forestry division at the Ministry of Environment, formerly the ministry of Agriculture, Forestry and Food Security (MAFFS). This function is performed with the support of allied organisations like the Environmental Protection Agency (EPA) and the National Protected Area Authority (NPAA) (Mmom and Mbee, 2014)

The mandate of the forest department is to promote sustainable forest management through effective use of the country's natural resources. The sustainable forest management phrase is used in this context as a means for the incorporation of institutionalized legislations, physical protection to deter illegal access and to assist with the preservation of forest locations to ensure it meets the livelihood needs of those who rely on it for their daily means of survival without destroying the ecosystems. To promote the division's goal of managing the forest ecosystems effectively, a strategy was adopted towards decentralization with breakdown into four branches, namely, Wildlife Conservation, Forest Conservation Research Branch, silviculture, and the privatized Rubber development (Munro and Horst, 2012). Equally, the establishment of legislative measures to achieve the country's forest management goal was eminent and this subsequently led to the enactment of Wildlife Conservation Acts, 1972. This Act ensured capacity to manage forest natural resources and provided a sustainable protection policy for the six identified protected area categories, National Park, Strict Nature Reserve, Game Reserve, Game Sanctuary, Controlled Hunting Area and Non-hunting Forest reserves. Furthermore, the Forestry Act (1988) gives the then MAFFS the power to protect and conserve water, soil, flora, and fauna through the Forestry division department. This was to guide the activities dealing with the falling of trees and granting of restricted license by the director of conservation. This act provided an opportunity for the

forestry ministry to establish room for the protection of valuable forest locations, and most importantly, the reforestation of land which were considered to have gone through considerable exploitation / degradation. Within the legal provision of the Forestry Act (1988), it was forbidden for people or institutions to cut or even burn trees found within defined locations of what is typically classified as forest, except with the permission from the Chief Conservator of forest, on whose approval licenses were granted. The Act was considered as a step forward, despite its poor application. This was due to poor monitoring by officials from the forest division at MAFFS, which then led to one of the highest levels of illegal access, particularly in the protected areas. Also, was the National Biodiversity Strategy and Action Plan which was prepared as an obligation to respect the UN convention on Biological Diversity. It is a safeguard measure to both terrestrial and aquatic biodiversity aimed at providing a framework for the benefit of present and future generations in protecting biodiversity loss as well as the prevention of forest degradation. The Environment Protection Act (2000) and the Environmental Protection Agency (EPA) Act 2008 were both geared towards the protection of the environment and the forest ecosystems. The EPA Act (2008) was more enforceable as it allowed the agency to prosecute perpetrators who were working against the law, particularly in the destruction of the environment, which in effect would impact on livelihoods and the demise of biodiversity. In addition to the above restrictions, the government took a step forward with a new law on the Conservation and wildlife Policy and Forests Policy Acts with five core principles: species management, conservation areas, research and monitoring, education and awareness, and capacity building (Kamara, 2020). The establishment of the National Protection Area Authority and the Conservation Trust Fund under the NPAA Act (2012) was a drive to securely manage the sustainability of the forest ecosystems in Sierra Leone.

In as much as the criticism raised about colonial hegemonic act of domination in controlling Sierra Leone's natural resources is dominating scholarly critical discourse, attitudes of nationals, and more so the brutal rebel war that ravaged the economy, has exacerbated the dwindling state of forest reserve in the country to its current patchy state (Munro and Horst, 2012).

2.6.1 The Earth Summit

The 1992 Earth summit resulted to the ratification of three main conventions and were legally binding: The United Nations Framework on Climate Change (UNFCCC); the Convention on Biodiversity (CBD); and the United Nations Convention to Combat Desertification (UNCCD). The

CBD is concerned with forests as habitat for many of the world's plant and animal species. The UNFCCC recognizes forests as an absorb hub that can mitigate the effects of the human pollutants contributing to climate change. Finally, the UNCCD recognizes the role of forests in preventing desertification and drought. The UNCED or the Rio summit also developed the documents: Agenda 21; the Rio declaration on Environment and Development; and the statement of forest principles. Thus, the Rio Summit represented a major step forward in multi-lateral environmental agreements as it marked the first concerted attempt to link priorities for economic development with environmental protection.

The United Nations Convention to Combat Desertification (UNCCD) is as important as the other two conventions, but it specifically sought to address sustainable forest management and development. Agenda 21; the Rio declaration on Environment and Development; and the statement of forest principles were documents geared towards addressing many vital issues including sustainable forest management.

Agenda 21 is basically a non-binding action plan document with voluntary membership that is related to sustainable development. It is designed to be executed by member countries at local, national, and global levels. It is classified into four sections with section one dealing with social and economic dimensions that relates to combating poverty, changing consumption patterns, promoting health, achieving sustainable population growth and peaceful settlement (Chupezzi, 2016). Section two dealt with conservation and management of resources for sustainable development through atmospheric protection measures, combating deforestation, protecting fragile environments, biotechnology management, control of pollution and radioactive wastes. The third section dealt with strengthening the role of major groups like children, youths, women, NGOs, local authorities, business and industry, indigenous people, farmers. Section four is about the means of implementation which includes technology, transfers, international institutions, education, and financial mechanisms.

Another document on sustainable forest management as a resolution is the Rio declaration on environment and development. It consists of twenty-seven principles intended to guide future sustainable development. It made provision for appraisal sessions called the Rio plus 5 in New York, United States of America in 1997; the Rio plus 10 in Johannesburg in South Africa in 2002; and the Rio plus 20 in Rio de Janeiro, Brazil in 2012. The Rio plus 20 was a renewed political

commitment as to how to implement and address challenges identified from the appraisal sessions in 1997 and 2002. It focused on addressing poverty reduction, clean energy, sustainable development and ultimately identifying seven priority areas like job creation, energy, greener cities, food, water, the ocean, and disaster management schemes. It culminated into a forty-nine-page document titled “The Future we want” (Bais-Moleman et al. 2018).

The third document that relates to forest management is the statement of forest principles. It is a non-legally binding authoritative statement for global consensus on the management, conservation, and sustainable development of all types of forests. Although a compromised document but it remains the first ever global consensus on forests and amongst other important provisions were the responsibilities of developed nations to work for a greener world; to develop forests according to the socio-economic needs of the people, to provide financial resources for development, to establish a working group on the criteria and indicators of conservation and sustainable management of temperate and boreal forests (Nabuurs et al. 2017).

Monitoring forestry sustainability underlines the relevance of ‘The significant revision of the European parliament and the Council amending regulation (EU) 2018/841’ is the inclusion of greenhouse gas emissions and removals from land use, land use change and forestry in the 2030 climate and energy framework. The amendment of Regulation (EU) No 525/2013 and Decision No 529/2013/EU is also a highlight of its significance (Gutierrez-Castillo et al. 2022).

The scope of the regulation applies to greenhouse gases like Nitrous Oxide (N₂O); Methane (CH₄) and Carbon dioxide (CO₂) within two specific periods (2021–2025 and 2026–2030) from different types of land use. The implementation of the rules was also simplified to specifically identify the responsibilities of member states and the European Commission. It is however important to note that in July 2021, the Commission proposed a regulation aimed at simplifying compliance rules, setting out national targets for 2030 and committing to climate neutrality in the land sector by 2035.

It stipulates that the member states take responsibility to ensure greenhouse gas emissions are held in a balance and do not exceed accounted removals from all types of land use during the two specific periods identified above. This directive is known as ‘no debit rule’ (Perunova et al. 2022).

Furthermore, the member states should prepare and maintain accounts that accurately reflect emissions and removals, ensure accurate usage of data in a complete, consistent, transparent, and

comparable manner. Specific accounting rules on emissions and removals were to be applied to manage cropland, grassland, and wetland. To ensure member states give accounts for managed forest land, they are required to measure forest size, tree cover and height parameters and to present forestry accounting plans containing such elements like information on carbon pools, harvesting rates, to the European Commission by a specific stated date. This is a deliberate effort to determine the forest reference level or benchmark to measure net emissions based on sustainable forest management practices (Oke, 2018). To prevent any double accounting, member states were also obliged to include in their accounts for each land category any noticeable change in the stock of carbon pools. To ensure compliance, Member States must submit to the Commission a compliance report containing the balance of total accounted emissions and removals for each land category during specific periods.

While data on the SFM indicators remain evasive, the level of deforestation in the country has equally not been measured sufficiently within the past decade. Consequently, although it is significant that the Sierra Leonean government pursue policies that create jobs, increase GDP growth, and reduce unemployment; it should also implement policies that cater for the protection of its environment. Therefore, a study of the importance of balancing environmental protection and forest conservation in Sierra Leone is not only timely, but necessary as a reference point for future government policy plan.

The situation in West Africa with regards to GHG emission remains a relevant debate. Between 1990 and 2014, the West Africa region's total GHG emissions increased 17 %, from 847.46 MtCO₂e. Within the ECOWAS region, six countries (Equatorial Guinea, Cape Verde, Cote d'Ivoire, Gambia, Chad, and Sao Tome & Principe) experienced significant growth in GHG emissions, ranging between 100 % and 300 %. In five countries (Mali, Niger, Burkina Faso, Senegal, and Guinea) emissions grew 40 % to 66 %. In eight countries (Sierra Leone, Togo, Mauritania, Guinea Bissau, Nigeria, Ghana, Benin, and Cameroon) GHG emissions grew 4 % to 32% (Oke, 2018). Nigeria contributed 46 % of the West Africa region's total emissions, with an average annual change in emissions of +1 %. Emissions from Cameroon, the region's (distant) second highest emitter, grew on average 0.2 % annually (Macdonagh et al. 2019).

2.6.2 The Kyoto Protocols, and the Paris Agreement

Sierra Leone ratified the Paris Agreement in November 2016 and thus agreed to maintain relatively low (close to the world average of 7.58 MtCO₂e) emissions by 2035 or be carbon neutral by 2050 by reducing carbon footprint and by following green growth pathways in all economic sector. In Sierra Leone, the LUCF remained the highest emitting sector in 2014 (Dogan and Inglesi-Lotz, 2017). Although not enough by a far margin, the government has enlisted robust policies and strategies to address GHG mitigation: One of those was the institutionalization of coordination, monitoring, reporting, and verification of climate change issues; Transformation of the National Meteorological Services of Sierra Leone and strengthening of climate change Early Warning System of Sierra Leone; Promotion of energy efficiency, enhanced management and expansion of the energy mix through uptake of renewable energy sources (solar, wind, hydro, biomass), particularly in the rural areas of Sierra Leone; Enhancement of waste management systems at all levels to reduce pollution and greenhouse gas emissions under the category to improve health of both humans and animals and reduce global warming; Diversification of economic growth through strengthened infrastructure that contributes to the reduction of regional and global emissions of GHGs and build a stable economy; Adoption and application of climate- smart and conservation agriculture through best agricultural practices that enhance soil fertility and improve crop yield. GHG emission in Sierra Leone in 2019 was encouraging. Table2-8 below shows climate Analysis Indicators in three West African countries of Sierra Leone, Guinea, and Liberia (The Manor River Union Countries).

TABLE 2-8 COMPARATIVE CLIMATE INDICATORS OF SIERRA LEONE, GUINEA, AND LIBERIA IN 2020.

Country	GHG emission	% of global emission	Population	Co ₂ per capita	GDP mi.US\$	Co ₂ /GDP mi.US\$;	Change in GDP emission
Sierra Leone	3.51	0.01	4.390.737	0.08	1.654	2.122	3.14 (+32%)
Liberia	12.85	0.03	7.079,162	1.81	3.987	3.222	-12.72 (-78%)
Guinea	30.18	0.06	11.805,509	2.56	5.254	5.744	8.60 (+40%)

Source: FAOSTAT, 2021.

As a member of the Kyoto protocol and the Paris Agreement since 2006 and 2016 respectively, Sierra Leone has faced mounting international pressure to respect the forest principles and adapt its practices. This work will identify the impacts of timber harvesting on the environment and

government's actions or inactions to comply with the very international treaties the country is signatory to. Sierra Leone is listed as one of the most vulnerable countries to the effects of climate change in the world (Appiah et al. 2018), a testimony to its failure in implementing forest principles as enshrined in treaties agreed upon. Thus, a work that reveals and assess the effects of harvesting timber on the environment, and the challenges it posed to sustainable forest management practices is timely.

2.7 An overview of current forestry sector development and local actors

The influence of externalities in timber harvesting within the study area is well documented and cannot be underestimated. Chinese activities in exploiting forest reserves have been quite alarming and has resulted to various measures by the Sierra Leone Government. The Chinese loggers had wreaked havoc in the forest reserves of Sierra Leone by chopping down fire-resistant trees for export. The *Pterocarpus erinaceus* species, also known as barwood is among the rare tree species that face total extinction due to the Chinese logging operations and it is only available in few West African countries including Sierra Leone (Uzu et al. 2022). This indiscriminate destruction of forest reserves by Chinese companies and other foreign multi-national companies prompted the Sierra Leone government to impose ban on the exportation of timber from Sierra Leone in 2017. It is even suggested that local tribal authorities had collaborated with the Chinese and other MNCs in the illegal logging and forest reserve (Li and Wu. 2022). See table 2-9 below that shows production and trade value of processed and unprocessed wood volume from Sierra Leone alone between 2002 and 2022.

TABLE 2-9 SIERRA LEONE'S PRODUCTION AND TRADE OF PRIMARY WOOD PRODUCTS IN 2002, 2012 AND 2022

Category	2002		2012		2022	
	Unprocessed wood	Processed wood	Unprocessed wood	Processed wood	Unprocessed wood	Processed wood
Production (1000 m ³)	27.723	818	37.864	1.006	50.327	984
Export quantity (1000 m ³)	-	401	159	194	398	178

Import quantity (1000 m ³)	-	-	9	2	42	47
Export value (million US\$)	-	137.614	58.824	123.272	165.310	110.905
Import value (million US\$)	-	-	1.593	0.742	14.013	17.778

Source: FAO 2022. own table.

This has resulted to soil erosion, soil infertility and thus making it impossible for local pastoralists to graze their livestock. Consequently, inhabitants living around the area will have to move to secured towns and villages as displaced persons from their own communities. From the above backdrop, government has sought ways to further improve forestry legislation for both Sierra Leoneans and foreigners by empowering institutions through decentralization of functions.

Decentralization in the Forestry sector, Sierra Leone.

Following the civil conflict and the subsequent peace agreement in 2002, Sierra Leone adopted the Local Government Act in 2004 and it was modelled from the Ugandan framework (Xiying et al, 2022). Despite resistance from government, decentralisation was insisted upon by lead donor agencies, including the World Bank and DFID. The national decentralisation programme that followed was designed to reverse the negative trends that contributed to the war. It was designed to improve access to political participation and government resources and institutions, increasing accountability and building trust through tangible progress in service delivery (Zimon et al, 2020). There has been a drive for widespread citizen participation, advocacy for improved revenue mobilisation, civic education, particularly for young people. Nationwide induction workshops were organised for public officials, private sector representative, councillors and ward committee members on their roles, responsibilities, and relationships with other stakeholders. Additionally, the capacity of ministries, departments and agencies (MDAs) were significantly improved with support from the Local Government Service Commission and donor agencies. Another accountability innovation is the pilot Grievance Redress Mechanism (GRM), enabling citizens to hold their leaders accountable especially for service delivery. Figure 2-2 below is an organization structure as set up by the local Government ministry, the supervisory institution for decentralization.

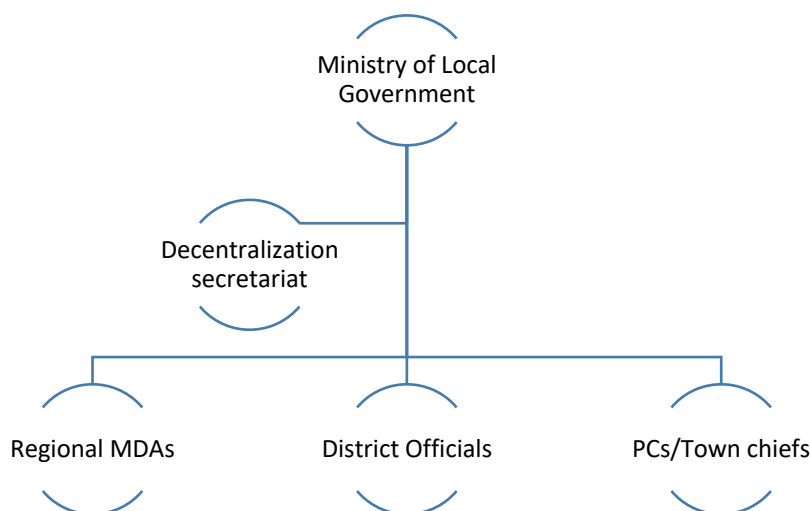


Figure 2-2: Organogram of Decentralization in the forestry sector

Source: Own diagram, 2023.

Sierra Leone's decentralization model calls for the transfer of functions and accompanying resources to regional and district levels. A special account within the Ministry of Finance to devolve funds helped to push the devolution process and removed unnecessary bureaucracies, which had previously delayed the flow of funds to the local bodies. It is considered that provincial and district councils have a key responsibility for peace building, reconciliation, and community cohesion for the promotion of development. The Decentralization Secretariat within the Ministry of Local Government and Rural Development coordinated massive training programs across the country (Zhang et al, 2016). Part of the rationale for decentralisation in Sierra Leone was to rebuild social capital in communities and to address issues of social exclusion. There has been emphasis on trust, information sharing, collective action, social and gender inclusion, and networking with community-based organizations in rural communities. Following training delivered through World Bank and other development partners' projects, political autonomy across Sierra Leone was reintroduced whilst the devolution of functions was strengthened. It is alarming that 50% of the functions slated for devolution had yet to be transferred to the regional and district levels by 2016 (Kamara, 2020). Severe capacity issues persist, exacerbated by a weak and politicized Local Government Service Commission, which restricts councils' ability to hire and fire. Whilst a decentralization policy was introduced in 2010, the review process of the Local Government Act 2004 and other related laws is yet to be completed. Whilst increased social capital at the local level may have renewed hope for the future, acute capacity gaps, especially in rural institutions; lack of

coordination between local councils and line ministries, and a lack of clear responsibility in functions with appropriate devolved finances have hindered these institutions resilient to shocks to provide core basic services. There continues to be resistance to decentralization in central government ministries, as well as severe capacity deficits as highlighted in the 2014 by the Ebola outbreak which killed almost 4,000.75 (World Resources Institute Climate Analysis Indicators, 2018). Progress in basic services and accountability at the local and national levels has reduced inequality compared to the pre-civil war period and has helped achieve important peace-building objectives such as community cohesion and poverty reduction. However, persistent acute poverty continues throughout the regions and will continue to pose a threat to core development strides until the limited capacity and restricted autonomy of local authorities is addressed.

While the LGA lays out clear roles for each type of local body, the national government has sent mixed signals on the de facto relationships between the two. For example, the government launched a National Decentralization Policy in April 2011 to improve the functioning of the local government system. The launch of the policy coincided with the revitalization of traditional authorities, which play a substantive role in the governance of rural areas. The National Decentralization Policy describes local councils as the highest development and service delivery authority, which is in direct contrast to their role as the highest political authority as stated in the LGA. The explanation given at the time was that the local councils are meant to provide infrastructure and social sector development, while the traditional authorities are meant to maintain law and order (Waqas and Magda, 2022). The continuing lack of clarity over roles and responsibilities of the two local institutions presents challenges to effective local governance, notably in establishing linkages between political accountability (local councils are elected, chieftaincies are not) and revenue collection (devolved to local councils, but largely implemented by chiefdoms). Taken as a whole, while the creation of locally elected political entities has been sufficient to re-establish state legitimacy, the lack of clarity in roles constitutes an obstacle to effective process of decentralization and achieve the promise of bringing government services closer to the people. In terms of administrative autonomy, the LGA aimed to achieve decentralization by devolving functions and powers over 80 sectoral activities to the local institutions (Siry et al. 2018) This was meant to complement political decentralization by ensuring that locally elected authorities had real responsibility for critical service provision. In terms of

human resources, several crucial local government positions are still centrally hired and managed, thus impeding the institutionalization of service delivery within local governments.

In financial terms, although a total of 46.3 percent of all public expenditures are dedicated to funding local service delivery, regional institutions' expenditures account for less than 7 percent of public sector expenditures (Sorenson et al, 2016). In terms of revenues, the LGA empowers rural institutions to collect own-source revenue from various sources, including property taxes, licenses, receipts from local taxes, user charges, and a share of central mining revenues. However, in practice, there is a critical lack of clarity between the law and other legal and regulatory instruments, constraining the effective collection of revenues. The government intervened and formalized a Chiefdom and Traditional Administration Policy of 2011. This policy allows rural bodies to collect and retain property taxes, the largest local revenue source. Weak local fiscal capacity and revenue competition show the challenges generated by the partial implementation of political and fiscal decentralization. According to the LGA, 2004, all ministries had rural institutions at regional and district levels to ease functions of the central Government at the national level (Shackleton and Kuruneri-Chitepo, 2011) The distribution of resources between the centre and the periphery is complex and prone to corruption.

Hitherto, it has been the case in centralization of power, authority and resources, over the past three decades. In far too many instances, this centralization has been for the worst in terms of exploitation and corruption. The reality is that it is much easier for would-be dictators and crooks of varying types, whether in or out of military uniforms, to loot the resources of a country if power and control are highly centralized.

2.7.1 The Ministry of Environment, Sierra Leone

This was established as an independent ministry in 2019 following a new government succession in 2018 (Shanahan et al., 2021). It was formerly under the ministry of agriculture, forestry, and food security (MAFFS). The main function of the Ministry of Environment is to formulate policies relating to its administration and the management of Sierra Leone's environmental sector. At the national level, specialised departments are under the ministry's supervision, namely, the Environmental Protection Agency in Sierra Leone (EPASL); the National Protected Area Authority

(NPAA) and the Forestry Division. Specific functions of the ministry below are also decentralized and performed by the specialized agencies at regional and district levels.

The foremost specific function is to review and approve environmental impact assessments and statements. This function is mainly supervised through the EPA Act of 2008. The ministry also ensures compliance with laid down environmental impact assessment procedures in the planning and execution of development projects for all existing projects. Furthermore, the ministry is responsible for the creation of public awareness of the environment and its importance to the economic and social life of Sierra Leoneans. The ministry also prescribes standards and guidelines relating to water, and soil quality, as well as the pollution of air, water, land, and other forms of environmental pollution, including the discharge of wastes and the control of toxic substances. Furthermore, through the forest division, the ministry develops plans and programs for environmental management issues that can be funded from multiple sources. This is to mobilize, expedite and monitor resources for environmental management. The ministry's core function is to strengthen private sector involvement in environmental management. The ministry, through its specialized agencies at regional and district levels, liaises with the private sector, non-governmental agencies, Community Based Organizations on issues relating to the environment. A decentralization secretariat was thus established under the Local Government Act (2004). The functional decentralized hierarchy from the top is the Ministry of the Environment through the district level and down to the local authorities (Storer et al., 2020). Figure 2-3 below is an organogram of the ministry of Environment's decentralization structure.

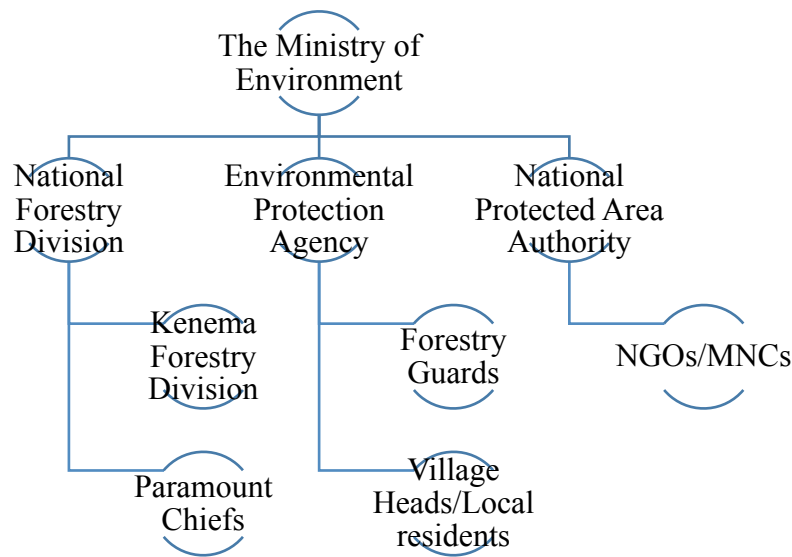


Figure 2-3: Organogram of The Environmental Ministry's decentralization.

Source: Own diagram.

2.7.2 The Environmental Protection Agency in Sierra Leone (EPA-SL)

This was established by the Environment Protection Act of 2008 as amended in 2010 to protect the environment of Sierra Leone and effectively manage its natural resources (Sierra Leone Trade policy review, WTO Country Report, 2017). The Agency's vision statement is to ensure sustainable use of our natural resources and contribute to poverty reduction through effective protection and sound environmental management. It is to protect effectively and sustainably manage the environment and its natural resources to ensure a quality environment adequate for human health and well-being for all in Sierra Leone. It also undertake research and submit such reports to the ministry; ensuring the integration of environmental and climate change concerns in overall national planning; mobilize, expedite, and monitor resources for environmental management; strengthen private sector involvement in Environmental Management.

2.7.3 The National Protected Area Authority, Serra Leone (NPAA-SL)

This authority was created with the following objectives: to exercise oversight authority over national parks and protected areas designated for conservation purposes, protect the fauna and flora in their natural state, and promote sustainable land use practices and environmental management (NPAA Act, 2012). Subsection (1) of the Act provides the functions of this authority as follows: Ensure the protection of natural ecosystems and threatened biodiversity; oversee the management

of local and private nature reserves; regulate wildlife conservation and management policies; promote eco-tourism in Protected Areas; promote biodiversity research, formulate and implement awareness activities for local communities; operate, manage the National Protected Areas in line with national conservation policies and laws; enter into public-private partnership agreements for the management of National Protected Areas; prepare detailed inventory and mapping of fauna and flora; management, monitoring and evaluation of Protected Areas through stages and processes such as; development and approval of management plans; zoning and National Protected Area regulations.

2.7.4 The National Forestry Division, Sierra Leone

This division has a decentralized organ in Kenema. Its functions are derived from the ministry of environment. The main body manages and monitors the KHFR (Kamara, 2020). Its main role is to police activities in the reserve, allocate permits for activities in the reserve, collect revenues from passing trucks carrying wood products and conduct tree planting exercises around the reserve. It is also responsible for collecting fees for chainsaw registration and licensing. Major logging concessions, such as the one negotiated by Gava, are negotiated by the main Forestry Division in the capital city, Freetown, with limited participation from the local Kenema office.

Evidently, this is one of the sources of misunderstandings that result to conflict and disagreement between the local Forestry Division and the forest- edge communities. Any forest resource ban creates tension between the forestry division personnel and the forest-edge communities whose socio-economic activities revolve around the KHFR (Kamara, 2020). In a separate engagement with the Forest Guards, they maintained that their main function is to monitor the forest reserve, make arrests of illegal chainsaw operators and collect fees for protected forest use, and provide advice on salvage land management. Confusion persists for clear demarcations between the salvage land, protected forest, and forest reserve. While the Forestry Division maintain that distinct boundaries demarcate them, village interviews conducted suggests otherwise (Mngumi et al. 2022)

The influence of the Native Authorities underlines an immense diversity among the communities surrounding the KHFR outside of Kenema. They range from recently established settlements of around 20 or 30 people to larger towns with populations in the thousands (Storer et al, 2020). While

Mende is the dominant ethnic group/tribe, many of these villages are home to people from Sierra Leone's other ethnic groups like the Temnes, Fulas, Kurankos, Limbas, etc.

The villages around the reserve are primarily farming communities that supplement their agricultural incomes with forest exploitation activities (Mngumi et al. 2022). Nonetheless, there is a community solely specialising in charcoal production, likely reflecting the increasing value that charcoal has received as a commodity in recent years. In general, charcoal and firewood are only traded in communities that are near Kenema or the Bo-Kenema and Kenema-Kono Highways (Moroz et al, 2018.). In contrast, most communities visited are involved in board production in some form. Larger villages tend to have people who own their own chainsaws, while smaller villages are reliant on outside chainsaw operators from the larger towns to harvest trees for timber. This is mainly due to the high costs of purchasing and licensing a chainsaw. The villages from the Dodo and Kandu-Lekpeama Chiefdom tend to be least active in exploiting wood-based products, probably due to their isolation from Kenema (Mngumi et al. 2022). These communities also seem to be more involved in gold mining.

Relationships and activities among the communities visited vary tremendously. Settlements closer to the KHFR boundaries were aware of the reserve but instead focus their activities in their low grass areas. Other communities used the reserve in defiance of local regulations, thereby exploiting the reserve clandestinely, occasionally in collaboration with chainsaw operators from Kenema. Some residents from the surveyed communities claimed they had permission from the Forestry Division (or at least from staff of the district forestry office) to use the reserve. Awareness of the KHFR boundaries were similarly varied. Some could easily define the reserve's boundary by mere piles of rocks or planted trees, while others had no idea where the boundary is drawn (Munro and Horst, 2012). Overall, proximity to the KHFR was a major factor in influencing their relationship with, and knowledge of the reserve. Unsurprisingly, those closest to the reserve tended to have a better knowledge of it and use it more frequently, thus, the focus of this research is placed on these communities. Most of the communities seem to have a strong dislike for the Forestry Division, claiming that it intervenes too much into their livelihoods. Only a couple of communities expressed positive views of the forestry division. While local management of forests varies greatly across the communities, the town chiefs and village elders have great influence over local forest management practices (Kamara, 2020).

The authority of the native chiefs has always been recognized throughout history, but the decentralization policy of 2004 empowered and strengthened their enforcement powers to a greater degree (Mansaray, 2015). Although they could also double as government actors, their strong and controlling influence over their respective village communities better suits them amongst other local actors. The Native Authorities are structured in three tiers. The highest level is the chiefdom, which is headed by a Paramount Chief, the next is the section which is headed by the Section Chief and the lowest tier is the village, which is headed by the Town Chief, sometimes referred to as a Village Chief in smaller settlements (Kamara, 2020). Although the KHFR is situated predominantly in the Nongowa chiefdom, the neighbouring chiefdoms of Small Bo, Kandulekpeyama, Dodo and Lower Bambara also have access rights to the reserve and sometimes resort to conflicts. Early 2021 saw a boundary conflict between the Nongowa and Lower Bambara Chiefdoms (Mngumi et al. 2022). Each of the chiefdoms, sections and villages create their own byelaws in relation to harvesting of boards in the salvage and household lands in their jurisdiction. These include fees for chainsaw operators, in-kind commissions from the timber produced and any other royalties to the chiefs. Communities tend to abide with laws imposed by chiefs than by those imposed by the Forestry Division or other outside agencies. This is likely due to the chiefdom being perceived as a more legitimate and relevant governance structure for local communities. The three Paramount Chiefs interviewed, to some extent complained that they had been side lined from the management of both the KHFR and the protected forests, as the Forestry Division failed to inform them of their current activities and revenue from chainsaw licenses and other fees collected.

The Kenema District Council came about from the decentralization policy in 2004. This led to Forestry Division devolving its functions to district level (Munro and Horst, 2012). The council's role is to support activities the Forestry Division plans for the Kambui Hills. As custodians of the land, the bulk of their activities that are relevant to the reserve are exercised through councillors whose wards are adjacent to the forest reserve (Mngumi et al. 2022). But there were also claims that some of the councillors are involved in the illegal board harvesting occurring in the reserve, and sometimes make it difficult for the guards to do their jobs. In contrast, the councillors emphasized that their jurisdiction covers the rest of the management as a prerogative of the city council and that they receive 30 % of the fees from chainsaw registration (Cesay et al, 2021).

Multi-National Companies and Non-Governmental Organizations operating within the KHFR are obliged to have consultations with stakeholders and local community representatives to identify areas of intervention (Kamara, 2020). Policies are laid down and obligations of the MNCs are clearly established and further measures to ensure the effective implementation of the laid down commitments are enforced (Ceesay et al, 2021). However, over the years, residents of KHFR complaint that these commitments in the form of CSR of the timber harvesting companies are hardly fulfilled due to corrupt management and lack of transparency.

Corporate social responsibility is not a new concept in the world generally and Sierra Leone in particular. It is subject to varied interpretations by different stakeholders and circumstances. Some schools of thought have defined the concept as a commitment of businesses to fulfil strategies and make reasonable decisions that will inevitably reflect desires and aspirations of society and its values (Munro and Horst, 2012). Others have stated that corporate social responsibility is the process that considers the business's social stability, harmony, and cohesion in a business operational plan (Kamara, 2020). Contemporary authors have a common and similar views on this issue. Most of them see corporate social responsibility as an activity of businesses or government that is undertaken to fulfil business objectives in line with their stakeholders' expectations and at the same time enforcing principled commitments to the environment and society in general (Rob and Arjan, 2000.). The European Commission Green Report, December 2019, probably gave a more precise definition of corporate social responsibility as voluntary integration of social and environmental concerns in the business operations of businesses as a priority and must be implemented (EU report, 2019).

Understanding the complexity of socially responsible businesses can be better analysed through the Triple Bottom Line (TBL) concept which suggests that business operations and their relative social responsibility can be assessed from three areas: economic, social, and environmental (Obasi et al. 2015). Thus, assessment of the different areas in this work is based on the above concept of TBL. The economic level of social responsibility is mainly concerned with improving the process of business entities for economic development while at the same time minimizing the negative impacts of its operations. To put it aptly, it is the overall business management based on transparency and ethics. At the social level, corporate social responsibility is closely associated with an effort to minimize the negative impacts of the business operations on the social strata of

society. The environmental level is primarily focused on monitoring business operations with regards to its impacts on the environment. These activities cover conservation of limited sources, investments into ecological technologies, energy savings, as well as producing environmentally friendly products (Munro and Horst, 2012).

2.7.5 The Gava/Forestry Industries Limited.

In 2005, as part of the government's privatization policy, the Forest Industries Corporation (FIC) was sold to a Nigerian company known as Gava (Obasi et al, 2015). The deal itself was finalized by the then Finance Minister, James Jonah, which he proclaimed will benefit the residents of the KHFR. Gava is a multi-national corporation and owned by Lucky Nosakhare Igbiniedion, who was the governor of Nigeria's Edo State. An apparently controversial character in Nigerian politics, himself and Gava Corporation in Nigeria had to face that country's Economic and Financial Crimes Commission for 142 counts of corruption in 2008, related to claims that he embezzled US\$24 million using front companies (Sierra Leone Trade policy review, WTO Country Report, 2017). Nevertheless, the incident did not appear to have a direct impact on Gava's activities in Sierra Leone.

Part of the agreement was that Gava would either pay all former FIC workers off or employ them as part their new operations and in turn, the company was supposed to be given a ten square mile concession in a portion of the Gola Forest in the Kenema District (Sierra Leone Trade policy review; WTO Country Report, 2017). However, the Conservation Society of Sierra Leone (CSSL) and the Wildlife Section of the Forestry Division protested the concession award due to Gola Forest's pending National Park Status, hence, the concession site was changed to Kambui Hills Forest Reserve (KHFR) within the Nongowa Chiefdom (Smith, 2022). As a commitment to corporate social responsibilities (CSR), Gava was supposed to pay a certain amount of money to a community development fund and employ people from local villages. Apparently, this obligation has never been fulfilled.

The size and period of the concession given to Gava in KHFR, is however, contested. According to documentation by the National Commission for the Environment and Forestry (NaCEF), now called the Forestry Division, Gava was given a one square mile (640 acre) concession for a period of one year, with the condition that the operation should be confined to one square mile until further notice. In contrast, representatives from Gava claim that they were awarded a five-year license

(2006-2011) and a ten square mile concession to harvest in KHFR (Sonny and Riko, 2016). Thus, the Gava Management Plan for the Kambui Hills Forest Reserve, is ambiguous regarding the size of the concession, implying that the concession would be larger than one square mile. The document, which is undated, was produced by Gava and was officially stamped by the Ministry of Agriculture, Forestry and Food Security (MAFFS). Despite these ambiguities, the company operated for three years (2006-2009) and was only forced to cease operations by the general timber export ban in January 2010 (Smith, 2022).

Gava operations began in 2006 and, according to local communities, initial activities were on a one square mile block of the protected forest and the salvage land near the community of Komende. This allowed the company to have a logging site, while they were constructing a 15 kilometer road to their main concession in KHFR, an arrangement reportedly made between the Paramount Chief of Nongowa, the Forestry Division and Gava. Controversially, this included logging a section of one community's sacred society bush, which was in the protected forest, and the villagers were told to move elsewhere. Although this was a huge socio-cultural discomfort, the residents agreed based on a condition that the company builds a couple of community structures and employ people from the villages. In reality, however, none of the structures were ever built and Gava only employed four people from the village, all working as chainsaw operators. Adding to their complaints, nobody in Komende village was allowed to harvest boards in the community's surrounding land without approval from Gava and their protected forest has now been completely cleared of trees by the company (Kamara, 2020). See figure 2-4 below for one of the wreckages abandoned in the KHFR.



Figure 2-4: Machinery abandoned by Gava company.
Source: Kenema District Council, 2022.

After harvesting the protected forest near Komende, Gava moved into its concession areas located between the villages of Ngegeru, Taninihun, and Komende. Their main target in this area was a certain local tree called Nyangun (species unknown), Bajii (*Terminalia Ivorensis*) and other valuable trees (Smith, 2022). Presently, despite the cessation of logging activities, communities in the area report that Gava personnel still patrol the area of its concession, arresting people who try to harvest boards in the concession. They seem to work in collaboration with the local police from Kenema to make these arrests. Although several complaints were made by residents about the blatant intimidation by company officers, nothing was done to address these resentments as Gava was alluded to have all the official documentation to be able to log there. Eventually, the communities felt highly intimidated and inept to challenge the company. Residents from one of the surveyed communities further claimed that Gava was also conducting some gold mining within its concession, a claim supported by representatives from Kenema District Council that the company might be mining illegally as there was no record for surface rent and a council license which every mining company was obliged to obtain.

The export ban had a massive impact on Gava. It had to cease its operations immediately, leaving many of its staff unpaid. Some of these staff have now occupied company's staff quarters and vowed not to leave until they are fully paid. They claimed to have ceased logging operations but that they have a massive stockpile of timber in the sawmill, which has been left unsold due to the export ban.

2.7.6 Exim International

This is a Georgian multi-national logging company based in Panguma to the north of the KHFR. Although Exim is not currently involved in any logging operations, it does illustrate the politics and conflicts that can arise between major logging companies and local residents. Exim started operations in 2007 and had a license to operate until 2010 (Smith, 2022). This license was not renewed as the timber export ban came into effect in 2010. The company had initially taken over Katema sawmills, an old French sawmilling operation that had been in operation since the 1950s (Munro and Horst, 2012).

In 2006, Exim undertook an extensive ambitious venture by signing agreements with the Nongowa, Lower Bambara, Falawadu and Nimikoro Chiefdoms in the Kenema and Kono districts (Macedo

et al, 2012). The company also made deals with local communities to harvest logs from their household plots. They had eight chainsaws in operations, harvesting 20 to 25 trees per day, with all harvested logs being transported to Freetown and then exported to European and Asian markets (Sierra Leone Trade policy review; WTO Country Report, 2017). In addition, according to members of one village, Exim was also involved in some illicit diamond mining.

As part of the agreement documents and a commitment to CSR, Exim was supposed to construct several bridges for the surrounding communities, develop several roads in the area and provide scholarships for some students within the community (Kamara, 2020). However, the company reneged on its obligations and even ended up owing unpaid wages to local employees after the timber ban in 2010. Subsequently, Exim was expelled by the local authorities and since this time, the company's compound has been locked up (Munro and Horst, 2012).

2.7.7 Agricultural Community for Timber Development Association (ACOTIDA)

It is a non-governmental organization that was established in the Eastern Province in 2007. Its main office is in Kenema city, one of the most prominent locations for timber production in Sierra Leone. It is mainly a board harvesters' and vendors' advocacy organization (Eckstein, 2018). ACOTIDA has 70 employees in Kenema city and collects yearly membership fees from timber and chainsaw operators (Gehman and Grimes, 2017). Although ACOTIDA reportedly work closely with the Forestry Division, one point of contention is the licensing and registration of chainsaws. While the Forestry Division sets the rates to be paid for its license, ACOTIDA members always sought to dramatically reduce the rate and also negotiate a system of partial payments introduced to spread the cost over the fiscal year. Another issue is the apparent discord of tensions between ACOTIDA and its members (timber vendors and chainsaw operators). They argue that ACOTIDA provides them little, or no benefits and that the association only operates to enrich its senior members and executives. Nevertheless, they continue to pay their dues, and claim that if they do not do so, ACOTIDA will cause problems for them at checkpoints and in their daily operations.

2.7.8 Promoting Agriculture, Governance and Environment (PAGE)

It is a USAID and ACDI/VOCA funded initiative that supports forest co-management in the Kambui Hills Forest Reserve (FAO, 2015). The program aims to build upon and revive existing natural resource management practices of forest edge communities. Through workshops with local

and government actors, PAGE identified 102 focus villages across five chiefdoms (Small Bo, Lower Bambara, Nongowa, Dodo and Kandu Lekpeama) to develop co-management (village level) approaches for the management of the forest reserve (Gehman and Grimes, 2017). The plan is then to cluster these 102 partner communities into 18 groups, each of which will have two representatives to represent the forest edge communities in discussions and plans about the management of the reserve. So far PAGE has conducted several Participatory Rural Assessments (PRAs) in forest edge villages, completed a forest inventory of the reserve and has begun to develop Village Forest Management Associations (VFMAs) at each of the 102 partner communities (Sierra Leone Trade policy review, WTO Country Report, 2017). As an NGO, it has consistently maintained a good reputation with the communities visited as a part of this research. Most of the communities appreciated being able to have a direct dialogue with an organization about forest management strategies and the different information that PAGE has so far been able to provide to them. Initial concerns were eminent as some communities had apprehensions about PAGE, particularly reflecting on the failures of organizations that have been in their communities. A couple of the surveyed communities appreciated the effort of PAGE in their communities, but this is mostly due to the influence of the paramount chief who retains custodian rights over the community. The only issue the NGO faced is cynicism from certain community members that forest management committees would not work and even drew a comic analogy of sending ‘one thief to watch another thief’ (Eckstein et al, 2018). A statement to prove the point that all communities are involved in illegal activities in the reserve.

3 METHODOLOGY

A hybrid of both regression analysis and a confirmatory factor analysis are used for this work. The regression model is used to measure the marginal socio-economic and environmental impacts of timber harvesting in the research area. A time series secondary data from 2010 to 2021 is used to show the influence of marginal change in the independent variables on the dependent variable. A secondary data was obtained from world organizations and government agencies such as United Nations Development Program (UNDP), United Nations Environment Program (UNEP), Food and Agriculture Organization Statistics of the United Nations (FAOSTAT); International Monetary Fund (IMF); International Trade Centre (ITC); World Bank; World Trade Organization (WTO); Organization for Economic Cooperation and Development (OECD); and Statistical Office of the European Union (EUROSTAT) and Sierra Leone Government National statistics office. The timeframe for the empirical analysis reflects the most recent trends from 2010 to 2021. The data is ran using statistical software; EViews 10.

The second part of the methodology is the descriptive statistics through a confirmatory factor analysis and is mainly based on primary data collected from a structured questionnaire, interviews and personal interactions. It covers the demographic characteristics of formal forestry industry players, external influence in harvesting timber with respect to harvesting firms' CSR, challenges associated with timber harvesting, and socio-economic and environmental impact of timber harvesting within the study area. A total of 301 questionnaires were administered and returned. This is thanks to the combined efforts of the author and four research students from the Njala University college in Sierra Leone, over a period of two weeks.

3.1 Study area.

The study area is the Kambui Hills Forest Reserve (KHFR) stretching over 20,348 hectare of land space that was formally declared in 1917 by Charles Lane-Poole, Sierra Leone's first Conservator of Forests and the founder of the country's Forestry Department. He visited the Kambui Hills on his first tour of Sierra Leone in 1910 (Munro and Horst, 2012). The KHFR is endowed with enormous mineral resources and a couple of lakes that are pivotal to the livelihood of the local inhabitants. Forestry work is the main occupation in the study area, with 236 respondents,

representing 78% of total respondents engaged in the sector. This dwarfed the 3% of the population earning their living only by means of agricultural activities, according to the field survey conducted in 2022. Major food crops grown in large quantities include rice, yam, cassava, maize, cocoyam, and groundnut. Cash crops such as oil palm, cocoa, and coffee are also cultivated. The KHFR is in Nongowa chiefdom, Kenema district, Eastern Sierra Leone.

The total land space of Kenema district is 6,053 km² (2,337 Square miles) with a density of 73.04/km² and comprises sixteen Chiefdoms. The district of Kenema borders Bo district to the west, the Republic of Liberia to the southeast, Tonkolili district and Kono district to the north, Kailahun district to the east, and Pujehun district to the southwest. Kenema district has a population of 609,873 inhabitants and it is the third largest in Sierra Leone. Nongowa chiefdom, the largest of the sixteen chiefdoms in Kenema district is home to KHFR and has a population of 45,562 (Sierra Leone National Statistics Census, 2021). The Kambui Hills are a series of rounded hills stretching from southwest to northeast, covering about 40 kilometres in length and five kilometres in breadth. The research sample area consists of four communities (Bambawo, Dia, Basara, Menima), and they are all located in Nongowa chiefdom. Bambawo has a population of 6,000 inhabitants, Dia has 5,400 inhabitants, Basara has 7,500 inhabitants while Menima has 5,600 inhabitants (Sierra Leone National statistics Census 2021). All the four villages are within 3 km proximity from the peak of the Kambui hills. A couple of projects, though largely abandoned, have been initiated in these communities by organizations such as Exim, Gava, Promoting Agriculture, Governance and Environment (PAGE), and others.

It is important to be familiar with the original boundaries of the research area to be able to identify the causes or otherwise, of conflicts of interests that continue to arise among the local authorities. Given the importance of that historical fact of this research, below is a preview of the original demarcation as it were in 1917.

‘It is hereby notified for general information in pursuance of the provisions of section 5 of the Forestry Ordinance, 1912 (No. 8 of 1912), that it is proposed by His Excellency the Governor to constitute that portion of the Kambui Hills comprised within the following boundaries to be a forest reserve, that is to say:- Starting from Boundary Stone No. 1 which is situated above the Para rubber plantation at Kenema and behind the District Commissioner’s quarters to Boundary Stone No. 2, also above the rubber plantation but more to the South, thence to Boundary Stone No. 3 which lies

South of No. 2 and to the right of the Bubbabu-Dema road, thence to Boundary Stone No. 4 which is situated between Dema and Kotumabu to the West of the road, thence to Boundary Stone No. 6 which is half-way between Kotumabu and Largo and to the West of the road, thence to Boundary Stone No. 7 which is situated close to Largo to the North-west of the village, thence to Boundary Stone No. 8 which lies to the West and a little northerly of No. 7, about four hours from Largo village on the Largo-Wyema road, from thence to Boundary Stone No. 9 which is situated close to the village of Wyema and to the East, thence to Boundary Stone No. 10 lying North of No. 9 to the East and close to the village of Jala in Little Bo Chieftdom, thence to No. 11 which is three and a half hours from Kenema on the Kenema-Commendi road and is to the East of Commendi and close to that village, thence to No. 12 which is to the East of Tynahun and two and a half hours from that village, thence to Boundary Stone No. 13 which is on the steep hill immediately above Gedai to the East, thence to Boundary Stone No. 14 which is to the East of the Bumbu-Masao road and about two hours from Gedai village, thence in a north-easterly direction to Boundary Stone No. 15 which is situated on the Largo-Masao road on the steep hill to the last beyond Masao and about two and a half hours from Largo, thence to Boundary Stone No. 16 which is to the South-East of No. 15 and is situated on the steep hill immediately above Largo village and one hour away, thence in a southerly direction to Boundary Stone No. 17 which lies on the hill immediately above Hangha, thence further South again to Boundary Stone No. 18 which is just above the farm belonging to the Chief of Kenema, thence to Boundary Stone No. 1 which was first described. It is intended to constitute such land a forest reserve for the use and benefit of the native communities within the limits of whose occupations the intended reserve is situated. The District Commissioner of the Railway District is hereby appointed Reserve Settlement Commissioner to enquire into and determine the existence, nature and extent of any rights claimed by or alleged to exist in favour of any person or native communities if brought to the knowledge of such Reserve Settlement Commissioner affecting the lands or the rights in or over the lands, which it is proposed to constitute a forest reserve' 2nd January, 1917 (Munro and Horst, 2012). See figure 3-1 below for the map of Gola forest showing the KHFR.

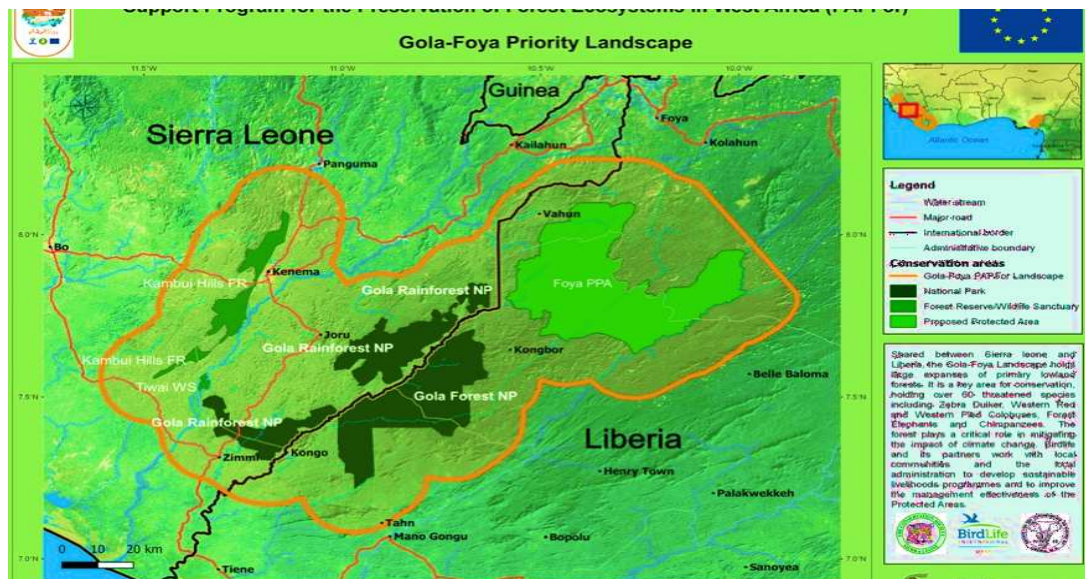


Figure 3-1: Map of Gola forest showing Kambui Hills Forest Reserve.

Source: The Kenema District Forestry Office Archives, 2022.

3.2 Methodological framework for descriptive statistical analysis

The variables to be analysed using the descriptive statistical analysis are mostly unmeasurable and therefore employed the Confirmatory Factor Analysis (CFA). Moreover, illegal foraging and lack of concise data hub, coupled with perceived reluctance of the locals to fully corporate with data gathering ventures, might limit any genuine quantification effort at getting a reliable result. The CFA through Principal Component Analysis (PCA) varimax statistical analysis suits the methodology used. The goal of factor analysis is to explain the covariance relationships among the variables in terms of non-measurable random factors. A wide range of multivariate variables may exist, and to reduce the dimension or multi-variate data set to an orderly structure, factor analysis is appropriate (Obasi et al, 2015). It is basically a technique of describing groups of highly correlated variables by a single underlying construct or factor that is responsible for the observed correlations. Once the groups of correlated variables are identified, they can be easily interpreted and labelled. For the specific purpose of this research, the PCA will be used instead of the Common Component Analysis (CCA) although both techniques often lead to similar results. CFA procedure using the principal component model with varimax rotation will be employed in grouping the socio-economic and environmental effects of harvesting timber into major components. Only variables with factor loading of 0.5 and above, are used. There is a very high reliability of factor analysis models in social science studies and has been widely used by several authors. To employ it for this

research is not a novelty. Chukwuone and Okorji (2008) applied factor analysis to investigate the willingness to pay for systematic management of community forests for conservation of non-timber forest products in Nigeria's rainforest region while Ajake and Enang (2012) applied factor analysis to determine the demographic and socio-economic attributes affecting forest ecosystem exploitation and management in the rural communities of Cross River State, Nigeria. Similar research by Obasi et al (2015), also applied CFA to examine the Environmental and socio-economic effects of timber harvesting in Ebonyi State, Nigeria.

3.3 Design and Pre-testing of questionnaire.

The questionnaire is designed in accordance with guidelines recommended by Babbie and Mouton (2010). It contains both open-ended and close questions, the questions asked were clear and simple and with no double meanings. The questionnaire was developed in consultation with the supervising professor from Czech Republic and a professor from Sierra Leone.

The questionnaire was pre-tested to collect data from a study population. Five people each from two communities (Bandama and Serabu), just outside the study area were asked to complete the questionnaire. The answers were examined to see if respondents understood the instruments and whether respondents were reluctant to answer some questions. Necessary revisions were made and the corrected questionnaire was used for data collection. This is to avoid problems such as lack of clarity in the instruments of the questionnaire, which can threaten the validity and reliability of the results. It is a necessary corrective measure that enables the researcher to reconstruct some of the wordings of the questions, add a couple of other relevant instruments that might have been left out.

3.4 Sampling procedure

The sampling techniques adopted for selecting the respondents is the multi-stage sampling technique. Four communities were purposely selected for a structured questionnaire. The selection process was based on a number of criteria: a) The size of the population, each selected town must have five thousand and above inhabitants; b) The proximity of the towns to the KHFR, each of the town must be 3 kilometres or less close to the forest peak; c) Each selected town must have one or more established timber products enterprises like sawmills or furniture manufacturing establishment d) Each selected town must also possess a local administration or a farmer's

association. The aim here is to have a proper representation of the locals, the administration and timber product businesses, all views must be reflective from the data collected through the questionnaire. Moreover, this selection process will avoid the problem of ghost towns that behaves random sampling technique. It will further prevent the problem of unequal representation as smaller towns without recognized local administrations and established timber products enterprises or farmers association are left out. The relevance of the proximity criteria is to avoid selecting targets that have little or nothing to do with the KHFR, the closer the towns to the forest the more reliant they are on the forest for their daily livelihoods. A total of 301 questionnaires were administered among the four selected communities. The questionnaires were distributed among the established timber harvesting enterprises, farmers' associations, and the residents.

3.5 Identification and definition of variables used in the descriptive statistical analysis

The following variables are used for the PCA: Loss of biodiversity, Climate change/global warming, Destruction of immature trees and non-wood forest products, Migration of wildlife, Natural hazards; flooding, soil erosion, soil encroachment, Disruption of hydrological cycle of water catchment area, Income generation through taxes, Provision of employment opportunities, income and expenditure, Provision of energy for social activities such as cooking, heating, Provision of recreational facilities for the locals, Provision of clean air for the locals, High cost of farm labour, Disputes and crisis over land and compensation and firms CSR, challenges and perceptions. Selected variables described below were highly correlated and are used for the CFA.

3.5.1 High labour costs

Each unit of timber has some number of jobs and income associated with its production. Besides, the more timber harvesters within the study area, the higher the demand for labour which subsequently results to high labour cost. Thus, high labour cost is a key variable used in the descriptive analysis of timber harvesting economic impact assessment within the study area. In a study on employment and wage impacts of timber harvesting and processing in the United States, by Sorenson et al. (2016), emphasis was placed on timber product output and federal employment data, regional estimates of employment per million cubic feet of timber harvested, referred to as employment direct response coefficients (DRCs), and wages per worker in 16 sectors of the US

primary forest products industry were developed. The results confirmed an inconsistent employment and wage brackets among the sectors, the lowest being the harvesting sector, followed by the processing and manufacturing sectors. What this indicates is that, even though harvesting of timber within the study area results to rising labour costs, there are other sectors like the processing and manufacturing sectors within the supply chain that pays better wages.

3.5.2 Loss of income and revenue

This is another key variable used in the descriptive statistical analysis to assess the economic impacts of timber harvesting in the study area. Corruption is so rampant that forestry officials prefer to allow illegal loggers rather than issuing harvesting license. It became so bad that government revenue from the study area plummeted by 60% in 2010 alone (Munro and Horst, 2012). However, the loss of revenue from timber harvesting within the study area is not only limited to illegal loggers. Periodical logging bans and drop in price of wood products are also contributing factors. In a work published by Sonny Mumbunan and Riko Wahyudi (2016), titled Revenue loss from legal timber in Indonesia, the authors opined that the link between forest revenue administration and under performance of revenues from legal timber is a huge problem for governments. This article analyses revenues from the timber royalty and reforestation funds, two important forest non-tax revenues in Indonesia whose tropical forest had been under threat of extinction, particularly from commercial timber logging. These findings thus agree with the result that tax exemptions from royalties or natural rents in the forestry sector leads to government revenue loss, as in Indonesia.

3.5.3 Increase cost of wood and timber products

As a variable in the descriptive statistical analysis of the economic impacts of timber harvesting, it relates to the direct consequence of high labour cost. The higher the labour cost of producing timber, the higher the prices of timber and wood products. In a work by MacDonagh et al. (2019), an examination of logging production, costs, innovation, and capacity was performed. The conclusion to their work is that logging production rates increased relative to the decrease cost of production per ton. Consequently, the result affirmed, this led to a drop in the prices of wood logs and wood products. It even got better with innovative harvesting techniques; the research

concluded. This supports the relative significance of price of wood products in evaluating the impacts of timber harvesting on residents.

3.5.4 Silting of rivers and disruption of Hydrological cycle

The silting of rivers in the forest are the consequence of poor forest management. Loggers and miners are not closely monitored to ensure they comply with waste management principles and therefore resort to dumping all of the wood particles and dirt in the rivers, thereby causing real disruptions in the river flow and creating an artificial dumping site convenient for them. Water quality and yield are often affected by tree harvesting. Moreover, the indiscriminate tree felling in hilly areas reduces the ability of soil to absorb water, thereby inducing high volume of run off. Should forest management be effective, forestry monitoring officers can avert such situations. The effect of destruction of forest cover on catchment water balance and runoff dynamics depends on the climate, topography, soil and the type of forest (Villela et al. 2019). It is obvious that the removal of forest cover during tree harvesting predisposes the environment to destructive actions of wind and rainstorm too. Eroded soil is often deposited in rivers and fishponds thereby causing siltation and contamination of such water body.

3.5.5 Damaging of trees and non-wood products

Felling of trees for firewood has been recognized as one of the factors responsible for environmental degradation in arid and semi-arid parts of Africa. Most times, illegal loggers fell trees with no regards to their maturity, thereby damaging non-wood products in the process. This is particularly common in Sierra Leone where forest protection is ineffective and riddled with corrupt officials. When trees in an area are removed, the rate of evapotranspiration is disrupted. This affects precipitation and cause desertification. The average precipitation is 30% lower in deforested areas of the Kambui Hills Forest Reserve than in forested areas (Mansaray et al, 2015). This agrees with the conclusion that the indiscriminate felling of trees aggravated the effect of wind erosion and deposition of sand dunes in northern part of Nigeria, thereby enhancing desertification (Gwandu et al. 2018). From 2002 to 2022, Sierra Leone lost 35.9 ha of humid primary forest, making up 1.9% of its total tree cover loss in the same period. Total area of humid primary forest in Sierra Leone decreased by 13 % (Munro and Horst, 2012). The impact of deforestation is a chain of serious consequences ranging from soil erosion to siltation of river bodies. Devastating

destruction of farmland and houses have been reported in areas where indiscriminate tree felling had taken place as well (Okon, 2018).

3.5.6 Loss of Biodiversity

The tropical forest provide habitat for two-thirds of all identified terrestrial species (Owuor et al., 2021). These plant and animal species serve as source of food, medicine, and fuel in most rural communities in developing countries. Timber harvesting adversely affects the population and variety of plant and animal species in the forest. The removal of forest cover during logging has in some instances resulted in the scarcity or extinction of many important plant and animal species. Some wild animals have also been observed to migrate from areas where tree cover was removed to undisturbed vegetations. Some plant and animal genetic resources that could be used in producing new pharmaceuticals or traditional medicine are lost because of the destruction of forest cover. Rainforests house 80 % of the world's terrestrial biodiversity, which is the foundation of life on earth (FAO, 2017). The report also states that one-quarter of all modern medicines are derived from rainforest plants. If these tropical forests are destroyed, we not only lose some of our most critical species, but also the potential for discovery of many others. One of the most dangerous and unsettling effects of deforestation is the loss of animal and plant species due to their loss of habitat as 70 % of land animals and plants live in forests. The trees of the rainforest that provide shelter for some species also provide the canopy that regulates the temperature. Deforestation results in a more drastic temperature variation from day to night, much like a desert, which could prove fatal for many inhabitants. Thus, this variable is crucial in evaluating the impacts of wood harvesting in the study area.

3.5.7 Disappearance of vegetation

Destruction of forest cover during tree harvesting results in the loss of vegetation, which the plant cover gives to the soil (Gwandu et al. 2018). Timber harvesting also interrupts the nutrient cycle of the forest, promotes nitrification, and increases nutrient leaching thereby leaving the topsoil impoverished and susceptible to poor growth of plants that provides the necessary vegetation. Primary forest erodes at an average rate of 12 tons per hectare while open vegetation in deforested areas of upland topography erode at the rate of about 84 tons per hectare per year (MacDonagh et al. 2019). Moreover, when the trees are cut the soil is eroded to an extent that there is no loam soil

available that can support the growth and the development of these trees. In other words, Soil erosion in the country has also been noted as one of the threats to food security because the soil nutrients that would have otherwise supported wheat, rice, and maize farming are depleted (Cheng et al. 2019). There have been reports of accelerated erosion that resulted in extensive gullies in various deforested parts of Nigeria due to the disappearance of vegetation (Kajembe et al, 2000). This trend adversely affects the aquatic ecosystem and disrupt biodiversity by killing fish and other organisms. To put in a chronological order, Sierra Leone had 5.62 ha of tree cover, equivalent to 77 % of its land area and 0.14% of the global total in 2000 (Munro and Horst, 2012). The same report noted that, in 2010, the top 2 regions that represent 74 % of all tree cover, the Northern region had the most tree cover at 2.42 ha compared to an average of 1.36 ha in the southern region. In 2020, Sierra Leone had 5.92 ha of land above 10 % tree cover, extending over 81.7 % of its land area (Okon, 2018). This trend of events underlines the significance of this variable in the evaluation of timber harvesting within the study area.

3.5.8 Climate change and global warming

This is one of the key variables used to assess the impact of timber harvesting on the environment using descriptive statistical analysis from primary data. In addition to the loss of habitat, the lack of trees also allows a greater amount of greenhouse gases to be released into the atmosphere. Healthy forests absorb carbon dioxide from the atmosphere, acting as valuable carbon sinks. Deforested areas lose that ability and release more carbon. These processes trigger climate change and global warming. The environmental pollution problems in forest industries include solid waste disposal, air emission of toxic and non-toxic particulates, veneer dryer emission and glue waste disposal. It is worthy to note that between 2001 and 2020, forests in Sierra Leone emitted 41.8MtCO₂e/year and removed -35.8Mtco₂e/year. This represents a net carbon source of 6.00MtCO₂e/year (Kamara, 2020). During this same period, an average of 41.8Mt per year was released into the atmosphere because of tree cover loss in Sierra Leone. In total, 920Mt of co₂e was emitted, and an average of 28.7Mt of co₂e was observed in areas where the dominant drivers of loss resulted in deforestation, according to the same report.

3.6 Data collection and model specification for descriptive analysis

Data for the study is collected from the study area using a set of structured questions grouped in a pre-tested questionnaire. Information gathered included the perceived effects of timber harvesting on man and the environment. The questionnaire had three sections namely: Socio-economic characteristics of key timber harvesting stakeholders; level of environmental effects, economic and social losses incurred because of timber harvesting; identification of the challenges encountered by the formal forest players in harvesting timber and firms CSR. The questionnaire was administered by the researcher with the assistance of the Dean and faculty members of the School of Natural Resources at Njala University, Sierra Leone, and the forestry office in Kenema district. They were quite familiar with the terrain and the people. The exercise lasted for two weeks starting from February to March, 2022.

3.7 Demographic characteristics of respondents

The questionnaire is designed with instruments relating demographic characteristics about the respondents in the study communities. For instance, gender, religion, income, employment sectors, educational levels, marital status, household head gender, expenditure, environmental threats, socio-economic challenges, and few others. These demographic characteristics of respondents are processed using CFA, and expressed in frequencies and percentages in the results, See tables 4-1 and 4-2 for the breakdown of the above characteristics. Further analysis relating to the executive respondents from the timber harvesting firms can be seen in figures 4-1 to 4-9.

3.8 Methodological framework of regression analysis model

The regression analysis model is done using secondary data from UN affiliated databases. It shows that any marginal change in the independent variables reflects a change in the dependent variable. It uses a time series data set (2010 to 2021), from world organizations and government agencies such as United Nations Development Program (UNDP), United Nations Environment Program (UNEP), Food and Agriculture Organization Statistics of the United Nations (FAOSTAT); International Monetary Fund (IMF); International Trade Centre (ITC); World Bank World Development Indicators; World Trade Organization (WTO); Organization for Economic Cooperation and Development (OECD); and Statistical Office of the European Union (EUROSTAT) and Sierra Leone Government National statistics office. Further data from books,

journals, statistical bulletins, publications from Web of Science (Thomson Reuters), Scopus (Elsevier), Consortium of International Agricultural Research Centre Library, internet search engines, and other sources were used. The identified variables specifically showed wood harvesting impacts on the environment, the economy, and the social fabrics of Sierra Leone. Unlike the variables used in the descriptive statistical analysis, the variables used here are measurable and includes Primary literacy Level, Carbon emissions, Inflation, Unemployment, Gross Domestic Product, Exports, and Natural Rent. The acquired data used in the models was ran using statistical software, EViews 10.

3.9 Identification and definition of variables in regression analysis

3.9.1 Natural Resource Rent.

For natural forest resources like wood products, natural rents are the percentages paid to the custodians of the forest as a rent for a specific land space where timber will be harvested. However, in real terms, it is supposed to be calculated as the difference between the price of a commodity and the average cost of producing it. A research work by Mansaray et al (2015) agrees that economists often make a sharp distinction between profits and rents. Profits are said to be Returns on Capital risk. For instance, if the standard cost of extracting a barrel of oil is 7 US\$ and the market price for a barrel of oil is 120 US\$, then the 7 US\$ covers the cost of the labour, capital, and the cost of the risk on returns to enterprise. Therefore, the difference between 120 US\$ of the barrel of oil and the 7 US\$ is 113 US\$ and that is the rent (Obasi et al, 2015). Ethically, the right thing to do is to ensure that the rent on natural resources goes back to the citizens. For most companies, rents are indistinguishable from profits, they are all swept into the concept of profits. There is a need therefore to separate tax regime for resource extraction from manufacturing, for example, and in this process, three basic instruments are advised to be employed: minority equity stake in the company, taxation of profits and taxation on royalties. This is why African countries, including Sierra Leone tend to derive less for natural rents than the real amounts to be paid by MNCs involved in harvesting wood products. Lack of research by the Sierra Leone Government to know the value of the resources, coupled with corruption are major factors for such inadequacies.

Accounting for the contribution of natural resources to economic output is important in building an analytical framework for sustainable development. It is important to note that when countries use such rents to support current consumption rather than to invest in new capital to replace what

is being used up, they are, in effect, borrowing against their future (Mansaray et al, 2015). Thus, natural rent is a key variable used in the regression model to evaluate the impact of timber harvesting in Sierra Leone. Time series data from 2010 to 2021 is used here to evaluate the influence of natural rents on the socio-economic and environmental impacts of timber harvesting in the study area.

3.9.2 Exports

Wood exports from Sierra Leone is another relevant variable used in the regression analysis to evaluate the relationship between exports and the impact on the economy. In 2019, Sierra Leone exported 9,770,683 tons of wood to China, 6,080,402 tons to Vietnam and 2,892,332 tons to India. This accounted for US\$20.3 million in trade balance for only three destinations (UNDP, 2022). It is not a novelty to use exports as a variable to determine the impact of timber harvesting on the environment. A study by Waqas et al. (2022) sought to investigate how factors such as forest wood product exports, energy use, urbanization, and gross domestic product (GDP) per capita affect environmental performance. This study analysed annual data for 31 countries from 2001 to 2020 using various econometric methodologies including the generalized method of movement (GMM) approach and the Hausman test for fixed and random effects. The results show that forest wood product exports, urbanization, energy consumption, and GDP significantly contribute to increasing CO₂ emissions and deteriorating environmental performance. Time series data from 2010 to 2021 is used here to evaluate timber exports from the study area.

3.9.3 Gross Domestic Product.

In a scientific research paper by Kamara and Su (2018), a computer analytics system is used to get a clear picture of the relationship between the economic growth (GDP) and the primary forest cover in three different countries in South America. The study established a strong relationship between GDP and the change in the primary forest cover of the three countries. Thus, the use of GDP as variable for forest resource impact assessment is well supported with scientific literature. It is a key economic indicator that measures the monetary value of goods and services provided within that area. It is calculated based on private consumption + gross private investment + government investment + government spending + (exports – imports). In Sierra Leone, forestry provides more than 8 % of the GDP. Timber export from the study area significantly contributes to the national GDP. Time series data from 2010 to 2021 is used here to evaluate the contribution of timber value to the national GDP.

3.9.4 Unemployment.

From research conducted by Perunova and Zimmermannova. (2022), titled ‘Analysis of forestry employment within the bio-economy labour market in the Czech Republic’. Unemployment consequences relating to health and lifestyle were used as variables. A questionnaire for forest workers within the age brackets of 20 to 49 years was administered to determine the employment status. The paper concluded that unemployment among construction workers is to some extent dependent on lifestyle, health, and job satisfaction in addition to age, marital status, and unemployment history. For forest workers, unemployment is less determined by individual factors but rather by prevailing socio-economic circumstances such as cultural lifestyle and other forest management principles effective implementations. This conclusion was earlier supported by Munro and Horst (2012), in a study titled ‘predictors and consequences of unemployment in construction and forest work.

Unemployment is the share of the labour force that is without work but available for and seeking employment. Sierra Leone unemployment rate for the age bracket of 15 to 24 years in 2021 was 5.33%, a 0.14% increase from 2020. For the same age bracket, the unemployment rate for 2020 was 5.20%, a 0.55% increase from 2019. The unemployment rate for the age bracket of 25 to 40 years in 2021 was 10.77 %, a 0.33% increase from 2020. The same age bracket accounted for 10.44% in 2020, a 0.82 % increase from 2019. This is a serious problem in the country, thus a key variable considered for the regression analysis. The overall national unemployment rate is alarmingly above 50 % from 2017 to 2021 (IBRD, 2021). Time series data from 2010 to 2021 was used to evaluate unemployment in the study area.

3.9.5 Inflation.

The wood logging sector is vital to the wood supply chain supporting both the sustainable management of forest resources and the prosperity of the forestry industry. The increase in the cost for labour and associated costs such as fuel, leads to high cost of wood products as well. This is investigated and analysed by Gutierrez-Castillo et al. (2022) in research titled ‘The impact of rapid fuel cost inflation on Arkansas’ logging sector’ It delved into the impact of rapid fuel cost inflation from February to May of 2022 on the logging sector based on a survey sent to 430 logging and hauling firms, with a direct comparison between the January and May operating costs and production quantities. Results showed that logging and hauling firms paid about 50 % more per litre of diesel fuel in May than in January. The overall timber production declined in May concerning January, wherein 50% of the firms significantly lowered their production by 16.1%.

The compounding consequence is a lower wood product supply, ultimately leading to higher price inflation. This conclusion aligns with the work done by Cronin and Pandya (2009), who opined that changes in logging costs and wood products are factual reasons for inflation of wood product prices.

In Sierra Leone, the inflation rate for consumer prices moved over the past 42 years between -3.3 % and 68.7%. From 2022 to the present, the inflation rate has moved from 27.2 % to 56% as calculated by National Statistics Sierra Leone. From 2022 to 2023, the average inflation rate went from 31.1 % per year to 56.81 percent in May of 2023. Time series data from 2010 to 2021 is used here to evaluate the impact of inflation on timber products in the study area.

3.9.6 Carbon emissions.

Carbon dioxide emissions or CO₂ emissions stemming from the burning of fossil fuels and the manufacture of cement. This includes carbon dioxide produced during consumption of solid, liquid, and gas fuels. The largest contributor to carbon emissions are fossil fuels, coal, oil and gas, and they represent nearly 90 % of all carbon emissions. It can be described as a colourless, odourless, and non-poisonous gas formed by combustion of carbon and in the respiration of living organisms. The rampant cutting of trees for timber and energy use in rural areas contribute greatly to carbon emissions. Carbon dioxide released into the atmosphere through human activity such as the burning of fossil fuels like coal or gas has been on the increase in Sierra Leone. Thus, this work evaluates time series data from 2010 to 2021. To understand the problem in Sierra Leone, it is important to note that carbon emission rose from 6.51 MtCO_{2e} in 1990 to 9.31 MtCO_{2e} in 2020 (Wei et al., 2023).

3.9.7 Primary enrolment and literacy Level.

Literacy rate is the percentage of people ages 15 and above who can both read and write with an understanding of a short simple statement about their everyday life. Sierra Leone's literacy rate for 2021 was 47.70 %, a 4.49% increase from 2018. Between 2004 and 2018, Sierra Leone's literacy rate moved from 34.83 % to 43.21 %, a 9.17% increase in fourteen years period (Kamara, 2020). The rapid literacy rate increase from 2018 to now can be credited to the ruling government in power. Their flagship program is human capital development, with free quality education for all, is a welcome and positive policy going forward. Time series data from 2010 to 2021 is used here to evaluate primary enrolment in the study area.

3.10 Data collection and model specification in Regression analysis

Given the theme of the research, it is important to estimate the socio-economic impact of timber harvesting in Sierra-Leone by using a multiple regression method. This is done by formulating three hypotheses, each representing the social, economic, and environmental factors. Therefore, carbon emission, GDP and primary enrolment are captured as dependent variables for the three models representing environmental, economic, and social impacts respectively, whereas exports, natural rents, unemployment, labour, inflation, and percentage of GDP are used as explanatory variables in the following models.

Model 1:

$$ENV_t = \beta_{11}x_{1t} + \beta_{12}x_{2t} + \beta_{13}x_{3t} + \varepsilon_{1t} \dots\dots\dots(1)$$

Where ENV represents Environmental factor proxied by carbon emissions as the dependent variable. Carbon emission is measured in kiloton (kt). The independent variables include X_1 =natural rent, X_2 =timber exports and X_3 =gross domestic product (GDP). Where the β 's are the coefficients that explain the impact of the independent variables. $\beta_{11}X_{1t}$ is the natural resource rent, measured in US dollars (US\$). It is the amount of money paid for a given space of forest land after cost and profits are deduced. $\beta_{12}X_{2t}$ is the value of timber exports, measured in US dollars (US\$). it is the amount of money derived from a total export of timber products from the study area. $\beta_{13}X_{3t}$ represents GDP, measured in percentages. It is the percentage contribution to the national GDP from timber products from the study area. ε_{1t} represents error terms.

Model 2:

$$ECO_t = \beta_{21}x_{1t} + \beta_{22}x_{2t} + \beta_{23}x_{3t} + \beta_{24}x_{4t} + \varepsilon_{2t} \dots\dots\dots(2)$$

Where ECO is the Economic factor proxied by GDP as the dependent variable. GDP is measured in percentages. It is the percentage contribution to the national GDP from timber products from the study area. The independent variables include X_1 natural rent, X_2 exports of timber, X_3 labour productivity and X_4 is the rate of inflation. Where the β 's are the coefficients that explain the impact of the independent variables. $B_{21}X_{1t}$ represents natural resource rent, measured in US dollars (US\$). It is the amount of money paid for a given space of forest land, after cost and profits are deduced.

$B_{22}X_{2t}$ is the value of timber exports, measured in US dollars (US\$). It is the amount of money derived from the total export of timber products from the study area. $B_{23}X_{3t}$ stands for labour productivity, measured in US\$. It is the wages received from labour inputs in the production of timber products from the study area. $\beta_{24}X_{4t}$ stands for inflation, measured in percentages. It is the percentage increase in prices of timber products in the study area. ε_{1t} represents error terms.

Model 3:

$$SOC_t = \beta_{31}x_{1t} + \beta_{32}x_{2t} + \beta_{33}x_{3t} + \beta_{34}x_{4t} + \varepsilon_{3t} \dots\dots\dots(3)$$

Where SOC is the social factor proxied by primary school enrolment as the dependent variable. This is measured by numerical numbers of primary school enrolment per year.

The independent variables include X_1 =natural rent, X_2 =timber exports, X_3 =labor productivity and X_4 is the level of unemployment. Where the β 's are the coefficients that explain the impact of the independent variables. $B_{31}X_{1t}$ represents natural resource rent, measured in US dollars (US\$). It is the amount of money paid for a given space of forest land, after cost and profits are deduced. $B_{32}X_{2t}$ is the value of timber exports, measured in US dollars (US\$). It is the amount of money derived from the total export of timber products from the study area. $B_{23}X_{3t}$ stands for labour productivity, measured in US\$. It is the wages received from labor inputs in the production of timber products from the study area. $B_{34}X_{4t}$ is unemployment, measured in percentages. It is the percentage increase of unemployment resulting from timber harvesting in the study area. ε_{1t} represents error terms.

Assumptions. The following assumptions are observed for the multiple regression models:

- ❖ Carbon emission represents the environmental impact as the dependent variable in the model.
- ❖ GDP represents the economic impact as the dependent variable in the model. It also represents explanatory variable in the other two models.
- ❖ Primary enrolment represents the social impacts as the dependent variable in the model.
- ❖ The time series data used is from 2010 to 2021 and are limited to the age brackets between 15 and 24 years for all variables in the regression models.
- ❖ The model did not violate heteroskedasticity test as the variables are constant.

4 RESULTS AND DISCUSSIONS

4.1 Descriptive statistical analysis

The first half of this chapter presents and discusses the results for the descriptive analysis of the socio-economic and environmental impacts of timber harvesting within the study area. Firstly, It presents results of the demographic characteristics of respondents in the study area, expressed in percentages and frequencies. The demographics includes age, household size with gender identities, literacy, employment status, income, expenditure, and educational expenditure. Furthermore, respondents from formal forestry industry players such as the harvesting firms, farmers associations, NGOs and community based organizations were requested to answer a section of the questionnaire. The results highlighted the outcome of timber harvesting firms' initiatives towards corporate social responsibilities, the process of forming their CSR goals, consultations on their objectives, time span of their operations and how to sustain the implementations of these objecties agreed with the residents within the study area. The results in this section are derived from the primary data collected from a total of 301 questionnaires administered within the four communities in the study area, including 40 representatives from the formal forestry players as identified above. See table 4-1 below where the demographic characteristics of respondents in frequencies and percentages are presented.

TABLE 4-1 DEMOGRAPHIC CHARACTERISTICS OF THE RESPONDENTS IN FREQUENCIES AND %

Demographic characteristics	Frequency	Percentage
Sex		
Female	147	48.84
Male	154	51.16
Total	301	100
Level of education		
None	1	0.33
Basic	240	80.1
Secondary/High School	52	17.33
Tertiary	7	2.33
Total	300	100

Marital status		
Single	25	8.31
Married	261	86.71
Widowed	15	4.98
Total	301	100
Religion		
Christianity	23	7.64
Islam	250	83.06
Traditionalist	28	9.30
Total	301	100
Occupation		
Agriculture	9	2.99
Forestry	236	78.41
Petty trading	37	12.29
Craftmanship	13	4.32
Salaried worker	2	0.66
Others	4	1.33
Total	301	100
Gender head		
Female	160	53.16
Male	141	46.84
Total	301	100
Occupation of household head		
Agriculture	3	1
Forestry	237	78.74
Petty trading	29	9.63
Craftmanship	12	3.99
Salaried workers	9	2.99
Others	11	3.65
Total	301	100

Source: Field survey, 2022

The sociodemographic characteristics of the respondents are expressed in frequencies and percentages in the table above. The survey shows that the number of male respondents was just three percent dominant over the number of female respondents, 154 male respondents constituting 51.16% of the respondents and females constitutes minority of the respondents (147) constituting

48.84% of the respondents. The result further revealed that majority of the respondents have elementary education, and this represents 80.1% of the total respondents. Only 0.33% of the respondents have no formal education and these were in the minority among the respondents. The findings also revealed that most of the respondents were married, representing 86.71%, followed by those who were single (8.31%). This is an indication that majority of the heads of households were married. It is also observed that, most of the respondents (83.06%) were Muslims while the religious minority of the respondents (7.64%) were Christians. However, respondents who were Traditionalists also make up of 9.3% of the respondents. It is also evident from the table above that most of the respondents are engaged in forestry work (237 respondents), constituting 78.74% of the respondents (farmers, loggers and so on) and those who are salaried workers constitutes minority of the respondents constituting 0.66% of the respondents. See table 4-2 below for the household characteristics of the respondents.

TABLE 4-2 HOUSEHOLD CHARACTERISTICS OF THE RESPONDENTS

Variable	Mean	Standard deviation	Minimum	Maximum
Age	42.1	12.7	1	89
Household size	7.2	3.6	1	26
Males in Household	3.6	2.3	0	19
Females in Household	3.8	2.0	0	16
Age below 15 years	3.1	0.2	0	12
Below 15 years males	1.6	1.5	0	8
Below 15 years females	1.7	1.3	0	5
Age 15-65 years.	3.9	2.5	0	16
15-65 males	2.0	1.7	0	10
15-65 females	1.9	1.5	0	10
Above Age 65 years	0.4	1.0	0	9
Above 65 years males	0.1	0.7	0	9
Above 65 years females	0.4	1.2	0	11
Total of Literates.	2.7	2.3	0	12
Male Literates.	1.6	1.7	0	11
Female Literates.	1.5	1.5	0	16
Total of employed	2.5	0.2	0	15
Total unemployed	3.9	2.6	0	16
Monthly expenditure	23.3	0.5	1	12

Monthly income	51.6	0.3	1	15
Education Expenses	19.7	2.0	0	8

Source: Field survey, 2022.

The results in table 4-2 above show that the average age of the formal forestry worker is 42.1. The standard deviation (SD) is 12.7. Out of 7.21 mean score for household size, there is 3.6 standard deviation. The difference in the mean between the females and males implies that females dominate males in a household. The working age bracket provides an interesting result as the average mean of household members between the age of 15 and 65 years is 3.9 with an SD of ± 2.51 . The age brackets of 1 to 15 and 65 years form the dependent group within the households. The average mean of people who are literates is 2.7 with an SD of 2.3. Out of the mean average of 2.7 literates, 1.6 were males and 1.5 were females. Further, the average number of people employed has an average mean of 2.5 with an SD of 0.2 while average mean of people unemployed is 3.9 with an SD of 2.6. It also shows demographics such as age, household size, gender characteristics, literacy, employment status, income, expenditure, and educational expenses, expressed in frequencies and percentages.

Principal Component Analysis (PCA)

Here, the socio-economic and environmental impacts are expressed using factor analysis, subject to the given condition, that not all variables selected have the factor loadings of 0.5 and above. See table 4-3 below, illustrating CFA of socio-economic and environmental impacts.

TABLE 4-3 CONFIRMATORY FACTORS ANALYSIS OF SOCIO-ECONOMIC AND ENVIRONMENTAL IMPACT.

Factors and variables	Factor loadings	Mean score	Eigen values	% of variance
Environmental Impacts			1.08	0.12
Silting of Rivers and lakes	0.7	4.3		
Damaging of Trees and NWPs	0.6	3.8		
Loss of biodiversity	0.9	3.9		
Loss forest cover	0.6	4.1		
Disappearance of vegetation	0.5	4.1		
Climate change and global warming	0.6	4.1		
Economic Impact			1.80	0.20

High labor cost	0.5	4.4		
Government revenue loss	0.9	3.8		
Increase cost of Wood Products	0.7	3.9		
Social impact			1.63	0.19
Land compensation and disputes.	0.8	3.8		
Rural-Urban migration.	0.9	4.0		

Source: Field survey, 2022.

4.1.1 The Environmental impact

It has Eigen value of 1.08 and explains about 12 percent of the total variance. The items within this factor all have loading exceeding 0.5 which indicate that they are determining factors for environmental impact of timber harvesting. Among the environmental factors, silting of rivers and lakes had second highest mean score which means that silting of rivers and lakes is a key factor that contribute to environmental impact of timber harvesting. This explains the dominating activities of farming and timber logging without any available provision for residuals and wastes, thus making the rivers and lakes a dumping ground. Loss of biodiversity was the most important factor that contributes to impact of timber harvesting with factor loading of 0.9. This result agrees with the findings of Sampson et al. (2019) who found that Silting of rivers and lake, damaging of immature trees and non-forest wood products, reduction in soil fertility and crop output, loss of biodiversity, disappearance of vegetation, climate change and global warming, and soil erosion affects the environment negatively. The result further shows over-dependence on timber resources and other non-wood forest resources by the rural dwellers, resulting in scanty forest vegetation and poor forest sustainability. This result also concurs with Adam et al. (2022), who opined that excess reliance of the locals on timber resources results in a heavy decline of forests ecosystem.

The variables that loaded high under factor analysis for the environmental effects were Silting of lakes (0.7); Damaging of trees and Non-wood products (0.6); Loss of biodiversity (0.9); Loss of Forest cover (0.6); Disappearance of vegetation (0.5) and climate change (0.6). They indicated areas that have significantly affected the study area in terms of their environmental well-being. A lot of these effects seems obvious upon visitation to the study area. There is an increased draught, water stress, poor yield of crops arising from flooding, crisis, whereas others like global warming and species extinction are likely to manifest in the future. It is quite evident from the result obtained that utilizing forest timber products unsustainably are self-defeating. It ends up impoverishing the

people, and in most cases, violates treaties that the country is signatory to. This position agrees with Uzu et al. (2022), Kamara and SU, 2018) who opined that if timber exploitation is left uncontrolled, it will eventually lead to factors that will increase GHG emission exponentially.

4.1.2 The economic impact

The economic impact of timber harvesting accounts for 20 percent of the total variance. In this area, high cost of labour is the most important factor that contributes to economic impact of timber harvesting. In the same vein, taking the economic effects into consideration, high cost of farm labour, and loss of income and revenue by government all have economic effects in the study area. The rampant timber harvesting activities in the area has led to a diversion of the labour force from farming activities. Thus, it becomes relatively expensive to hire labour from the timber production sector because it ensures higher income comparatively to farming activities which paid less. The income loss of government revenue is due to the poor management of the forest itself, illegal logging activities and endemic corruption amongst forestry guards, forestry administrators, the police, city and district councils, the residents themselves and the chiefs. This result also conforms to the findings of Sampson et al. (2019).

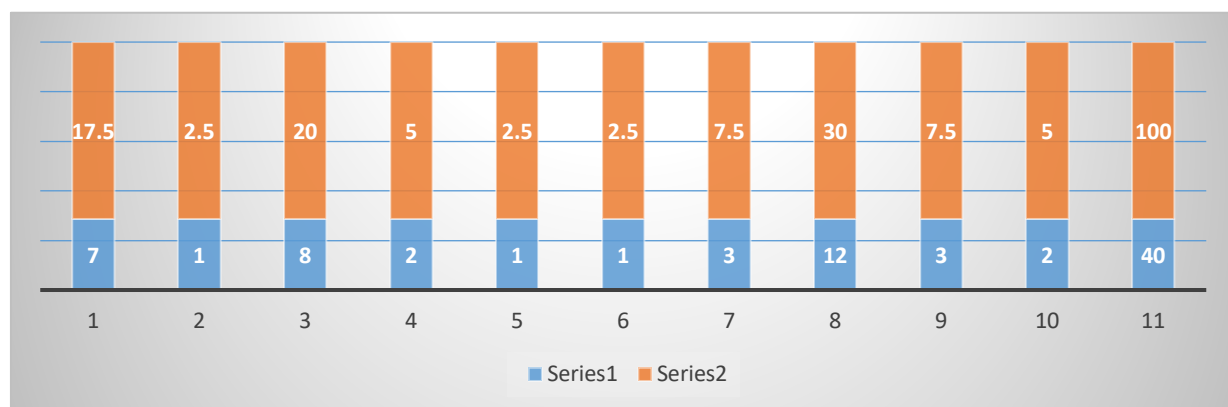
4.1.3 The social impact.

This contains items like occurrences of disputes and crises over land and compensation, and rural-urban migration. This factor explains 19 percent of the variance. The most important item under this construct is rural-urban migration. This is evident in the study area as most young active residents are moving from their communities to the cities where there are more job opportunities. Should the timber harvesting firms fulfil their CSR, in terms of jobs, infrastructural development, and other facilities as promised in their commitment manifestoes, apparently, the narrative of rural-urban migration might be different, and it will also affect the high cost of farm labour because there will be abundant labour force. In the same vein, taking the social effects into consideration, occurrences of dispute and crises over land and compensation as occurrences between two communities of Bambawo and Basara in the study area, and rural-urban migration. This dispute was over a small piece of land of 1km in size. The dispute carried huge interest because the most sought-after trees that is in demand by forestry firms are found on this particular strip of land. It was only settled with the intervention of government and a buffer zone was created wherein the disputed land space remained the property of the government and neither party have access to it.

These were also factors established by Sampson et al. (2019) that contribute to social impact in the study area.

4.1.4 Firms' compliance to SFM Principles

The Data used here is collected from 40 respondents (10 from each of the 4 surveyed communities) and were represented by top executives from the timber harvesting firms, forestry division, farmers associations and NGOs. This section discusses timber harvesting with respect to firms compliance to SFM principles through their corporate social responsibilities (CSR). The discussions include laid-down policies of CSR towards forestry, visions and objectives of CSR towards forestry, how CSR policies were initiated, duration of implementation, estimated life span of the CSR, and measures towards its sustainability. The aim of this analysis is to assess the level of compliance by timber harvesting firms to the core elements of sustainable forest management principles as enshrined in the documents of the Earth Summit held in Brazil, 1992. See figure 4-1 below that shows firms' CSR policy formulation processes.



Key:

1 = Firms' specific CSR initiatives. 2 = management policies. 3 = Community sensitization. 4 = workers consultations. 5 = Firms and government consultations corporations. 6 = Community consultations. 7 = Community sensitizations. 8 = Policy consultation. 9 = stakeholders' consultations. 10 = Joint consultations. 11 = Total number of respondents

Figure 4-1: Firms CSR Policy formulation processes.

Source: Field survey, 2022.

Results from figure 4-1 above reveals consultations between the harvesting firms, the government of Sierra Leone and the residents within the study area. 12 respondents (30%) indicates that the laid down policies were based on policy consultations with other stake holders while 8 respondents

(20%) suggests that their CSR laid down policies were based on community sensitizations. 7 respondents (17.5%) confirms that their policies were based on their own CSR specific initiatives, as enshrined in their operational guidelines. 2, 3 and another 2 respondents (5%, 7.5% and 5%) confirm that their policies were based on workers consultations, stakeholders' consultations, and joint consultations, respectively. Only 1 respondent each (2.5%), indicates that their CSRs were formulated based on community consultations, corporations among firms and government policies. See Figure 4-2 below for the policies of firms' CSR.

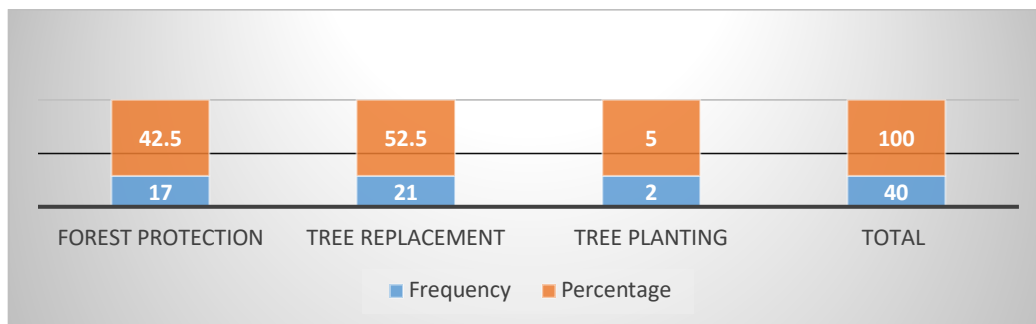
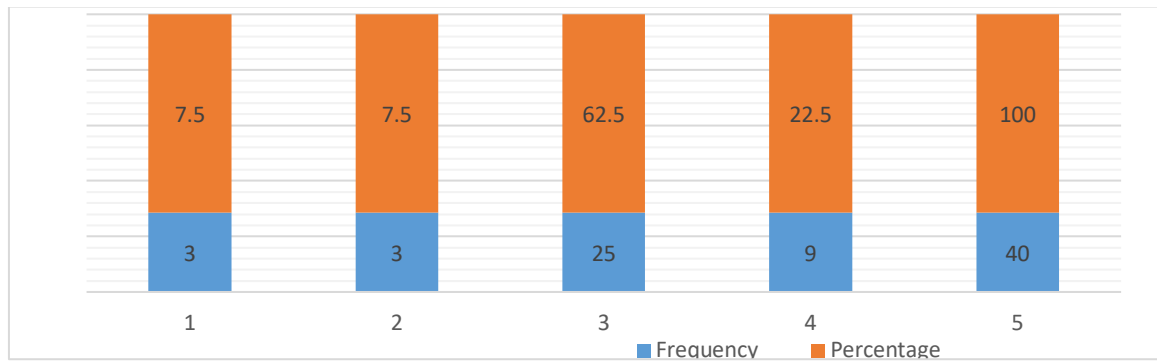


Figure 4-2: Laid-down policies of firms CSR.

Source: Field survey, 2022.

Figure 4-2 shows laid down policies of firms and other stakeholders are decisions agreed upon after initial consultations. Out of the 40 respondents representing MNCs and other stakeholders, 17 (43%) suggest that protecting the forest from illegal loggers was a main laid down policy decision agreed upon in consultations. Another policy agreed upon is tree replacement, which means there should be a tree planted for each tree cut down. 21 of the respondents, which is 52.5% confirmed that their respective firms agreed with the residents to maintain and sustain forest cover. As few as 2 respondents (5%) confirm tree planting projects as a policy to sustain the forest. However, it is not clear whether the respondents did not confuse tree replacement and tree planting. See figure 4-3 below for firms' CSR objectives.



Key: 1= Reforestation; 2 = Mitigate illegal logging 3 = Residual recycling 4 = Policy sustainability; 5 = Total

Figure 4-3: Firms CSR Objectives

Source: Field survey, 2022.

Figure 4-3 provides the vision and objectives of MNCs and other stakeholders. 25 respondents (62.5%) agreed that they had the vision and objective to introduce timber waste management system by recycling the residues. This is to avoid dumping these wastes in the lakes. This is a factor that contributes to climate change because the lakes will eventually dry up permanently. 3, 9, and another 3 respondents representing 7.5%, 22.5%, and 7.5% confirms reforestation, sustainability projects and mitigation of illegal logging, respectively., as the visions and objectives of their firms' CSR approach. Sampson et al. (2019), Munro and Horst (2012) also agreed with the results above and confirms that reforestation and mitigating illegal logging are essential objectives for MNCs involved in timber harvesting. See figure 4-4 below for the duration of firms' operations.

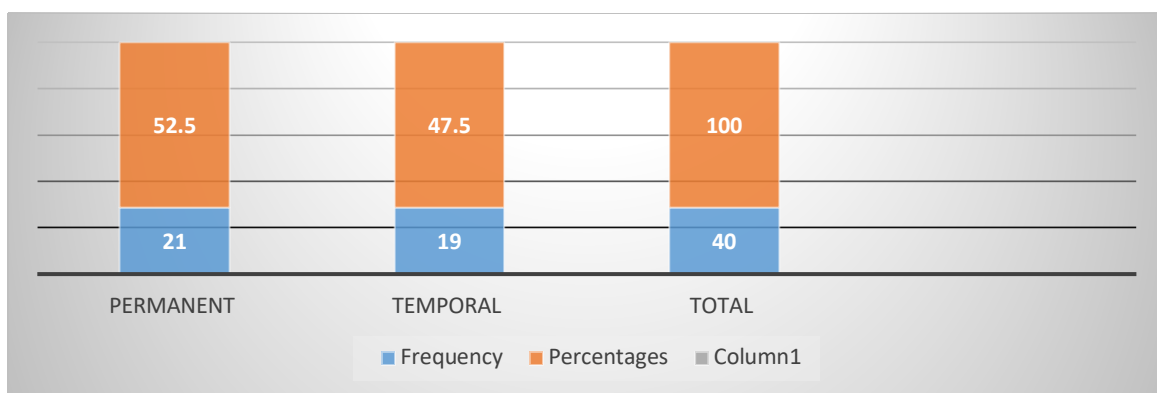


Figure 4-4: Duration of Firms Operations.

Source: Field survey, 2022.

Figure 4-4 above shows results for the duration of firms operations in the study area. 21 respondents, representing 52.5% indicated that their firms' CSRs operations were permanent whereas 19 respondents (47.5%) confirms temporal duration for their firms operations in the study

area. This result is in line with the findings of Kamara and Su (2018), Uzu et al. (2022) who opined that MNCs timeframe for their operations are permanent in most cases but also have those who prefer to operate on temporal basis. See figure 4-5 below that shows the life span of firms' CSR.

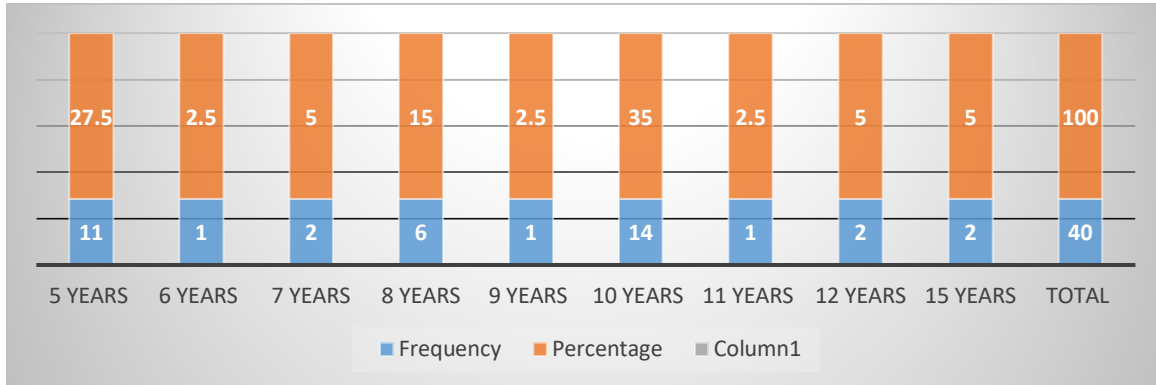
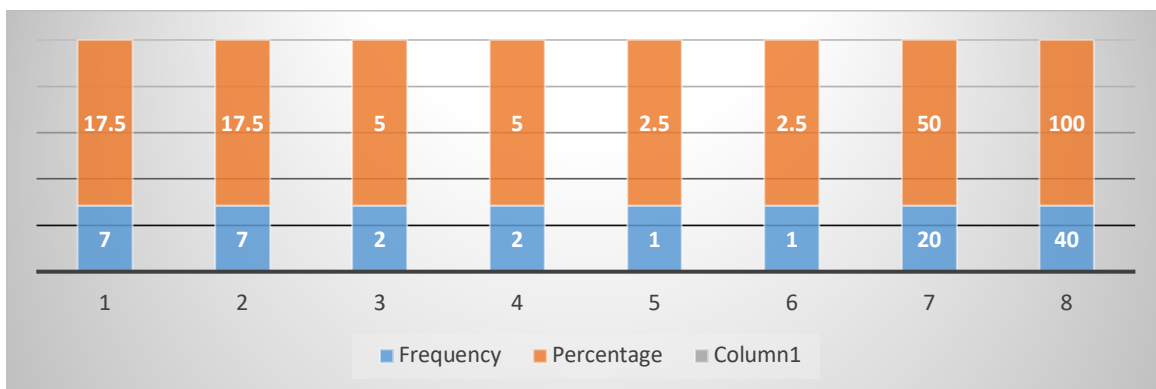


Figure 4-5: Life span of Firms CSR.

Source: Field survey data, 2022.

Figure 4-5 shows interesting results. It indicates the period for each CSR projective initiative from 15 years to 5 years. More firms (14 respondents representing 35%) confirmed that their CSR projects will last for 10 years and only 2 respondents each (5%) indicated that their projects will last for 15, 12, and 7 years. 2 respondents each (5%). 1 respondent each (2.5%) indicates the duration of their projects for 11, 10 and 6 years. 11 respondents (27.5%) and 6 respondents (15%) suggested 5 and 8 years respectively. See figure 4-6 for the measures taken to ensure CSR sustainability.



Key: 1 = Educational level; 2 = Forestation; 3 = Joint Sensitization; 4 = Surveillance; 5 = Workers Consultations; 6 = Workers Sensitization; 7 = Workers Education; 8 = Total

Figure 4-6: Measures to ensure CSR Sustainability.

Source: Field survey, 2022.

Majority of the respondents representing 50% indicated that workers education is the measure taken to ensure sustainability of CSR projects. Again, 7 respondents representing 17.5% cited education and forestation as measures to ensure sustainability of the CSR projects. Only 1 respondent each, representing 2.5% stated workers consultation and workers' sensitization as measures taken to ensure sustainability of the CSR project. This result conforms with the findings of Munro and Horst (2012) who identified consultations with stakeholders and regular sensitization as effective strategies to ensure sustainable forestry management.

The results from the frequency and percentage analysis suggests that majority of the respondents representing 33.88% indicated the rise in farming cost as one of the impacts they encountered because of timber harvesting in their communities. Also, 90 respondents representing 29.90% indicated scarcity of available labour supply as a huge social impact due to timber harvesting, 58 respondents representing 19.26% indicated a decrease in their incomes as a resultant impact of timber harvesting while 51 respondents representing 16.94% suggested high expenses. See figure 4-7 for the social challenges faced by KHFR residents.

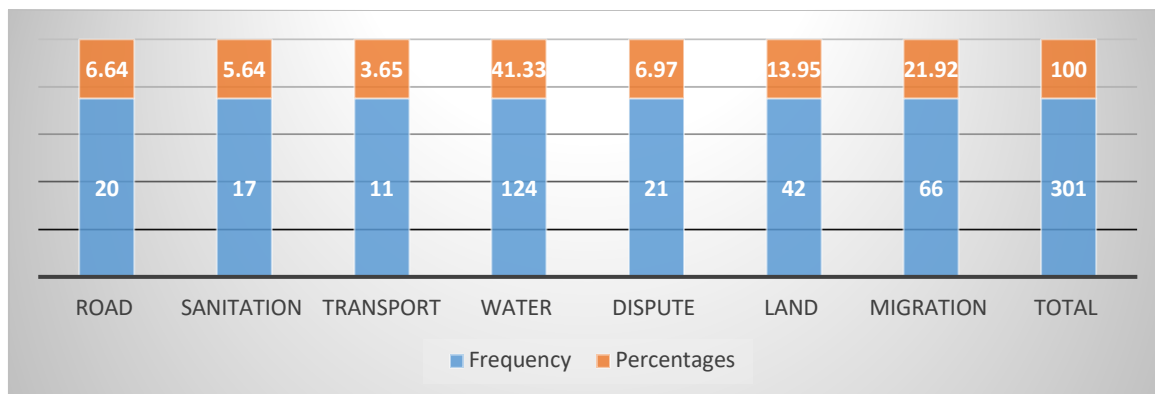


Figure 4-7: Social challenges

Source: Field survey, 2022.

The chart above indicates that most of the respondents representing 41.33% indicated that unavailability or shortage of water is an impact encountered through hydrological disruptions encountered from harvesting timber. Also, 21 respondents representing 6.97% indicated dispute while 66 respondents representing 21.92% indicated migration. Another 20 respondents, representing 6.64% suggested poor road network while 17 respondents representing 5.64% indicated sanitation. Further 11 respondents representing 3.65% viewed transportation as the social challenges associated with timber harvesting in their communities. Thus, the most pressing social

impact is unavailability or shortage of water followed by the huge rural-urban migration. The result is in line with the findings of Sampson et al. (2019) who also identified challenges facing timber harvesting were poor road network, high tariff, transportation of products and capital-intensive nature of the harvesting process. See figure 4-8 below for the economic challenges faced by residents of the KHFR.

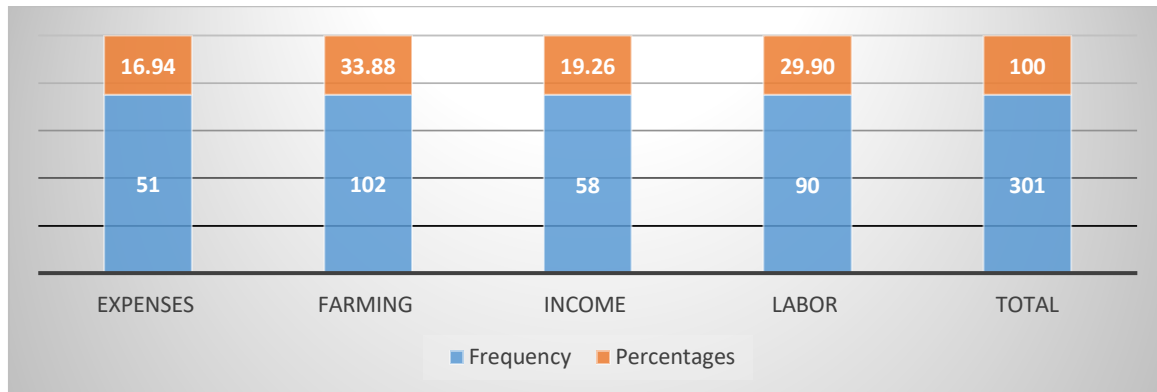
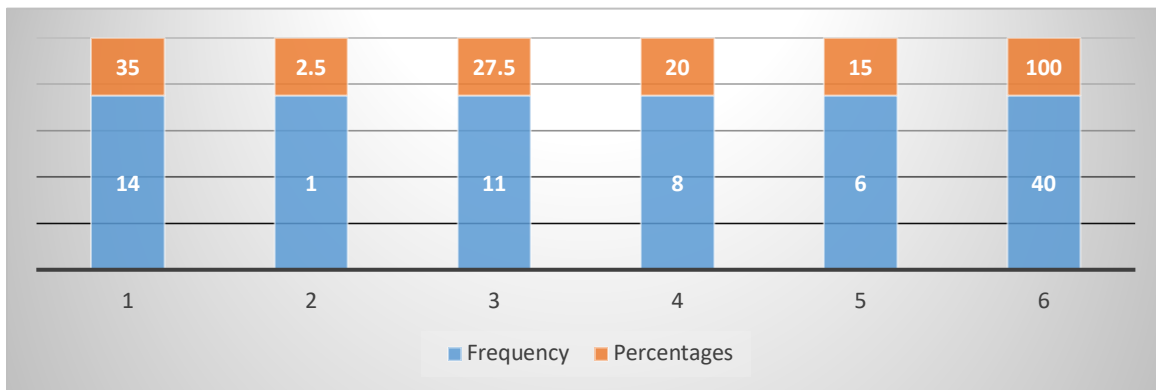


Figure 4-8: Economic challenges

Source: Field survey, 2022.

It is relevant to discuss the environmental challenges encountered by the timber harvesting firms and other major players. It is a highlight of the challenges faced and expects to mitigate them. The results are presented in Figure 4-9 below.



Key: 1 = Climate; 2 = Deforestation; 3 = Pollution; 4 = Weather; 5= Windy; 6 = Total

Figure 4-9: Environmental Challenges

Source: Field survey, 2022.

The results from figure 4-9 above indicates that 14 of the respondents representing 35% indicated that climate change is one of the environmental effects felt in harvesting timber. Also, 11 and 6

respondents representing 27.5% and 15% cited pollution and stormy wind respectively, as challenges associated with harvesting of timber. However, only 1 respondent each representing 2.5% who are the minority indicated deforestation as a challenge associated with timber harvesting. Another common challenge associated with timber harvesting is weather as 8 respondents representing 20% cited it as a threat to their farming activities and livelihood. The result is in line with the findings of Sampson et al. (2019) who also identified that challenges facing timber harvesting were climate change which results to unfavorable or unexpected changes in the weather, pollution, and capital-intensive nature of the harvesting process. Therefore, the reoccurrence of these factors remains some of the serious impacts encountered in timber harvesting in the study area.

4.1.5 Discussions from the descriptive statistical analyses

The result from the descriptive statistical analysis agrees with similar conclusions of other relevant literature on the subject matter. A total number 40 top executives representing farmers' associations, timber harvesting firms, and NGOs serve as respondents during the primary survey. They responded to questions about harvesting firms' CSR and the processes leading to the completion of the specific objectives. These opinions formed part of the 301 respondents (10 executives from each of the four surveyed communities). The findings not only revealed violations of sustainable forest management principles but also identified the failure of timber harvesting firms to respect compliance with their core corporate social responsibilities (CSR) and timeframe.

The result suggest that more male children attend schools than their female counterparts, which supports the pattern in most African countries where male children get the advantage of attending schools than their female counterparts (Jackson et al. 2015). Furthermore, there is higher unemployment in the study area which supports the unemployment status of the country at national level (Sierra Leone National Statistics Census, 2021). The result also reveals that youths are typical individuals who engage in forestry jobs with an average age of 42.1, confirming the country's youthful population with 60% of the citizens below 45 years of age. The standard deviation (SD) at 12.7 seem high but this is due to the high variance used in the analysis. However, when compared to the mean, it is relatively small. Table 3 also reveals that there is less variation in the age of the

formal forestry workers, making the mean a reliable estimate of the respondents' age. These results are supported by Obasi et al. (2015), and Kamara and Su (2018).

Table 4 reveals a male-dominated population, mostly younger than 65 years. This is a huge boost to the forestry work sector, given that the active male population was reduced by a proportion of 1 to 3 during the 11 years civil conflict in Sierra Leone (Namuene and Egbe, 2022; Uzu et al., 2022). It is noted that there were more female heads of households than male, which could also be attributed to the fact that several households are headed by widows who lost their husbands during the civil crisis between 1991 and 2002. Lack of technological advancement in the forestry sector in Sierra Leone means that maximum strength is suitable for the energy-intensive forestry jobs. This compliments the result confirm more households headed by females while much of the forestry work were done by their male counterparts. It supports the complementary relationship between the social groups in the study area. Despite men being the traditional bread winners of the family, the female-headed households dominate the decision-making responsibilities. The educational level within the study area also suggests that the respondents were educated enough to understand the responses they provided and so can be trusted. The result also reveals the over-reliance of the residents on forest resources because more than 80% of the residents practice agricultural farming activities, timber harvesting and trade. These results agree with the findings of Munro and Horst (2012), the Sierra Leone National Statistics Census (2021).

The factor analysis suggests that farming and timber logging, without provision for residuals and wastes, has led to the rivers and lakes substituting as dumping sites. This is even worsened by rural dwellers' over-dependence on timber and other non-wood forest resources. These findings concur with Siry et al. (2018); and Namuene and Egbe (2022). They both opined that excess reliance of the locals on timber resources results in a heavy decline of the forests ecosystem and contributes adversely to silting of rivers and lakes, soil erosion, and reduction in soil fertility and crop output. All these factors will culminate to climate change and global warming. These concerns justify the timely EU intervention, providing expert resources and other compliance initiatives. In line with the Sierra Leone government's recent efforts to comply with binding international treaties, timber harvesting firms must consult relevant stakeholders to identify their core CSRs and how they intend to accomplish them within stated timeframes. Most harvesting firms, farmers associations and NGOs operating in the study area demonstrated that reforestation and mitigating illegal logging are

essential specific objectives of their firms. This conclusion agrees with the findings of Sampson et al. (2019), Munro and Horst (2012), who suggested that reforestation and mitigating illegal logging are crucial for sustainable forest management principles compliance.

Moreover, Obasi et al. (2015) also opined that harvesting firms should consult with the inhabitants of the study area. This corporation will better reveal their needs, rather than initiating abstract CSR projects in isolation. Result from figure 5 confirms that the time frame to complete the specific objectives gives the firms enough room to renege on them. This is particularly possible in a country with a huge illiteracy rate and corruption. Thus, the reoccurrence of these factors is one of the serious impacts encountered within the study area.

To adequately discuss the research questions, consider the results derived from the CFA. Although many variables were loaded under the PCA, only those with a 0.5 and above were considered to significantly explain the socio-economic and environmental impacts of timber harvesting in the study area. The social impact of timber harvesting from the CFA has Eigen value of 1.63 and explains 19 percent of the variance.

From the primary data obtained, the social impact of timber harvesting was mainly influenced by factors such as disputes and crises over land rent and compensation with a significant loading of 0.8, rural-urban migration with the most significant loading of 0.9.

This is evident in the study area as most young active residents are moving from their communities to cities with more job opportunities. Should the timber harvesting firms fulfil their CSR in terms of jobs and infrastructural development, the rural-urban migration narrative might differ. This result is supported by the findings of Sampson et al. (2019), who also concluded that rural-urban migration is often influenced by unemployment and lack of opportunities compared to urban areas.

Furthermore, from the primary data obtained, the economic impact of timber harvesting from the CFA has Eigen value of 1.80 and explains 20 percent of the total variance.

Result shows that the economic impact of timber harvesting was mainly influenced by factors like high cost of labor with a factor loading of 0.5, increased cost of timber products with a 0.7 factor loading, and loss of revenue by government with the most significant factor loading of 0.9. While other economic variables were employed for the PCA, only the above two variables satisfy the 0.5 threshold to be included in the CFA.

Discussing the Environmental impact, the result shows that it has Eigen value of 1.08 and explains 12 percent of the total variance. The variables that influenced the environmental impacts were; Loss of biodiversity with the most significant factor loading of 0.9, followed by silting of lakes (0.7); Damaging of trees and Non-wood products (0.6); Loss of Forest cover (0.6); Disappearance of vegetation (0.5) and climate change (0.6). There is an increased draught, water stress, poor yield of crops arising from flooding and crisis. Global warming and species extinction are likely to manifest in the future. It is quite evident from the result obtained that utilizing forest timber products unsustainably are self-defeating, and in most cases, violates treaties that the country is signatory to. This result agrees with Uzu et al. (2022), Kamara and Su (2018), who opined that if timber exploitation is left uncontrolled, it will eventually lead to factors that will increase GHG emission.

The regression analysis used other factors contributing to socio-economic and environmental impacts. These were natural rents and royalties; Gross Domestic Product; Unemployment; Inflation; Carbon emissions; Primary enrolment, and literacy Level.

The results from the CFA revealed the efficiency of education and sensitization. It reveals that majority of the respondents representing 50% indicated that workers' education is the measure taken to ensure sustainability of CSR projects, whereas 2.5% stated workers' sensitization as a measures taken to ensure sustainability of the CSR project. (see figure 7). The significance of education is further supported by the level of education expenses, which have a 19.7 mean value and 2.0 standard deviation. This result conforms with the findings of Munro and Horst (2012), who identified education and regular sensitization as practical strategies to ensure sustainable forestry management.

To reflect on the socio-economic activities of the respondents, it is evident from table 10 that most of the respondents are engaged in forestry work (237 respondents), constituting 78.74% of the respondents (farmers, loggers and so on) and those who are salaried workers constitutes a minority of the respondents constituting 0.66% of the respondents. Furthermore, the results in table 3 also show that the average number of people employed has an average mean of 2.5 with an SD of 0.2, while the average mean of people unemployed is 3.9 with an SD of 2.6.,

The question of local initiatives to ensure sustainability is quite relevant in the face of growing threats to deforestation. Figure 3 shows that 25 respondents (62.5%) agreed to introduce a timber

waste management system by recycling the residues. This is to avoid dumping these wastes in the lakes. This factor contributes to climate change because the lakes will dry up permanently. 3, 9, and another 3 respondents representing 7.5%, 22.5%, and 7.5%, respectively, confirm reforestation, sustainability projects and mitigation of illegal logging, as ways to ensure forest sustainability. Sampson et al. (2019), and Munro and Horst (2012) also confirm that reforestation and mitigating illegal logging are essential sustainability approaches in forest management. Furthermore, 50% of the respondents agreed that educating the workers is a viable measure to ensure sustainability of the forest. Also, 1 respondent each, representing 2.5% stated workers consultation and workers sensitization as measures taken to ensure sustainability of the CSR project. This result conforms with the findings of Munro and Horst (2012) who identified consultations with stakeholders and regular sensitization as effective strategies to ensure sustainable forestry management.

The hypothesis of the research was supported by the results from the CFA. The descriptive statistical analysis confirmed that the factors influencing environmental impacts from timber harvesting have a 1.08 Eigen values and explained 12% of the total variance. Similarly, the economic impacts from timber harvesting have 1.08 Eigen values and explain 20% of the total variance. Similarly, the social impacts from timber harvesting in the study area have 1.63 Eigen values and explain 19 % of the total variance. Thus, the hypothesis was supported entirely.

4.2 Regression model results.

This chapter presents and discusses the results of the regression models that show the socio-economic and environmental impacts of timber harvesting within the study area. It presents results from time series data (2010 – 2021) on selected dependent and independent variables. The selected variables include carbon emission, GDP, primary school enrollment, timber exports, inflation on timber products, unemployment, labor productivity and natural resource rents.

The results in this section are derived from secondary data obtained from UN-affiliated databases such as the FAOSTAT, ILO, UNESCO and others. Table 4-4 below are the results derived using Eviews 10 software for regression analysis.

TABLE 4-4 REGRESSION ANALYSIS RESULTS USING SOFTWARE EVIEWS 10.

Variable/s	Model 1: Carbon	Model 2: GDP	Model 3: Primary Enrollment
Natural rents	0.0067** (0.0026 et)	-0.1668** (0.0579 et)	0.0855 (0.2037 et)
Exports	0.0671 (0.0001 et)	-0.0095** (0.0375 et)	0.0001** (0.0139 et)
Gross Domestic Products	0.0266** (0.0090 et)		
Labor		0.3182** (0.0574 et)	-0.2019 (0.3509 et)
Inflation		-0.0273 (0.0416 et)	
Unemployment			6.0770*
Constant	-0.0596 (0.0538 et)	15.6823** (1.9433 et)	-13.6689(21.4351 et)

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels, respectively; using observations 2010-2021 (T = 12)

TABLE 4-5 COEFFICIENTS OF DETERMINATION FOR THE MODELS.

Prob(F-statistic)	0.0061	0.0042	0.0000
R-squared	0.5734	0.8596	0.9667
Adj. R-squared	0.4134	0.7794	0.9477
Normality	0.6528	0.0421	0.6583
Heteroscedasticity	3.6022	0.8963	10.0355
Serial Correlation	2.6383	4.4074	1.2576
Functional Form	0.0124	0.7297	0.1641
Cumulative sum	Stable	Stable	Stable
Cumulative sum ²	Stable	Stable	Stable

4.2.1 The environmental impacts

Table 4-6 below indicates that the dependent variable for environmental impact is proxied by carbon emission, and it is regressed to ascertain the influence of each of the following independent

variables: Natural Rent, Exports, and GDP. It further indicates that the amount of carbon emitted in the environment is influenced by the value of natural rent royalties received for a given forest space used for timber harvesting, the percentage amount contributed to the GDP, given the forest space and the exports value of timber harvested from a given forest space. To determine a statistically significant relationship between carbon emission and natural rent, the OLS estimation test was ran and the result suggests a p-value of 0.05.

TABLE 4-6 VARIABLES AND COEFFICIENTS FOR MODEL 1.

Variables	Coefficient	Standard Error	t-Statistic	Probability.
Carbon emission	-	-	-	-
Natural resource rent	0.0067**	0,0026	2,6305	0,0302
EXPORTS	0.0671	1,0051	0,6714	0,5209
GDP	0.0266**	0,0090	2,9407	0,0187

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12). Et denotes error term.

The results above suggest that the relationship between the variables is statistically significant. Thus, it supports the hypothesis that a change in natural rents will influence the amount of carbon emitted into the environment. To also determine a statistically significant relationship between carbon emission and GDP from the OLS estimation test, a p-value of 0.05 was obtained, which also separately supports the hypothesis that a change in the GDP will influence the amount of carbon emitted into the environment. There is also a significant relationship between the export as an independent variable and carbon emission, where a 0.0671 increase in export will lead to a unit change in carbon emission. Thus, the result confirms the first hypothesis that carbon emission impacting the environment is influenced by natural rents, GDP, and exports, separately or collectively. The model can be illustrated below.

Model 1: $Co_2 \text{ Emission} = \text{Natural rent} + \text{Export} + \text{GDP}$

$$y = 0.0067^{**} + 0.0671 + 0.0266^{**}$$

where y is the dependent variable, regressed by the independent variables of Natural rents, exports and GDP, respectively. See table 4-7 below for the environmental impacts' statistical tests and coefficients of determination.

TABLE 4-7 ENVIRONMENTAL IMPACT STATISTICAL TESTS OF SIGNIFICANCE AND COEFFICIENTS OF DETERMINATION

Dependent Variable	CARBON
Method	Least Squares
Date:	11/18/23
Time:	17:50
Sample:	2010 2021
Included observations	12
Prob(F-statistic)	0.0061
F-statistic	3,5837
R-squared	0.5734
Adj. R-squared	0.4134
Normality	0.6528
Heteroscedasticity	3.6022
Serial Correlation	2.6383
Functional Form	0.0124
Cumulative sum	Stable
Cumulative sum of square	Stable
Constant	-0.0596
S.E. of regression	0,0180
Sum squared residue	0,0026
Log likelihood	33,598

Mean dependent variable	0,1158
S.D. dependent variable	0,0235
Akaike info criterion	-4,933
Schwarz criterion	-4,7713
Hannan-Quinn criteria	-4,9928
Durbin-Watson statistics	1,6819

Table 4-7 above shows that the t-tests performed for this environmental impact model reveals an R^2 of 0.5734 with an adjusted R^2 0.4134. This shows that the independent variables in the model significantly explain 57 % of the dependent variable. The Normality test also determined that the data used in the research have a normal distribution. It shows 0.6528 (65 %) of normal distribution level. The Heteroscedasticity test measured at 3.6022 showed non-volatility in both dependent and independent variables, given a constant level of -0.0596.

Furthermore, serial correlation shows 2.6383 and thus indicates that the observations in this model are serially correlated. A functional form of 0.0124 also signifies a relevant relationship between the dependent and independent variables. The Durbin-Watson test reveals a positive autocorrelation between the variables as well. The Cumulative sum and sum of square tests were done to determine parameter stability. The tests show stability in the model. All other classical statistical tests and assumptions were satisfied, as seen from table 4-7 above.

4.2.2 The economic impacts.

See the result in table 4-8 below, which indicates that the dependent variable for economic impact is proxied by GDP, and it is regressed to ascertain the influence of each of the following independent variables: Natural Rents, Exports, labour participation and inflation. It further indicates that the contribution to GDP from timber harvesting is influenced by the value of natural rent royalties received for a given forest space used for timber harvesting, the labour input for a certain quantity of timber harvested, and the inflation of timber products over a given period. To determine a statistically significant relationship between GDP and natural rent, the OLS estimation test reveals a p-value of 0.05, suggesting that the relationship between the variables is statistically significant. Thus, it supports the hypothesis that a change in natural rents will influence the amount

contributed to GDP in the economy. To also determine a statistically significant relationship between GDP and export from the OLS estimation test, a p-value of 0.05 was obtained which also separately supports the hypothesis that a change in the timber export value will influence the amount contributed to the economy's GDP.

TABLE 4-8 VARIABLES AND COEFFICIENTS FOR MODEL 2

Variables	Coefficient	Standard Error	t-Statistic	Probability
GDP	-	-	-	-
Natural resource rent	-0.1668**	0,0579	-2,8818	0,0236
Exports	-0.0095**	3,7551	-2,5291	0,0393
Laour participation	0.3182**	0,0574	-5,5397	0,0009
Inflation	-0.0273	0,0090	2,9407	0,0187

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12)

There is also a significant relationship between labour input as an independent variable and GDP, where a unit change will lead to a unit change in contribution to GDP. However, the result shows a negative relationship between inflation and GDP, although the F-stat. Confirms a significant relationship between the two variables. Thus, the result suggests that the independent variables separately influence the dependent variable and confirms the second hypothesis that carbon emission impacting the environment is influenced by natural rents, GDP, and exports, separately or collectively. The model can be illustrated below.

Model 2: $GDP = \text{Natural Rent} + \text{Export} + \text{Labour} + \text{Inflation}$

$$y = -0.1668^{**} + -0.0095^{**} + 0.3182^{**} + -0.0273$$

where y is the dependent variable, regressed by the independent variables of Natural rents, export, labour and inflation, respectively. See table 4-9 below for the economic impacts' statistical tests and coefficients of determination.

TABLE 4-9 ECONOMIC IMPACTS STATISTICAL TESTS OF SIGNIFICANCE AND COEFFICIENTS OF DETERMINATION

Dependent Variable:	GDP
Method:	Least Squares
Date:	11/18/23
Time:	18:17
Sample:	2010 2021
Included observations:	12
Prob(F-statistic)	0.0042
F-statistic	10,71461
R-squared	0.8596
Adj. R-squared	0.7794
Normality	0.0421
Heteroscedasticity	0.8963
Serial Correlation	4.4074
Functional Form	0.7297
Cumulative sum	Stable
Cumulative sum of square	Stable
Constant	15.6823**
S.E. of regression	0,324823
Sum squared residue	0,73857
Log likelihood	-0,29959
Mean dependent variable	3,928333
S.D. dependent variable	0,691544
Akaike info criterion	0,883265
Schwarz criterion	1,085309
Hannan-Quinn criteria	0,808461

Table 4-9 above shows that the t-tests performed for this economic impact model reveal an R^2 of 0.8596 with an adjusted R^2 of 0.7794. This shows that the independent variables in the model significantly explain 86 % of the dependent variable. The Normality test also determined that the data used in the research have a normal distribution. It shows 0.0421 of the normal distribution levels. The Heteroscedasticity test measured at 0.8963 showed a non-volatility in both dependent and independent variables, given a constant level of 15.6823**.

Furthermore, serial correlation shows 4.4074 and thus indicates that the observations in this model are serially correlated. A functional form of 0.7297 also signifies a relevant relationship between the dependent and independent variables. The Durbin-Watson test reveals a positive autocorrelation between the variables as well. The Cumulative sum and sum of square tests were done to determine parameter stability. The tests show stability in the model. All other classical statistical tests and assumptions were satisfied, as seen from table 4-9 above.

4.2.3 The Social impacts

The result in table 4-10 below indicates that the dependent variable for social impact is proxied by primary enrolment, and it is regressed to ascertain the influence of each of the following independent variables: Natural Rents, Exports, labour participation and unemployment. It further indicates that students' primary enrolment is influenced by the value of natural rent royalties received for a given forest space used for timber harvesting, the labour input for a certain quantity of timber harvested, the export value of timber harvested and the unemployment level in the study area over a given period.

TABLE 4-10 VARIABLES AND COEFFICIENTS FOR MODEL 3

Variable	Coefficient	Standard Error	t-Statistic	Probability
Primary Enrolment	-	-	-	-
Natural resource rent	0.0855	0,2037	0,4196	0,6874
Exports	0.0001**	1,3951	5,8117	0,0007

Labour	-0.2019	0,3509	-0,5752	0,5831
Unemployment	6.0770*	1,0051	0,6714	0,5209

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12)

To determine a statistically significant relationship between primary enrolment and natural rent, the OLS estimation test reveals a positive relationship between the variables, thereby suggesting that it supports the hypothesis, separately, that a change in natural rents will influence the number of students enrolled in primary schools in the study area. To also determine a statistically significant relationship between primary enrolment and export from the OLS estimation test, a p-value of 0.05 was obtained, which also separately supports the hypothesis that a change in the timber export value will influence primary enrolment in the community. Although the result showed a negative relationship between labour input and primary enrolment, it does confirm from the prob(F-stat) in table 4-11 below that there is a high significant relationship between the independent variables and primary enrolment, where a unit change will lead to a unit change in the dependent variable as well. Similarly, the OLS test also reveals a positive relationship between unemployment and primary enrolment with a p-value of 0.1, which supports the hypothesis that a change in unemployment will result in a change in primary enrolment. These results suggest that the independent variables, separately and collectively, influence the dependent variable. This further confirms the third hypothesis that primary enrolment in the community is influenced by natural rents, exports, labour input and unemployment, separately or collectively. The model can be illustrated below.

Model 3: Primary enrolment = Natural Rent + Export + Labor + unemployment.

$$y = 0.0855 + 0.0001** + -0.2019 + 6.0770*$$

where y is the dependent variable, regressed by the independent variables of Natural rents, export, labour and unemployment, respectively.

TABLE 4-11 SOCIAL IMPACT STATISTICAL TESTS OF SIGNIFICANCE AND COEFFICIENTS OF DETERMINATION

Dependent Variable	PRIMARY ENROLLMENT
Method	Least Squares

Date	11/18/23
Time	18:20
Sample	2010 2021
Included observations	12
Prob(F-statistic)	0.0000
F-statistic	50,8693
R-squared	0.9667
Adj. R-squared	0.9477
Normality	0.6583
Heteroscedasticity	1.0355
Serial Correlation	1.2576
Functional Form	0.1641
Cumulative sum	Stable
Cumulative sum of square	Stable
Constant	-13.6689
S.E. of regression	1,56294
Sum squared residue	17,09944
Log likelihood	-19,1521
Mean dependent variable	14,24804
S.D. dependent variable	6,836727
Akaike info criterion	4,02535
Schwarz criterion	4,227394
Hannan-Quinn criteria	3,950546
Durbin-Watson statistics	2,464454

Table 4-11 above shows that the t-tests performed for the social impact model reveals an R^2 of 0.9667 with an adjusted R^2 of 0.9477. This shows that the independent variables in the model significantly explain 97 % of the dependent variable. The Normality test also determined that the data used in the research have a normal distribution. It shows 0.6583 of (65%) normal distribution level. The Heteroscedasticity test measured at 1.0355 showed non-volatility in both dependent and independent variables, given a constant level of -13.6689.

Furthermore, serial correlation shows 1.2576, thus indicating that the observations in this model are serially correlated. A functional form of 0.1641 also signifies a relevant algebraic relationship between the dependent and independent variables. The Durbin-Watson test reveals a positive autocorrelation between the variables as well. The Cumulative sum and sum of square tests were done to determine parameter stability. The tests show stability in the model. All other classical statistical tests and assumptions were satisfied, as shown in Table 20 above.

4.2.4 Discussions of the regression model results.

This work investigates the significant influence of specific variables such as: natural rents/royalty, value of timber exports, inflation of timber products, unemployment, and percentage contribution to GDP. It shows an empirical margin of influence of the independent variables on the dependent variable, collectively or separately. The research hypothesis were supported by the results from the regression analysis, as expressed in table 4-12 below.

TABLE 4-12 HYPOTHESIS RESULTS

Hypothesis	Prob(F-stat)	R^2	Adj. R^2	Serial correl.	Heteroscedasticity	Normality test	Results
H ₁	0.0061	0.5734	0.4134	2.6383	3.6022	0.6528	Supported
H ₂	0.0042	0.8596	0.7794	4.4074	0.8963	0.042	Supported
H ₃	0.0000	0.9667	0.9477	1.2576	10.0355	0.6583	Supported

The first hypothesis represents the environmental impacts of timber harvesting within the study area. The F. statistics of 0.0061 in the model suggests that all the explanatory variables jointly influence the dependent variable in the model. The R^2 and adjusted R^2 are 0.5734 and 0.4134 respectively. The estimated results show that the Adjusted R^2 accounted for the variability of the response variable in the model, while the R^2 shows that 57% of the independent variables do

explain the dependent variable, thus rendering the model valid. Assuming everything remains the same, the heteroscedasticity indicates a non-volatility in the model while the serial correlation validates the normal correlation among the variables. The normality test also confirms a normal distribution of the variables in the model.

Thus, from the result obtained, it can be confirmed that the independent variables used to explain the dependent variables are significant and the model is stable, suggesting therefore that it supports the null hypothesis that environmental impact (carbon emission) is a result of certain factors like natural resource rents, export value of timber products and the total value contribution to national GDP. According to the result obtained, these independent variables, separately or collectively, significantly influence the dependent variable. This result is supported by findings from Jing Luo et al. (2023), who concluded that harvesting wood product can play an important role in reducing carbon emissions while managing healthy forests. It is further supported by the annual report of the United States Department of Agriculture (2022), which suggested that the best way to explain the effects of forest management on the atmosphere is to consider what the atmosphere appreciates regarding carbon entering or leaving. Other works that supported this conclusion include (Appiah et al. 2018; Jordan et al. 2018; Caporale et al. 2021, Xiyang et al. 2022; Onofrei et al. 2022; Waqas et al. 2022; Wei et al. 2023), all concluded that there is a significant relationship between carbon emission and the independent variables of natural rents, exports, and GDP.

The second null hypothesis represents the economic impacts emanating from timber harvesting within the study area. The F. statistics of 0.0042 in the model suggests that all the explanatory variables jointly influence the dependent variable in the model. The R^2 and adjusted R^2 are 0.8596 and 0.7794 respectively. The estimated results show that the Adjusted R^2 of 78% accounted for the variability of the response variable in the model, while the R^2 shows that 86% of the independent variables do explain the dependent variable, thus rendering the model valid. Assuming everything remains the same, the heteroscedasticity indicates a non-volatility in the model while the serial correlation validates the normal correlation among the variables.

The normality test also confirms a normal distribution of the variables in the model. Thus, from the result obtained, it can be confirmed that the independent variables used to explain the dependent variables are significant and the model is stable, suggesting, therefore that it supports the null hypothesis that economic impact (GDP) is a result of certain factors like natural resource rents, the

export value of timber products, labor input and inflation in the prices of timber products. According to the result obtained, there was a negative relationship between some of the independent and dependent variables. However, with a p-value of 0.05 for almost all the variables, it can be confirmed that there is a significant statistical relationship between the variables, enough to separately or collectively, explain the significant influence on the dependent variable. This result is supported by findings from other literature (Dogan and Aslan, 2017; Bhatt et al. 2021; Assa, 2021; Namuene and Egbe, 2022), who all concluded that there is a compelling relationship between the GDP of a country and natural resource rents, export value, labor and inflation in timber harvesting.

The third null hypothesis represents the social impacts emanating from timber harvesting within the study area. The F. statistics of 0.0000 in the model suggests that all the explanatory variables jointly influence the dependent variable in the model. The R^2 and adjusted R^2 are 0.9667 and 0.9477 respectively. The estimated results show that the Adjusted R^2 of 95% accounted for the variability in the response variable in the model, while the R^2 shows that 97% of the independent variables does explain the dependent variable, thus rendering the model valid. *Ceteris paribus*, the heteroscedasticity indicates a non-volatility in the model while the serial correlation validates a normal correlation among the variables. The normality test also confirms a normal distribution of the variables in the model. Thus, from the result obtained, it can be confirmed that the independent variables used to explain the dependent variables are significant and the model is stable. This suggests that it supports the null hypothesis that social impact, represented by Primary school enrolment, is a result of certain factors like natural resource rents, export value of timber products, labour input and unemployment within the study area.

According to the result obtained, one of the independent variables shows a negative relationship with the dependent variable. However, with a p-value of 0.05 and 0.01 for two of the other independent variables, it can be confirmed that there is a significant statistical relationship between the variables, enough to separately or collectively explain the significant influence on the dependent variable. Findings from other literature support this result (Mansaray et al., 2015; Jegatheswaran et al., 2018; Arunachalam and Pandey, 2019; Memon et al., 2020; Shanahan et al., 2021; Owuor et al., 2021; Li and Wu, 2022), who all opined that there is a significant

relationship between education and such variables like natural resource rents, exports, labour, and unemployment in timber harvesting.

The regression analysis also answered the research questions about the main factors that influenced the socio-economic and environmental impacts from timber harvesting. The model identified factors that influenced the social impacts of timber harvesting. It confirmed that primary enrollment, which represents the social impact, is influenced by natural rent, export, labour, and unemployment factors. Similarly, the model confirmed that factors like natural rent, export, labour, and inflation influence GDP, representing economic impact in the second model. Regarding the environmental impact of carbon emission in the model, the result confirmed that factors like natural rent, export, and GDP influence it.

5 SUMMARY AND CONCLUSIONS

Findings from the regression analysis using time series data show that harvesting timber products within the study area has both positive and negative impacts socially, economically, and on the environment. The regression analysis results reveal that some independent variables have positive relationships with the dependent variables while some show a negative relationship. This implies that with any given quantity of timber harvested within the study area, there is either a positive or negative outcome on the social lives of the residents, their economic status and on the environment. For instance, the social status which is represented by primary school enrolments. Given all things to remain the same, a positive relationship exists between unemployment, natural resource rents, export quantity as independent variables, and the dependent variable (primary enrolment), which proxied the social impact. However, the result shows a negative relationship between labour participation and primary enrolment, which shows that a change in labour participation may not necessarily reflect a positive increase in primary enrolment. Collectively, however, with the probability (F.stat.) at 0.0000, there is a significant statistical relationship between the independent variables and the dependent variable. The model shows a normal distribution level, non-volatility, and a serial correlation between the variables. Similarly, the economic impact which is proxied by GDP shows a negative relationship between natural resource rents, exports, inflation as independent variables and the dependent variables. However, the independent variables separately showed a p-value of 0.5 and less, which means the variables are statistically relevant and can explain a relationship with the dependent variable. Labour participation as an independent variable showed a positive relation with the the dependent variable with a p-value of 0.05. Collectively, the economic impact model showed a probability (F.stat.) of 0.0042 and an R^2 of 0.8596 with Adj. R^2 at 0.7794, indicating that the model is statistically significant and 86% of the independent variables can explain the dependent variable. The result also reveals that there is a normal distribution level, non-volatility and a serial correlation between the variables. The environmental impact is proxied by carbon emission as the dependent variable. It showed a positive relationship with natural resource rent, export and GDP as independent variables with a p-value of 0.05 or less. This indicates that there is a significant level of relationship between the variables separately.

Collectively also, the model showed a probability (F-stat.) of 0.0061, an R^2 of 0.5734 and an Adj. R^2 of 0.4134 which indicates that the model is not only statistically significant, but that 57% of the independent variables can explain the dependent variable. There is also a normal distribution level, non-volatility and a serial correlation between the variables in the model.

The result from the CFA using primary data from the study area also supports the regression analysis outcome. The variables for the environmental impacts showed an Eigen value of 1.08 and explains 12 percent of the total variance with factor loadings of 0.5 for such variables like silting of rivers and lakes, loss of biodiversity, damaging of immature trees and non-forest wood products, disappearance of vegetation, climate change and global warming. Further results from the frequency and percentage analysis reveals that most of the respondents cited climate change and pollution as the main environmental impacts associated with harvesting timber. Also, some respondents cited deforestation, stormy windy conditions and inconsistent weather pattern as adverse impacts felt from timber harvesting in their communities. In similar vein, the economic impact from the result has an Eigen value of 1.80 and explains 20 percent of total variance with factor loadings of 0.5 and above. Thus, high cost of labour, loss of revenue by government, and increment in prices for wood products are identified as adverse economic effects of timber harvesting in the survey communities. The results from the frequency and percentage analysis suggests that majority of the respondents indicated the rise in farming cost in terms of scarcity of available labour supply, rapid decrease in their incomes as well as higher expenditure as economic impacts of harvesting timber within their communities. The result for the social impact followed a similar path as the environmental and economic impacts. It has an Eigen value of 1.63 and explains 18 percent of total variance with factor loadings above the threshold of 0.5. The social impact variables identified here includes land disputes and compensation crisis, and rural-urban migration. The frequency and percentage analysis also reveal that majority of the respondents cited hydrological disruptions, land disputes and compensations while some suggested urban migration, sanitation, transportation, and poor road network as the social impacts within their communities.

From the foregoing, it can be concluded that both the regression analysis using time series data and the descriptive statistical analysis from the primary data answered the research questions to a greater extent. The results also concord with the findings of Sampson et al. (2019) who identified challenges facing timber harvesting as poor road network, high tariff, transportation of products

and capital-intensive nature of the harvesting process. Again, the results concur with Obasi et al (2015), who opined that excess reliance of the locals of timber resources results to silting of rivers, soil erosion and heavy decline in the functioning of the forests' ecosystem are some of the adverse effects of timber harvesting. Furthermore, it can be deduced that results from the confirmatory factor analysis does support the results from the frequency and percentage analysis as well. This explains the validity and credibility of the findings of this work.

6 Policy Recommendations

The role of sustainable impact assessment in the face of inevitable socio-economic activities within the KHFR in Sierra Leone is more relevant now than ever and the need for an effective enforcement strategy at local, national, and global levels is even more crucial. SIA can help Government officials, forest managers and other stakeholders to select the best alternative system that will ensure forest sustainability. Furthermore, indicators and tools should be updated and adapted to new and evolving situations. SIA is an important input when defining best forest practices and the main driving factors for future development of sustainable forest principles. Findings from this work confirms that timber harvesting affects the socio-economic and the environmental well-being of the respondents in the study area in many ways than one. These effects have a direct or indirect consequence on global warming and climate change as they lead to increased GHG emissions. Timber products have served as a tool for sponsoring armed conflicts in Africa and this has not only claimed lives but has also reduced communities to the dangers of wind or catastrophic erosions leading to mudslide in Sierra Leone in 2017. Although the residents in the study area harvest these resources for immediate survival from poverty and hunger; the future will only reinforce the perpetual scourge of hardship. Thus, there is need for a holistic approach to resource management and sustainability. Hence, the study made the following recommendations.

It is also crucial that the Sierra Leone government hold MNCs and other commercial ventures responsible for unfulfilled policy promises. CSR represents a social contract between the communities and these business interests, and they must be executed. The 40 business interest representatives who were part of the 301 respondents were top level executives and does confirm that their firms' laid down policies were derived out of consultations with the communities in the study area and other stake holders. From the responses of the 40 business interest representatives,

it was largely confirmed that the environmental challenges in the area are climate change, pollution, weather inconsistency, deforestation and windy disturbances. This is a confirmation of failure, by themselves, and enough reason to hold them accountable.

It is also important that the government goes back to the policies of the 1990s and learn from them. GHG emission level was quite low and on average compliance rate compared globally between 1990 and 2002, according to the World Resources Institute Climate Analysis Indicators (2004). Although it is important that the Sierra Leonean government pursue policies that create jobs, increase GDP growth, and reduce unemployment, it should also implement policies that cater for the protection of its environment. While data on the SFM indicators remain evasive, the level of deforestation in the country has equally not been measured sufficiently in recent times. This is also in line with recommendations from Munro et al. (2012), and Kamara and Su (2018). Robust policies and effective implementation can help in this direction if the implications of timber harvesting on the environment and socio-economic well-being of residents in the concerned communities are to be addressed.

The Sierra Leone government should consider investing in training the forest guards and logistically support them. They can help to enforce rules and make subsequent arrest of offenders. Government must prove its effectiveness and capability to protect revenues and people's lives by ensuring that all timber harvesting companies, cooperative bodies, and private individuals acquire the necessary certification and license before entering the forest. This will go a long way in abating corrupt practices by potential users and timber landowners. Rigid penalties on wood companies must be effectively implemented and individuals who harvest these resources beyond agreed limits must be effectively prosecuted.

Government should ensure that all timber harvesting companies, cooperative bodies and private individuals respect all forestry rules, adhere to their laid down CSR policies. To abate the rampant corrupt practices, timber harvesting companies must acquire and tender their certification and license before entering the forest. Moreover, it is not enough for successive governments to issue timber exporting bans without banning its harvesting from the forests. As such, forestry guards should be empowered and the residents around the KHFR should be always sensitized, adequately educated about intended policies, and encouraged enough to see the benefits of maintaining the forest vegetation. This will allow the residents to be aware of the long-term implications of logging

on the environment and socio-economic well-being of themselves residing in the concerned communities.

PAGE's current co-management initiative is timely and necessary but it has great challenges ahead. Efforts are needed to reduce conflicts between the different actors, redefine boundaries, and clarify laws, before any management regime can effectively be implemented. The KHFR is an important natural resource for the region and its sound management is needed to ensure that the residents in the affected communities and various stakeholders can benefit from its presence.

There needs to be a clear clarification of the different laws governing the forest reserve, the protected forest and the salvage lands. Confusion around the laws appears to have allowed some forestry staff to be overzealous in their activities, regulating and charging fees for activities outside of their jurisdiction. The 1988 Forestry Act is very vague in terms of what their jurisdiction is, often leading to the perception that all forests (whether classified or private) should be the sanction of the Forestry Division. There needs to be a more effective top-down positive engagement between the forestry division and the resident communities. As noted during interviews, the forestry guards claimed that all the communities were thieves, stealing produce from inside the reserve, while the communities complained that the Forestry Division staff were overstepping their authority and putting unfair restrictions on their lives. Most of the villages complained that they are rarely visited by the forestry staff. Many of the communities requested a discussion to be initiated with forestry division staff so they could both obtain greater clarity about the forest reserve and its laws and raise their own concerns. Civic education is necessary, an applause for the current administration (2018 to 2028)

The most effective method for establishing and confirming the laws relating to KHFR and its surroundings at the community level would be to organise meetings at each of the villages with the communities, forestry staff and chiefdom authorities in attendance. During these meetings the laws governing the forest reserve, salvage land and protected forest can be discussed, byelaws agreed upon and recorded. The development of local participatory would also be useful in helping to communicate and clarify the different local land tenure regimes and existing byelaws. In relation to the 1988 Forestry Act and the historical production of the different land types, the natural regulatory regimes appear to be that the forest reserve should come under the authority of the Forestry Division, the Protected Forest should come under the jurisdiction of the Chiefdom

Authorities and the salvage land should come under the jurisdiction of the village level authorities. In this context, it should be made clear when forestry staff have the authority to intervene in non-forest reserve land. Naturally, this process would have to be linked with the clear re-demarcation of the forest reserve for it to be effective.

Most of the communities surrounding the KHFR had their lives and livelihoods devastated by the Sierra Leonean civil war. Many fled their towns during the war and have only returned in the last decade or so to start rebuilding their lives. In the post-civil war era, many have looked towards gold mining, charcoal harvesting and board production as supplements to their farming activities to bring security to their livelihoods and some of the communities have established themselves in the forest reserve for these activities. This is an awkward situation to deal with since, in the long term this pattern could have a negative environmental impact, if it is not managed and regulated correctly. Strategies are needed to help empower local communities, so that they can negotiate fairer prices for their commodities. Furthermore, some communities are harvesting valuable tree species for both international and domestic markets, and are only receiving token payments in return, thereby reducing natural resource rents useless. One factor causing these low prices is a lack of knowledge about the value of resources in local national and international markets. As suggested by Mansaray et al. (2015), there is always a greater chance of negative business transaction if there is a lack of adequate market information. Developing avenues to make this information more readily available at the community level could provide a strong bargaining position for these communities in their dealings with other actors in the commodity chains.

The development of alternative livelihoods is also a key issue. The main thing is the need for actual income earning options that is well suited to the local context. Some examples have already emerged in some of the communities and could be spread to others, such as one community providing food for the school feeding program and another that is involved in honey production. A couple of communities claimed that they were supposed to receive cassava processing equipment from Gava as part of the logging concession deal. Though this equipment never arrived, the communities indicated that they were still very keen to experiment with different opportunities such as this. The improvement of community infrastructure, notably schools and health clinics, is also vital for the long-term improvement of community livelihoods.

There is also a need for the oversight group to look into the activities of ACOTIDA. This organization collects substantial funds directly from its members as well as in the form of afforestation fees at Bandama checkpoint on the Bo-Kenema Highway. It is supposed to be an advocacy agency for timber and board producers. However, almost all its members expressed a strong dislike for the association, stating that they received no benefits as members. They opined that they only maintained membership as necessity for survival, because if they did not, ACOTIDA would make life in the board trade difficult for them. If the accusations are true, ACOTIDA, rather than serving as an advocacy organisation may be acting more like a protection racket, causing difficulties for those that do not join the association and pay fees. Although it is important to note that ACOTIDA is quite active in lobbying the Forest Division to have chainsaw fees reduced, which would no doubt bring great benefits for most of its members, however this was not openly recognised during interviews. Forestry staff also claimed that they have never received money from ACOTIDA for afforestation or seen the association partake in afforestation activities, bringing into question why ACOTIDA is allowed to collect afforestation fees at checkpoints. According to the 1989 Forestry Regulations, it is the Forestry Division that should be collecting this fee. Overall, the operations of ACOTIDA need a thorough investigation, in terms of the transparent use of the funds it receives, the deals it has made with forestry staff and the benefits it is providing for its members. Collective advocacy agencies for wood-based commodity traders are important but it appears that ACOTIDA is not properly fulfilling such a role.

The fees need to be restructured to be more reflective of the commercial value of the trade. One good strategy for restructuring is to work out a way to prevent average bribe that people pay and rather divert the payment to official government revenue accounts. This can act as a proxy of what an acceptable „fee“ might be. While this might be dramatically lower than the current official amount, if paid in full by all in the trade, it would still likely cause an overall increase in revenue for the forestry division, especially if the informal network of bribing is as extensive as the research respondents indicated. A more efficient revenue structure would benefit everyone, except perhaps those who are financially profiting from the bribes.

One problem relating to the above, is that in some areas the current legislations are too generic, in particular lumping board and timber (logs) production under similar regulation regimes. Their productions are notably different, the former is conducted by small-scale operators for the domestic

market, and the latter is generally conducted by large companies for the export market. However, many of the fees are homogenous for the two differing productions. This has caused problems. For example, the annual fees for operating chainsaws is probably a practical amount to charge companies (such as Gava, Exim and Capitol), but is unrealistically high for small-scale operators. Thus, the forestry division should consider establishing different fees for commercial and individual operators. This would allow the Forestry Division to have more specific controls of the differing productions and would help to reduce many clandestine logging activities by moving them into the realm of regulated fee paying operations. Most (illegal) small-scale operators indicated they are happy to pay fees for their operations if the fees that they are charged are priced. As mentioned earlier, the rates that operators are currently paying for informal fees can sometimes be a good proxy for establishing formal fee rates.

It is important to note that this industry could bring a lot of benefits if conducted correctly. Current harvesting techniques by individuals or companies alike, often involve the chain sawing of selective tree species, rather than employing clear felling methods. Such methods employed tend to be much more sustainable and have a lesser impact than those employed in most wealthy countries. Also, provided the trees are not fully stumped they generally have a good propensity to regenerate. The photo in Figure 9 is a good example of this, showing how much a logged area of forest can regrow in a three-year period. Thus, there is reasonable potential for the establishment of a sustainable commercial logging industry in Sierra Leone. However, it needs to be developed on a transparent basis and have sufficient institutional support with adequate mechanisms in place to ensure that economic benefits from the trade are realised properly at both the community and government levels.

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APPENDICES

Section A: QUESTIONNAIRE INSTRUMENTS ON HOUSEHOLD DEMOGRAPHICS

APPENDIX 1: Questionnaire

Bio Data

1. Community (Code)
2. Household code
3. Sex of respondent: [] 1=Female 0=Male
4. Age of respondent
5. Highest level of formal education of respondent: []
A. None B. Basic (Primary/JHS/Middle) C. SHS D.Tertiary
(Training college/Polytechnic/University)
6. Marital status of respondent: []
A. Single 2. Married 3.Divorced/Separated 4.Widowed
7. Are you the head of the household? (If man is away >6months/yr, then woman is Head)? A. Yes B.No

Household composition

8. How many people are you in your household? Total [] male [] female []
9. Number of persons with ages below 15 years. Total [] male [] female []
10. Number of ages between 15 and 65 years. Total [] male [] female []
11. Number of ages above 65 years Total [] male [] female []
12. Number of ages who are literate or attended school total [] male [] female []
13. Number of people in your household who are employed.....
14. Number of people in your household who are unemployed.....
13. Spouse years of education.....
14. Monthly household expenditure.....
15. Monthly income.....
16. Educational expenses per year.....
17. What is your religion [] 1= Christianity [] Islam [] Traditionalist [] others
18. What is your major occupation [] 1=Agriculture 2= Forestry (timber harvesting) 3=Petty Trading 4= Craftsmanship 5 = Salaried work (formal sector) 6 =other

Answer the following questions if you are not the household head

19. Gender of household head [] 1=Female 0=Male

20. Years of education of household head.....

21. What is the major occupation of the household head[] 1=Agriculture 2= Forestry (timber harvesting) 3=Petty Trading 4=Craftsmanship 5=Salaried work (formal sector) 6=other

SOCIAL, ECONOMIC AND ENVIRONMENTAL IMPACT OF TIMBER HARVESTING

I would like to know how much you agree or disagree with each of these statements related to the impact of timber harvesting in your locality. Kindly tick (√) your response for; Completely Disagree= 1, Disagree= 2, Neutral= 3, Agree= 4, completely Agree= 5

	1	2	3	4	5
22. Silting of rivers and lakes					
23. High cost of labour					
24. Occurrences of disputes and crises over land and compensation					
25. Damaging of immature trees and non-wood forest products					
26. Loss of income and revenue by government					
27. Reduction of soil fertility and crop outputs					
28. Loss of biodiversity					
29. Loss of forest cover					
30. Disappearance of forest cover					
31. Rural-urban migration					
32. High cost of living					
33. Increased cost of wood and timber products					

34. Climate change or global warming					
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GENERAL QUESTIONS

35. Is there a multinational company involved in the harvesting of timber your locality?

☐ yes ☐ no

36. If yes, how long has (have) the company be involved in the harvesting of timber in your locality?

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37. What are the challenges associated with the harvesting of timber in our locality ?

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38. What are the trading and distribution channels available for the timber in your locality ?

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SECTION B: INTERVIEW GUIDE FOR MNCS

APPENDIX 2. Background & MNC respondent's information

Please tick where appropriate and reflect you and your enterprise in which you operate.

1. What is your firm's line of business? Manufacturing organization ☐ Service organization ☐
2. How long has this firm existed/operated in the industry?
Under 5 years ☐ 6 – 10 ☐ 11 – 15 ☐ 16+ ☐

3. How many employees do your firm have?

0 to 5 employees ☐ 6 – 29 ☐ 30 – 99 ☐ 100+ ☐

4. Please indicate your **current position** in this firm. Owner-manager ☐ General

Manager ☐ Executive ☐ Other top management role ☐

5. Please indicate your **gender**. Male ☐ Female ☐

6. Please indicate your **age** (years). Less than 20 ☐ 20 to 29 ☐ 30 to 39 ☐

40 to 49 ☐ 50+ ☐

7. Indicate your Highest Educational Level. PhD ☐ Master's Degree ☐ First

Degree ☐ HND ☐ SSCE/MSLC ☐ Basic Education ☐ No formal education

☐

QUESTIONNAIRE INSTRUMENTS ON MNCS' CSR.

1. What are the laid-down policies of your firm in relation to Corporate Social Responsibility towards forestry?

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2. What are the visions and objectives of your firm in relation to Corporate Social Responsibility towards forestry?

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3. How was the firm CSR project initiated?

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4. Is the CSR project implemented temporary or permanent?

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5. What is the estimated life span?

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6. How is your organisation ensuring the sustainability continuity of the CSR implemented in the communities?

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7. What role do the other stakeholders play in ensuring the sustainability of CSR projects in the communities?

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APPENDIX 3; TABLE 1: Sierra Leone Forest Reserves

Name	Status	Area (km ²)	Comments
Bojeni Hills	Designated	7.38	
Dodo Hills	Designated	21.8	
Fabina	Designated	3.82	
Farangbala	Designated	12.60	
Freetown Water Works	Designated	11.34	
Gboi Hills	Designated	2.07	
Gola East Forest	Designated	228.00	Amalgamated, GRNP,2010
Gola West Forest	Designated	62.00	Amalgamated, GRNP,2010
Gola South Forest	Designated	11.60	
Gori Hills	Designated	79.40	
John Obay	Designated	2.07	
Kambui Hills	Designated	143.35	
Kambui South	Designated	8.80	
Kambui North	Designated	203.48	
Kandesuri	Designated	59.57	
Kasewe	Designated	12.24	
Kent Extension	Designated	6.44	
Kuru Hills	Designated	69.93	
Lalay Reserve	Designated	4.76	

Leicester Peak	Designated	0.52	
Lhel Hills	Designated	1.55	
Malal Hills	Designated	3.39	
Matete	Designated	8.50	
Moku Hill	Designated	1.46	
Moyamba Hills	Designated	1.89	
No.2 River	Designated	6.99	
Okra Hills	Designated	2.48	
Penisular	Designated	178.00	
Port Loko Reserve	Designated	2.16	
Singambe	Designated	2.99	
Sipende	Designated	8.29	
Tabé	Designated	3.25	
Tama	Designated	170.00	
Tobi	Designated	16.71	
Tonkolili	Designated	71.04	
Wara Wara Hills	Designated	10.41	
Waterloo	Designated	0.85	
Nimini South	Designated	25.90	

GRNP* Gola Rainforest National Park

APPENDIX 4 TABLE 2: Sierra Leone Game Reserves

Name	Status	Area (km ²)	Comments
Bagru-Moteva Creeks	Proposed	50.00	
Bumpe Mangrove Swamp	Proposed	49.21	
Kagboro Creek (Yawri Bay)	Proposed	50.00	
Kpaka-Pujehun	Proposed	25.00	

Sewa-Wanje	Proposed	100.00	
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APPENDIX 5; TABLE 3: Sierra Leone Game Sanctuaries

Name	Status	Area (km ²)	Comments
Bo Plains	Proposed	25.00	
Tiwai Island	Designated	12.00	

APPENDIX 6; TABLE 4: Sierra Leone National Parks

Name	Status	Area (km ²)	Comments
Gola Rainforest National Park	Designated	710.10	
Kuru Hills	Proposed	69.93	
Lake Mape/Mabesi	Proposed	75.11	
Lake Sonfon	Proposed	51.80	
Loma Mountains	Proposed	332.01	
Oufamba-Kilimi	Designated	808.13	
Wester Area Peninsula National Park	Designated	183.37	

APPENDIX 7; TABLE 5: Sierra Leone Non-hunting Forest Reserves

Name	Status	Area (km ²)	Comments
Kangari Hills	Designated	85.73	
Tingi Hills (Sankan Biriwa)	Designated	332.01	
Western Area	Designated	118.85	
Loma Mountains	Designated	176.88	

APPENDIX 8; TABLE 6: Sierra Leone Strict Nature Reserves

Name	Status	Area (km ²)	Comments
Bonthe Mangrove Swamp	Proposed	998.54	

Gola Forest West	Proposed	62.00	
Gola Forest North	Proposed	40.00	
Mamunta-Mayoso Sanctuary	Proposed	20.72	
Morgbai Proposed Gola North	Proposed	NA	
Port Loko Plains	Proposed	25.90	
Sulima Mangrove swamp	Proposed	25.90	
Wemago Gola East	Proposed	35.00	
Yelibuya Island	Proposed	38.85	

APPENDIX 9; TABLE 7: Sierra Leone Marine Protected Areas (MPAs)

Name	Status	Area (km ²)	Comments
Yawri Bay	Designated		
Sherbro River Estuary	Designated		
SL River estuary	Designated		
Scarcies River Estuary	Designated	694	

Reference: World database on protected areas: Sierra Leone UN Environment Programme World Conservation Monitoring Center, retrieved October 14th 2022

APPENDIX 10; TABLE 8: Global producers of industrial round wood, 2022.

Wood products	Producers	Quantity (1 000 m ³)
Industrial round wood	Global production	2011135.99
	Asia-Pacific	486050.29
	Latin America	246675.99
	Africa	79758.99
	EU+	390389.38
	Indonesia	80800.00

Brazil	142989.00
India	48154.00
China	180237.00
Nigeria	9810.00
South Africa	16604.49
Cameroon	5460.00

Source: ITTO, 'Annual Review Statistics Database', 2022.

APPENDIX 11; TABLE 9: Global importers of timber products, 2022.

Timber products	Importers	Quantity (1000 m ³)
Industrial round wood	Global imports	123504.09
	Asia–Pacific	57981.78
	EU	56217.92
	Latin America and the Caribbean	316.56
	China	43690.00
	india	4923.00
Sawn wood	Global Imports	138729.90
	Asia–Pacific	43514.04
	EU	35731.07
	Africa	3248.24

Plywood	Latin America and the Caribbean	4447.63
	China	26660.00
	Taiwan	1319.07
	Global Imports	29106.89
	Asia-Pacific	9271.57
	EU	6179.33
	Latin America and the Caribbean	1456.41
	Japan	2265.46
	Malaysia	1167.60

Source: ITTO Annual Review Statistics Database, 2022.

APPENDIX 12; TABLE 10: West African countries by forest cover, 2020.

CATEGORY	1	2	3	4	5	6
Status	Forest	Low	Modest	Adequate	High	Very high
	Crisis	Forest	Forest	Forest	Forest	Forest
		Cover	Cover	Cover	Cover	Cover
Forest cover (%)	0-5	6-10	11-20	21-30	31-50	51+
Countries	Mauritania	Togo	Mali	Burkina Faso	Ghana	Guinea-Bissau
	Niger	Benin	Cape-Verde	Nigeria	Liberia	
				Ivory Coast	Senegal	
				Gambia	Sierra Leone	
				Guinea		

Source: World Bank 2021a; World Bank 2021b.

APPENDIX 13; TABLE 11: Wood production in selected West African countries, 2020.

Country	Economic Freedom	GDP per capita	Fixed effect
Unprocessed			
Nigeria	0.00323	0.00009	17.540
Sierra Leone			15.297
Benin			15.309
Ghana			16.959
Gambia			13.262
Liberia			15.674
Guinea-Bissau			14.532
Processed			
Nigeria	-0.02350	0.00018	15.040
Sierra Leone			9.914
Benin			12.120
Ghana			14.477
Gambia			8.394
Liberia			12.532
Guinea-Bissau			10.517

Source: World Bank 2021a.

APPENDIX 14; TABLE 12: Selected West African countries, population, land area, forest cover, GDP per capita and economic freedom, 2022.

Forest cover category	Country	Population (millions)	Land area (000s km3)	Forest area (% of land area)	GDP per capita (\$)	Economic freedom score
3	Nigeria	235.9	923.8	23.7	5.363	58.7
3	Sierra Leone	8.2	72.3	35.1	1.794	51.7
4	Benin	11.1	114.8	27.8	3.432	59.6
4	Ghana	30.8	238.5	35.1	5.652	59.2
5	Gambia	2.3	11.3	24	2.321	58.8
5	Liberia	4.9	111.4	79.1	1.491	49.2

6	Guinea-Bissau	1.9	36.1	70.4	2.077	54.9
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Source: World Bank 2022.

Key: 1= forest crisis; 2: Low forest; 3: Modest forest; 4: Adequate forest; 5: High forest; 6: Very high forest

APPENDIX 15; TABLE 13: Selected West African countries, import, export and trade balance, 2002, 2012 and 2022.

Country	2002			2012			2022		
	Export	Import	Trade Balance	Export	Import	Trade Balance	Export	Import	Trade Balance
Nigeria	33.3	(278.3)	(245.0)	98.7	(994.1)	(895.5)	134.9	(1.102.1)	(967.3)
Sierra Leone	2.5	(6.8)	(4.3)	12.7	(7.8)	4.9	132.2	(15.5)	(116.7)
Benin	1.4	(15.9)	(14.6)	16.1	(17.0)	(0.9)	33.9	(18.1)	15.7
Ghana	137.6	(37.8)	99.9	178.8	(103.9)	74.9	290.0	(406.7)	116.8
Gambia	0.3	(2.6)	(2.3)	1.0	(8.2)	(7.2)	77.2	(11.4)	65.8
Liberia	213.9	(0.5)	213.3	5.5	(4.6)	0.8	23.6	(11.9)	11.7
Guinea Bissau	1.0	(0.2)	0.8	1.9	(1.2)	0.7	9.2	(1.4)	7.8

Source: World Bank 2022.

APPENDIX 16; TABLE 14: Comparative climate indicators of Sierra Leone, Liberia and Guinea, 2022.

Country	GHG emission	%Global emission	Population	Co2 per capita	GDP mi.US\$	Co2/GDP mi.US\$;	Change in GDP from emission
Sierra Leone	3.51	0.01	4.390.737	0.08	1.654	2.122	3.14 (+32%)
Liberia	12.85	0.03	7.079,162	1.81	3.987	3.222	-12.72 (-78%)
Guinea	30.18	0.06	11.805,509	2.56	5.254	5.744	8.60 (+40%)

Source: FAOSTAT, 2022.

APPENDIX 17; TABLE 15: Protected Areas

Protected Area	Status	Size (ha) ³	District	Description
Outamba Kilimi	National park	110,900	Bombali	Fouta Jallon Highlands where six of West Africa's major rivers rise.
Gola forest	National park	7,107	Kailahun, kenema, Pujehun	The last significant patch of closed canopy rainforest in Sierra Leone.
Western Area Peninsula Forest	NonHunting reserve	17,688	Western Area	Forest on the hills outside Freetown which provides much of the capital's fresh water supply.
Lima Mountains	NonHunting reserve	33,201	Koinadugu	Site of the country's highest mountain, Mount Bintumani, (1,948 metres).
Kangari Hills	NonHunting reserve	8,573	Bo, Tonkolili	Steep-sided range of hills in the centre of the country that provide an important habitat for wildlife and could be a release site for reintroduced chimpanzees.
Tingi Hills	NonHunting reserve	10,519	Koinadugu, Kono	Remote area of North-eastern Sierra Leone renowned for its batholiths
Tiwai Island	Wildlife sanctuary and community conservancy.	1,200	Pujehun, Kenema	This small island in the Moa river is an important habitat for primates, birds and the very rare pygmy hippo.
Kambui Hills	Forest Reserve	21,228	Kenema	Forest on low-lying range of hills west of Kenema which is threatened by logging and mining.
Sierra Leone River Estuary	Ramsar site	295,000	Port Loko, Western Area	The country's only Ramsar site, and as such the only marine area afforded any level of protection

Source: World Bank, 2006.

APPENDIX 18; TABLE 16: Sierra Leone's production and trade in primary wood products, 2002. 2012 and 2022

Category	2002		2012		2022	
	Unprocessed wood	Processed wood	Unprocessed wood	Processed wood	Unprocessed wood	Processed wood
Production (1000 m ³)	27.723	818	37.864	1.006	50.327	984

Export quantity (1000 m ³)	-	401	159	194	398	178
Import quantity (1000 m ³)	-	-	9	2	42	47
Export value (million US\$)	-	137.614	58.824	123.272	165.310	110.905
Import value (million US\$)	-	-	1.593	0.742	14.013	17.778

Source: World Bank 2022, own table.

APPENDIX 19; TABLE 17: Data for regression models.

	GDP (Bi.US\$)	Inflation (%)	Unemployment (%)	Labour Participation (%)	Exports (Mi. US\$)	Primary Enrolment (Ths.)	Natural Resource Rent (Ths.US\$)	Carbon (Mt Tns Per Capita)
2010	2,58	7,19	4,17	34,42	17,642	6,262	10,528	0,09
2011	2,94	6,79	4,29	33	18,540	8,043	10,986	0,08
2012	3,8	6,59	4,4	31,61	14,249	8,713	8,770	0,09
2013	4,92	5,52	4,51	30,25	12,960	8,891	7,226	0,11
2014	5,02	4,64	4,68	28,91	11,704	9,274	8,027	0,13
2015	4,22	6,69	4,74	28,36	12,483	12,469	9,571	0,15
2016	3,67	10,89	4,66	28,02	14,162	12,346	11,673	0,16
2017	3,72	18,22	4,66	27,68	14,477	13,883	11,936	0,12
2018	4,09	16,03	4,66	27,31	11,142	19,350	7,425	0,12
2019	4,08	14,81	4,65	26,91	13,087	20,839	8,555	0,12
2020	4,06	13,45	5,2	26,42	12,432	25,002	8,229	0,11
2021	4,04	11,87	5,33	26,59	15,970	25,952	9,037	0,11

APPENDIX 20;TABLE 18: Demographic characteristics of respondents in frequencies and percentages.

Demographic characteristics	Frequency	Percentage
Sex		
Female	147	48.84
Male	154	51.16
Total	301	100
Level of education		
None	1	0.33
Basic	240	80.1
Secondary/High School	52	17.33
Tertiary	7	2.33
Total	300	100
Marital status		
Single	25	8.31
Married	261	86.71
Widowed	15	4.98
Total	301	100
Religion		
Christianity	23	7.64
Islam	250	83.06
Traditionalist	28	9.30
Total	301	100
Occupation		
Agriculture	9	2.99
Forestry	236	78.41
Petty trading	37	12.29
Craftmanship	13	4.32
Salaried worker	2	0.66
Others	4	1.33
Total	301	100
Gender head		
Female	160	53.16
Male	141	46.84
Total	301	100
Occupation of household head		

Agriculture	3	1
Forestry	237	78.74
Petty trading	29	9.63
Craftmanship	12	3.99
Salaried workers	9	2.99
Others	11	3.65
Total	301	100

APPENDIX 21; TABLE 19: Household characteristics of the respondents.

Variable	Mean	Standard deviation	Minimum	Maximum
Age	42.1	12.7	1	89
Household size	7.2	3.6	1	26
Males in Household	3.6	2.3	0	19
Females in Household	3.8	2.0	0	16
Age below 15 years	3.1	0.2	0	12
Below 15 years males	1.6	1.5	0	8
Below 15 years females	1.7	1.3	0	5
Age 15-65 years.	3.9	2.5	0	16
15-65 males	2.0	1.7	0	10
15-65 females	1.9	1.5	0	10
Above Age 65 years	0.4	1.0	0	9
Above 65 years males	0.1	0.7	0	9
Above 65 years females	0.4	1.2	0	11
Total of Literates.	2.7	2.3	0	12
Male Literates.	1.6	1.7	0	11
Female Literates.	1.5	1.5	0	16
Total of employed	2.5	0.2	0	15

Total unemployed	3.9	2.6	0	16
Monthly expenditure	23.3	0.5	1	12
Monthly income	51.6	0.3	1	15
Education Expenses	19.7	2.0	0	8

APPENDIX 22; TABLE 20: CFA of socio-economic and environmental impacts.

Factors and variables	Factor loadings	Mean score	Eigen values	% of variance
Environmental Impacts			1.08	0.12
Silting of Rivers and lakes	0.7	4.3		
Damaging of Trees and NWPs	0.6	3.8		
Loss of biodiversity	0.9	3.9		
Loss forest cover	0.6	4.1		
Disappearance of vegetation	0.5	4.1		
Climate change and global warming	0.6	4.1		
Economic Impact			1.80	0.20
High labor cost	0.5	4.4		
Government revenue loss	0.9	3.8		
Increase cost of Wood Products	0.7	3.9		
Social impact			1.63	0.19
Land compensation and disputes.	0.8	3.8		
Rural-Urban migration.	0.9	4.0		

APPENDIX 23; TABLE 21: Results from regression analysis using EViews 10.

Variables	Model 1	Model 2	Model 3
	Carbon	GDP	Primary Enrollment
Natural rents	0.0067** (0.0026 et)	-0.1668** (0.0579 et)	0.0855 (0.2037 et)
Exports	0.0671 (0.0001 et)	-0.0095**	0.0001**

		(0.0375 et)	(0.0139 et)
Gross Domestic Products	0.0266** (0.0090 et)		
Labor		0.3182** (0.0574 et)	-0.2019 (0.3509 et)
Inflation		-0.0273 (0.0416 et)	
Unemployment			6.0770*
Constant	-0.0596 (0.0538 et)	15.6823** (1.9433 et)	-13.6689 (21.4351 et)
Software used is EViews 10			

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12)

APPENDIX 24; TABLE 22: Coefficients of determination for the models.

Prob(F-statistic)	0.0061	0.0042	0.0000
R-squared	0.5734	0.8596	0.9667
Adj. R-squared	0.4134	0.7794	0.9477
Test statistics			
Normality	0.6528 (0.7215 et)	0.042 (0.9792 et)	0.6583 (0.7195 et)
Heteroscedasticity	3.6022 (0.0654 et)	0.8963 (0.514 et)	10.0355 (0.5732 et)
Serial Correlation	2.6383 (0.1506 et)	4.4074 (0.0788 et)	1.2576 (0.361 et)
Functional Form	0.0124 (0.9145 et)	0.7297 (0.4258 et)	0.1641 (0.6994 et)
Cumulative sum	Stable	Stable	Stable
Cumulative sum ²	Stable	Stable	Stable

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12). Et denotes error term.

APPENDIX 25; TABLE 23: Variables and coefficients for model 1.

Variables	Coefficient	Standard Error	t-Statistic	Probability.
Carbon emission	-	-	-	-
Natural resource rent	0.0067**	0,0026	2,6305	0,0302
EXPORTS	0.0671	1,0051	0,6714	0,5209
GDP	0.0266**	0,0090	2,9407	0,0187

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12). Et denotes error term.

APPENDIX 26; TABLE 24:Environmental impact statistical tests of significance.

Dependent Variable	CARBON
Method	Least Squares
Date:	11/18/23
Time:	17:50
Sample:	2010 2021
Included observations	12
Prob(F-statistic)	0.0061
F-statistic	3,5837
R-squared	0.5734
Adj. R-squared	0.4134
Normality	0.6528
Heteroscedasticity	3.6022
Serial Correlation	2.6383
Functional Form	0.0124
Cumulative sum	Stable

Cumulative sum of square	Stable
S.E. of regression	0,0180
Sum squared residue	0,0026
Log likelihood	33,598
Mean dependent variable	0,1158
S.D. dependent variable	0,0235
Akaike info criterion	-4,933
Schwarz criterion	-4,7713
Hannan-Quinn criteria	-4,9928
Durbin-Watson statistics	1,6819

APPENDIX 27; TABLE 25: Variables and coefficients for model 2.

Variables	Coefficient	Standard Error	t-Statistic	Probability
GDP	-	-	-	-
Natural resource rent	-0.1668**	0,0579	-2,8818	0,0236
Exports	-0.0095**	3,7551	-2,5291	0,0393
Laour participation	0.3182**	0,0574	-5,5397	0,0009
Inflation	-0.0273	0,0090	2,9407	0,0187

Notes: The asterisks (*, **, ***) denote statistical significance at 0.1, 0.05, and 0.01 levels respectively; using observations 2010-2021 (T = 12)

APPENDIX 28; TABLE 26: Economic impact statistical tests of significance

Dependent Variable:	GDP
Method:	Least Squares
Date:	11/18/23
Time:	18:17
Sample:	2010 2021
Included observations:	12

Prob(F-statistic)	0.0042
F-statistic	10,71461
R-squared	0.8596
Adj. R-squared	0.7794
Normality	0.042
Heteroscedasticity	0.8963
Serial Correlation	4.4074
Functional Form	0.7297
Cumulative sum	Stable
Cumulative sum of square	Stable
S.E. of regression	0,324823
Sum squared residue	0,73857
Log likelihood	-0,29959
Mean dependent variable	3,928333
S.D. dependent variable	0,691544
Akaike info criterion	0,883265
Schwarz criterion	1,085309
Hannan-Quinn criteria	0,808461
Durbin-Watson statistics	1,888264

APPENDIX 29; TABLE 27: Variables and coefficients for model 3.

Variables	Coefficient	Standard Error	t-Statistic	Probability
Primary Enrolment	-	-	-	-
Natural resource rent	0.0855	0,2037	0,4196	0,6874
Exports	0.0001**	1,3951	5,8117	0,0007
Labour	-0.2019	0,3509	-0,5752	0,5831

Unemployment	6.0770*	1,0051	0,6714	0,5209
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APPENDIX 30; TABLE 28: Social impact statistical tests of significance

Dependent Variable	Primary enrolment
Method	Least Squares
Date	11/18/23
Time	18:20
Sample	2010 2021
Included observations	12
Prob(F-statistic)	0.0000
F-statistic	50,86933
R-squared	0.9667
Adj. R-squared	0.9477
Normality	0.6583
Heteroscedasticity	10.0355
Serial Correlation	1.2576
Functional Form	0.1641
Cumulative sum	Stable
Cumulative sum of square	Stable
S.E. of regression	1,562939
Sum squared residue	17,09944
Log likelihood	-19,1521
Mean dependent variable	14,24804
S.D. dependent variable	6,836727
Akaike info criterion	4,02535
Schwarz criterion	4,227394
Hannan-Quinn criteria	3,950546

Durbin-Watson statistics	2,464454
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APPENDIX 31; TABLE 29: Hypothesis results.

Hypothesis	Prob(F-stat)	R ²	Adj. R ²	Serial correl.	Heteroscedasticity	Normality test	Results
H ₁	0.0061	0.5734	0.4134	2.6383	3.6022	0.6528	Supported
H ₂	0.0042	0.8596	0.7794	4.4074	0.8963	0.042	Supported
H ₃	0.0000	0.9667	0.9477	1.2576	10.0355	0.6583	Supported

APPENDIX 32; TABLE 30: Sierra Leone trade balance and growth rates from 2009 to 2011.

Year	Export (\$'000)	import (\$'000)	Trade balance (\$'000)	Export growth rate
2009	9,919	5,439	4,480	43.1
2010	7,642	8,049	-407	-23.0
2011	8,540	8,116	424	11.8
2012	4,249	5,667	-1,418	-50.2
2013	29,607	5,998	23,609	596.8
2014	17,047	9,908	7,139	-42.4
2015	2,483	11,316	-8,833	-85.4
2016	41,629	8,589	33,040	1,576.6
2017	44,773	7,568	37,205	7.6
2018	111,428	8,185	103,243	148.9
2019	130,877	8,234	122,643	17.5
2020	124,328	8,005	116,323	-5.0
2021	159,702	8,175	151,527	28.5

Source: World Bank, 2020. Own table.

CONSENT FORM

I have had the purpose and nature of this study on the research titled 'Analysis of the socio-economic and environmental impacts of timber harvesting from the Kambui Hills Forest Reserve' explained to me in words and in writing. I have also had the opportunity to ask questions about the study to ensure my understanding of the research. It is specifically for academic purposes with the broader aim of contributing to national development. In line with the above objective, I have instructed all my subjects within the township of..... to honestly and faithfully answer all questions from the researcher. I voluntarily agree to share my own opinion and advise all my people to do same.

Signed..... Date. 20-2-22
Researcher

Signed..... Date 24/2/22
Town chief — Menims

APPENDIX 33; FIGURE 1: Sample of consent form signed by community heads.

Source: Field survey. 2022.



APPENDIX 34; FIGURE 2: Silt water infested with timber wastes.

Source: Field survey, 2022



APPENDIX 35; FIGURE 3: Soil erosion.

Source: field survey, 2022



APPENDIX 36; FIGURE 4: Disruption of hydrological circle.

Source: Primary survey, 2022



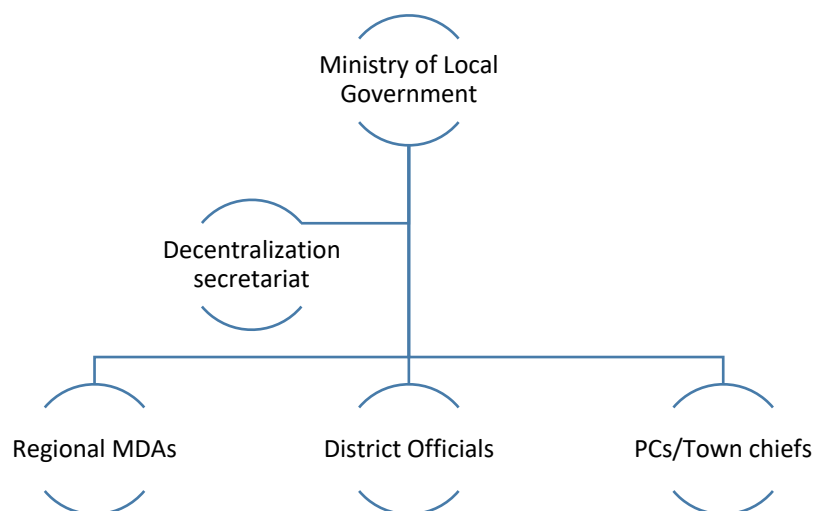
APPENDIX 37; FIGURE 5: Tree stump from timber harvesting.

Source: Field survey, 2022



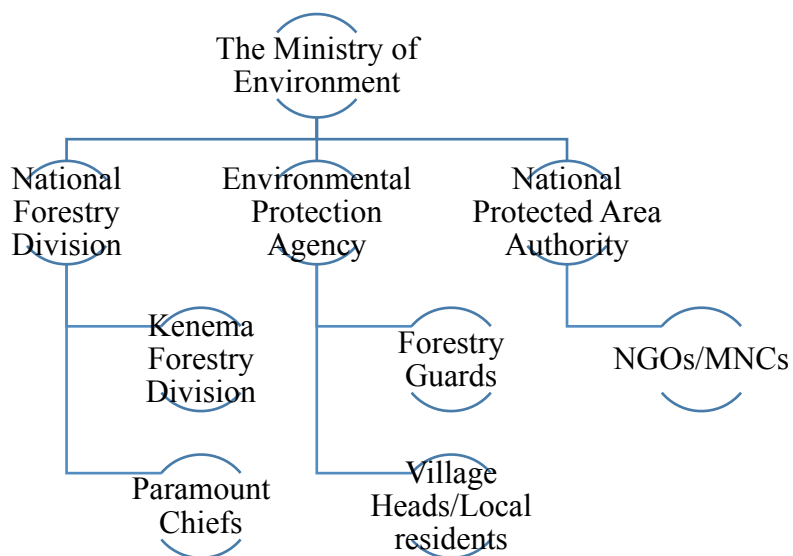
APPENDIX 38; FIGURE 6: Map of West Africa.

Source: World Bank, 2021.



APPENDIX 39; FIGURE 7: Organogram of decentralization.

Source: Own diagram, 2023.



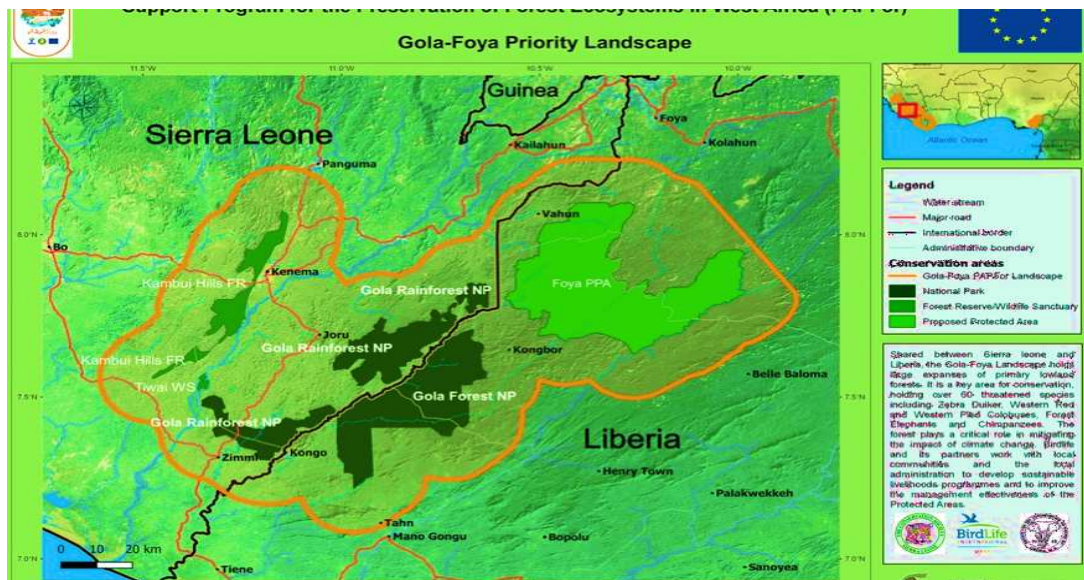
APPENDIX 40; FIGURE 8: Organogram of the ministry's decentralization.

Source: Own diagram



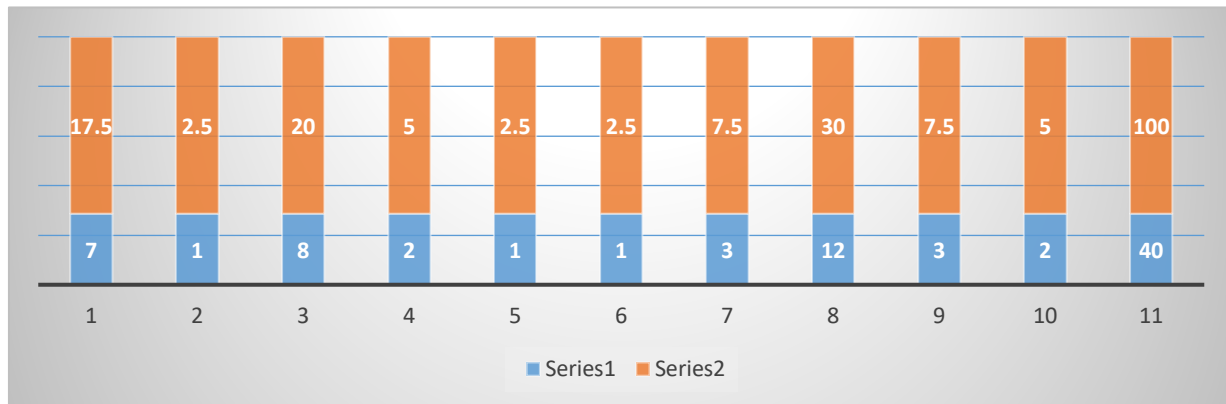
APPENDIX 41; FIGURE 9: Machinery abandoned by Gava company.

Source: Kenema District Council, 2022.



APPENDIX 42; FIGURE 10: Map of Gola forest showing KHFR.

Source: The Kenema District Forestry Office Archives, 2022.

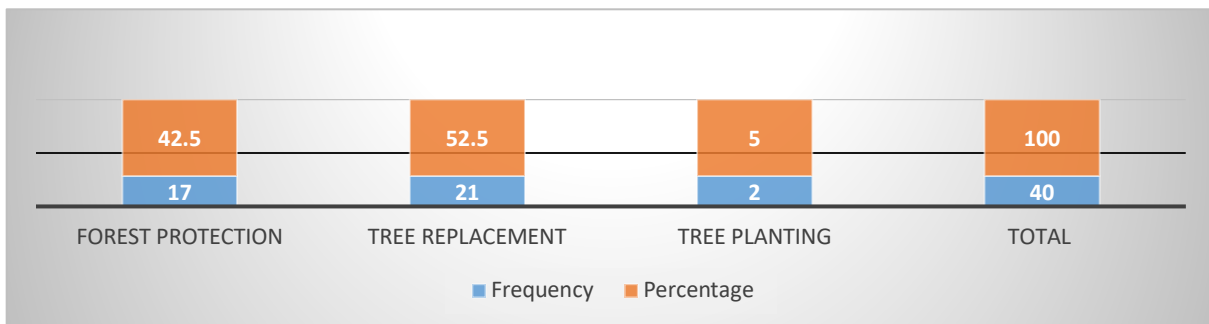


Key:

1 = Firms' specific CSR initiatives. 2 = management policies. 3 = Community sensitization. 4 = workers consultations. 5 = Firms and government consultations corporations. 6 = Community consultations. 7 = Community sensitizations. 8 = Policy consultation. 9 = stakeholders' consultations. 10 = Joint consultations. 11 = Total number of respondents.

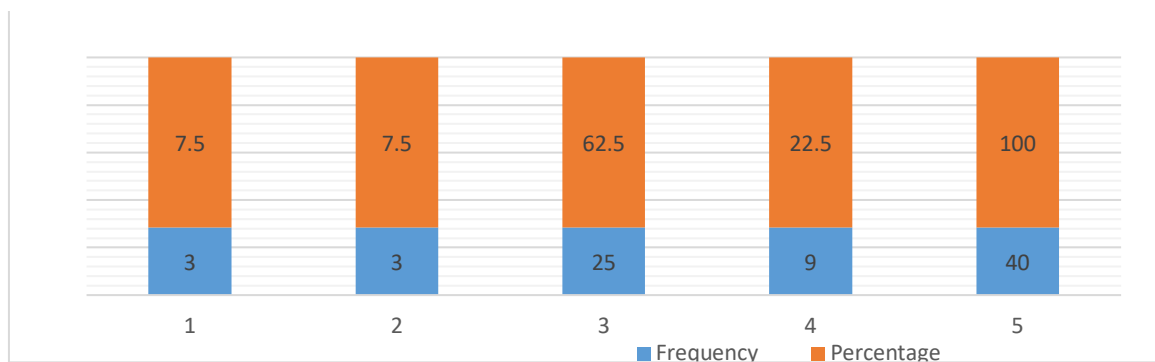
APPENDIX 43; FIGURE 11: Firms' CSR policy formulation processes.

Source: Field survey, 2022.



APPENDIX 44; FIGURE 12: Laid down policies of firms' CSR.

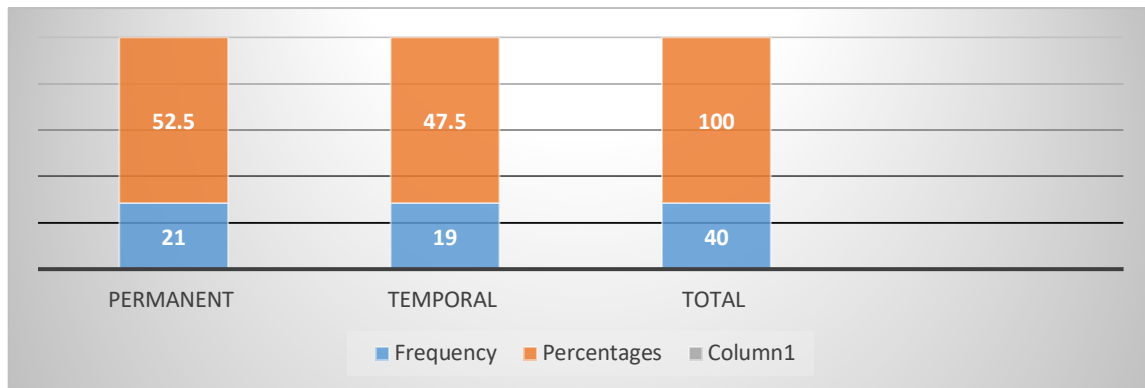
Source: Field survey, 2022.



Key: 1 = Reforestation; 2 = Mitigate illegal logging 3 = Residual recycling 4 = Policy sustainability; 5 = Total

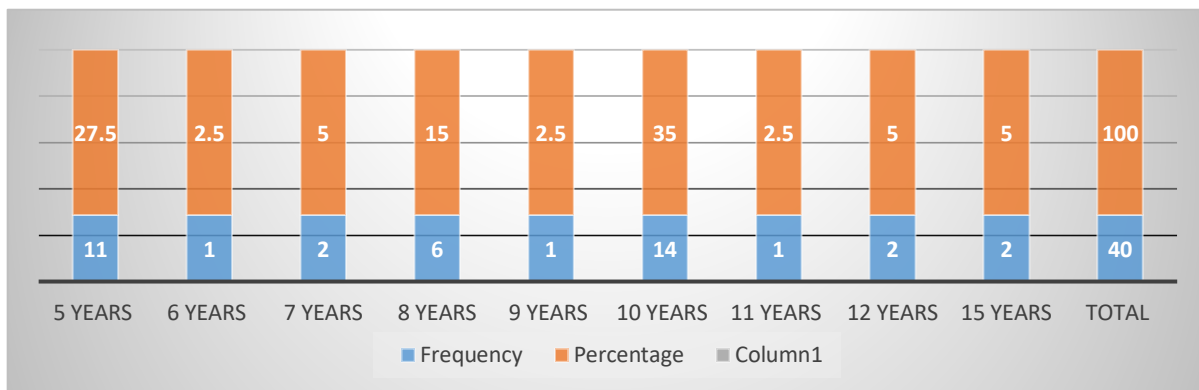
APPENDIX 45; FIGURE 13: Firms' CSR objectives.

Source: Field survey, 2022.



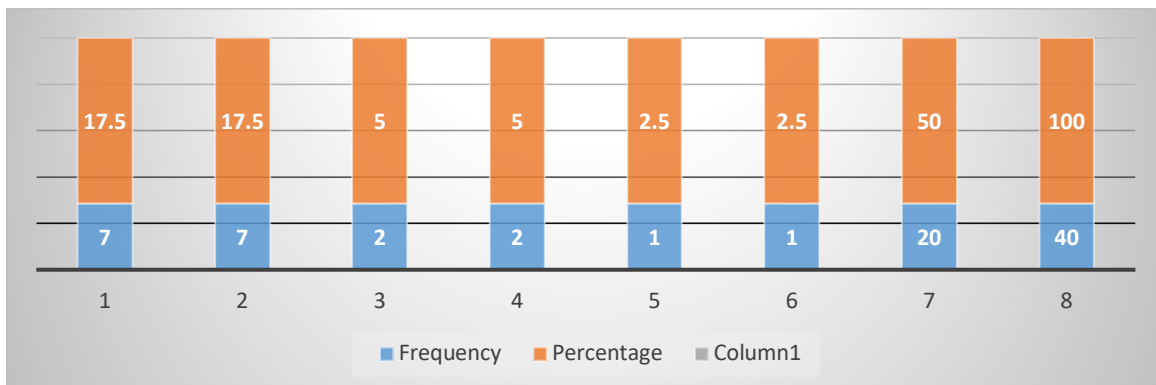
APPENDIX 46; FIGURE 14: Duration of firms' operations.

Source: Field survey, 2022.



APPENDIX 47; FIGURE 15: Life span of firms' CSR.

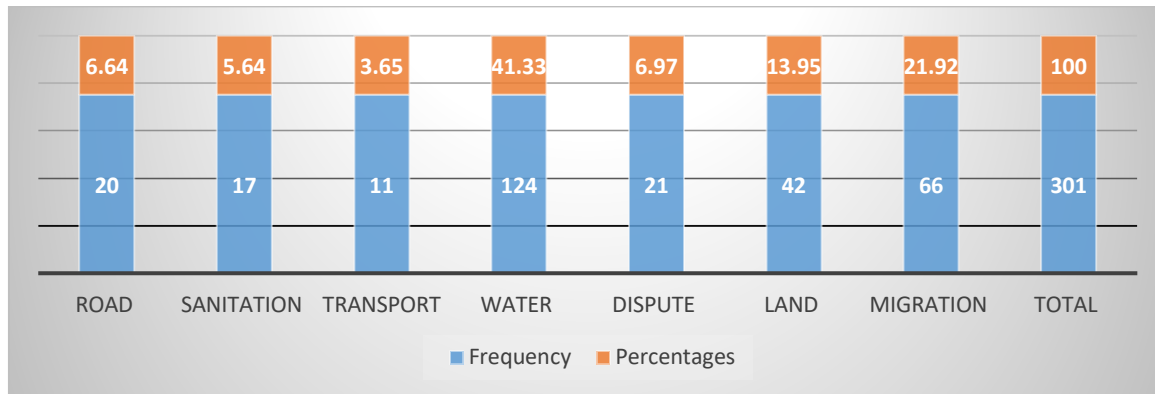
Source: Field survey data, 2022.



APPENDIX 48; FIGURE 16: Measures to ensure CSR sustainability.

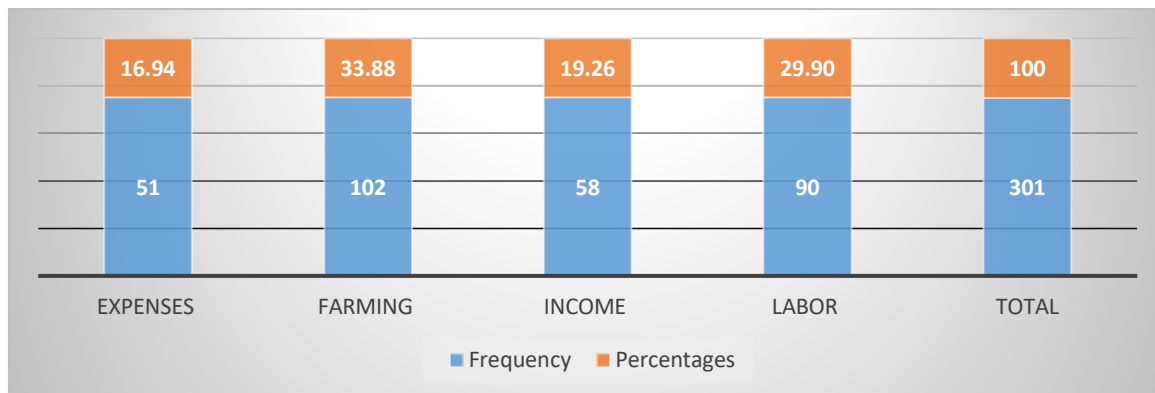
Source: Field survey, 2022.

Key: 1 = Educational level; 2 = Forestation; 3 = Joint Sensitization; 4 = Surveillance; 5 = Workers Consultations; 6 = Workers Sensitization; 7 = Workers Education; 8= Total



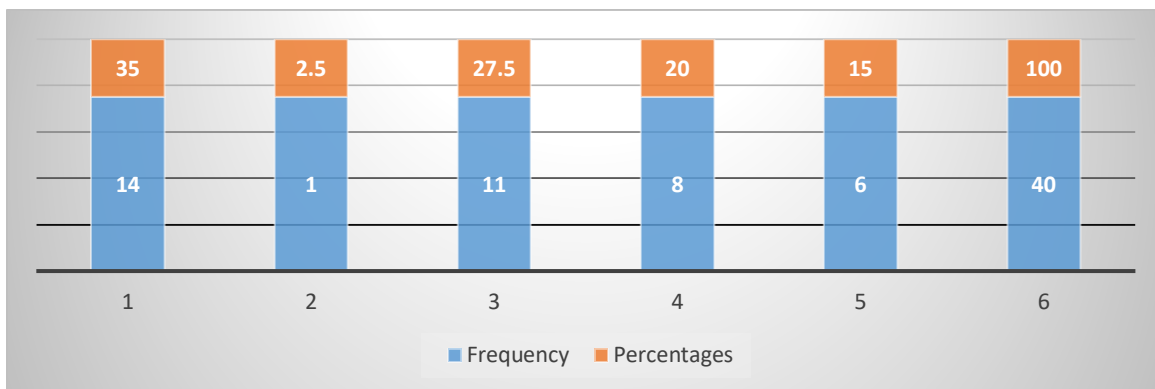
APPENDIX 49; FIGURE 17: Social challenges.

Source: Field survey, 2022.



APPENDIX 50; FIGURE 18: Economic challenges.

Source: Field survey, 2022.



Key: 1 = Climate; 2 = Deforestation; 3 = Pollution; 4 = Weather; 5= Windy; 6 = Total

APPENDIX 51; FIGURE 19: Environmental challenges.

Source: Field survey, 2022.

PUBLICATION

1. **Hlaváčková, P., Mansaray, S.K., Darkwah, S.A., and Zivelova, I. 2023.** Applying the European Union (EU) assessment initiative of forest sustainability in Africa: A case study of timber harvesting impact on the environment in Sierra Leone. Article in print, Journal of Forest Science, Open Access CAAS journals.
2. **Mansaray S.K., Adamu K.J., Darkwah S.A., Zivelova I. 2021.** Assessment of Beef Marketing and Quality Control within the Abeokuta Metropolis, Ogun State, Nigeria. *Int.J.Curr.Res.Aca.Rev.* 9(04), 39-52. doi: <https://doi.org/10.20546/ijcrar.2021.904.005>.
3. **Bamwesigye D., Doli A., Adamu K.J., Mansaray S.K. 2020.** A Review of the Political Economy of Agriculture in Uganda: Women, Property Rights, and Other Challenges. DOI: 10.13189/ujar.2020.080101
4. **Mansaray S.K., Maitah M., Hes T., Malec K. 2015.** Microfinance in Southeastern Sierra Leone. *International Business Management*, Vol. 9, number. 4, s. 498-507. ISSN: 1993-5250
5. **Mansaray S.K., Drarasová M., Afful C., Srnc K. 2015.** China's trade and investment with Africa: A study of Chinese influence in Sierra Leone. *International journal of current research and Academic review*. pp. 210-225, volume 3 number 3. ISSN: 2347 – 3215
6. **Hes T., Mintah S., Sulaiman H., Mansaray S.K. 2016.** Contours of Microcredit Mannequin: Socio-Economic Differentials of Debtors and Non-Debtors in Sierra Leone. *International Journal of Scientific & Technology Research*. Impact Factor: 3.023 Vol. 1 Issue 1. Ref. number: IJSTR-1216-15739.

1. CONFERENCES

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2. Sheku K. Mansaray, Czech University of Life Sciences Prague (CULS). Impediments to Microfinance Activities among Rural Agricultural Farmers in Southern Sierra Leone. International Scientific conference, 2016. Global and European Challenges for Food Production, Agribusiness and the Rural Economy 14 - 16 September, 2016, Prague, Czech Republic.

3. ERASMUS + STAY ABROAD:

1. 27th January to September 14th 2020 at The Federal University of Agriculture, Abeokuta, Ogun state, Nigeria.
2. 27th November 2021 to 27th March, 2022 at Njala University College, Sierra Leone.